

Random Topology

Higher-Order Connectivity from Preferential Attachment

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Stanford University
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Percolation...

even though this talk is not about percolation

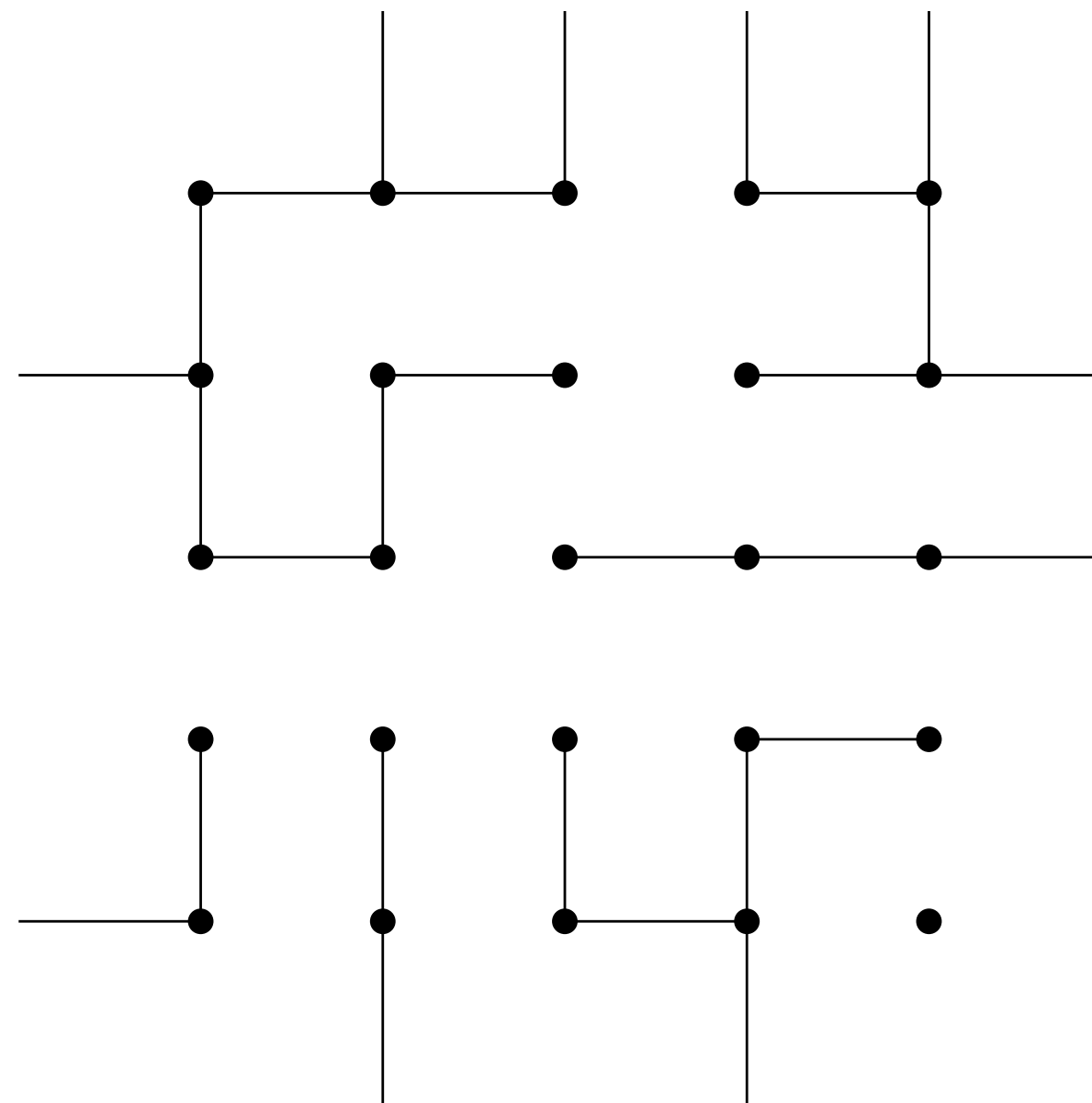
Percolation...

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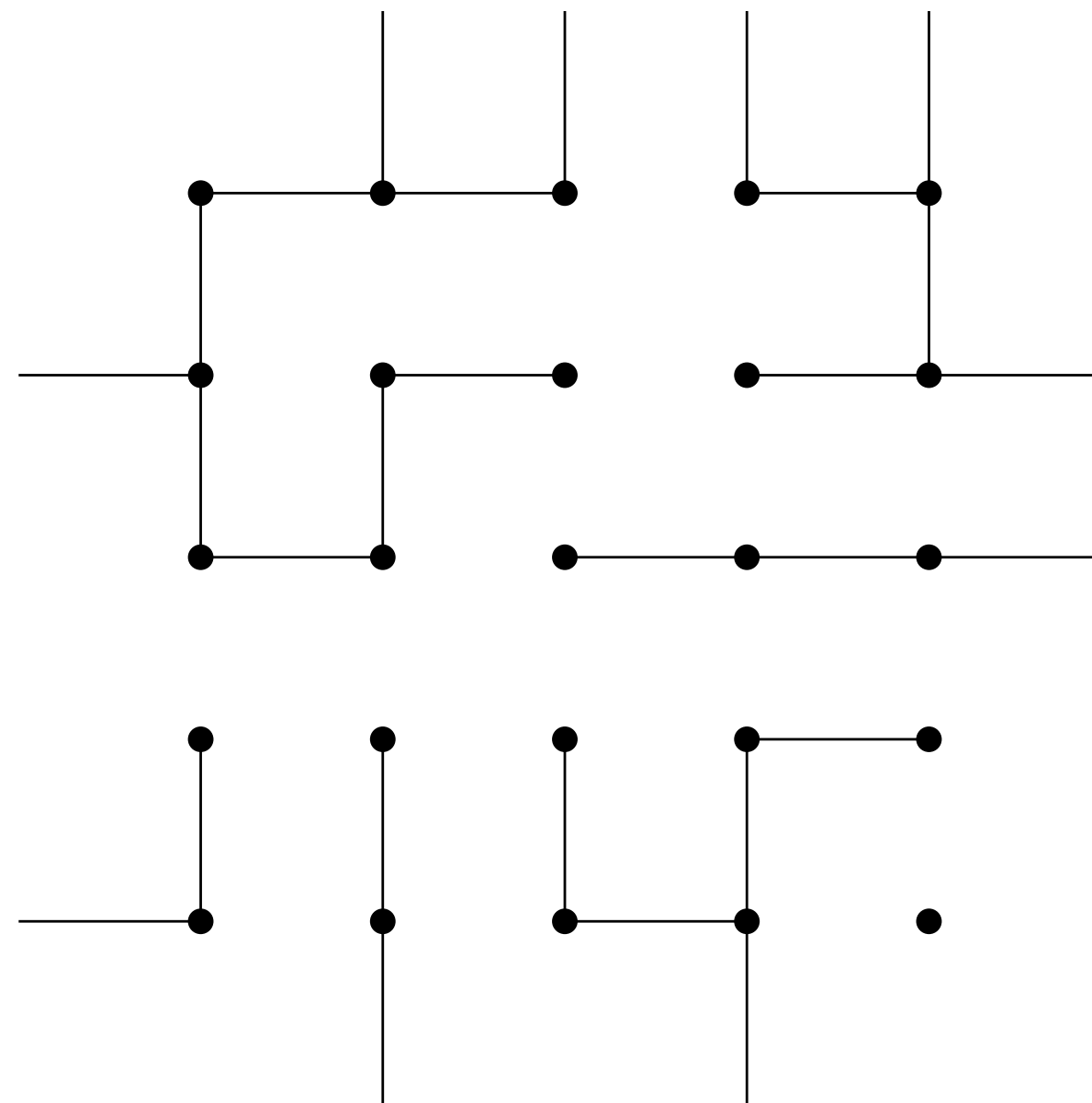
Percolation...

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Percolation...

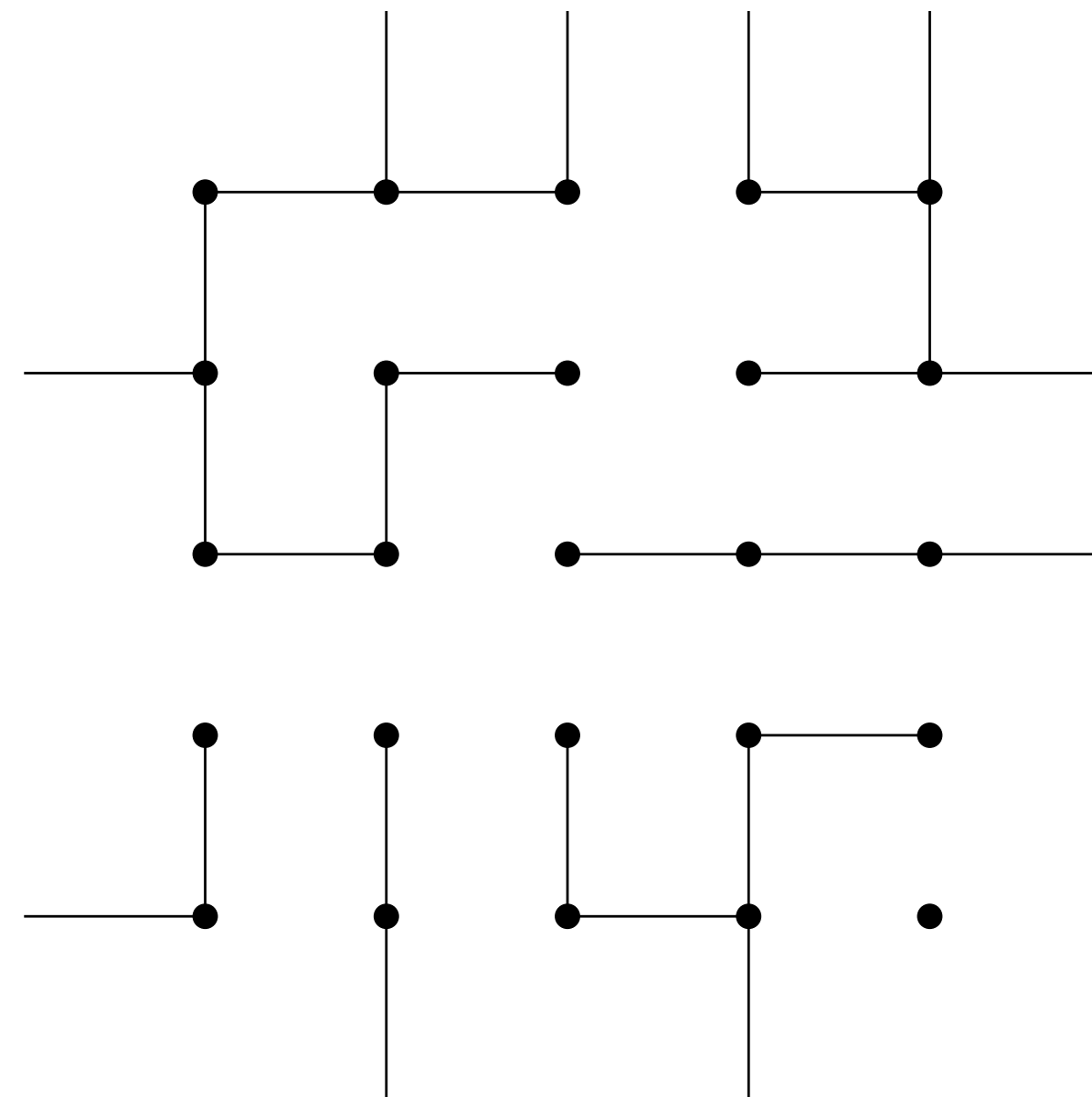
even though this talk is not about percolation



- Connectivity

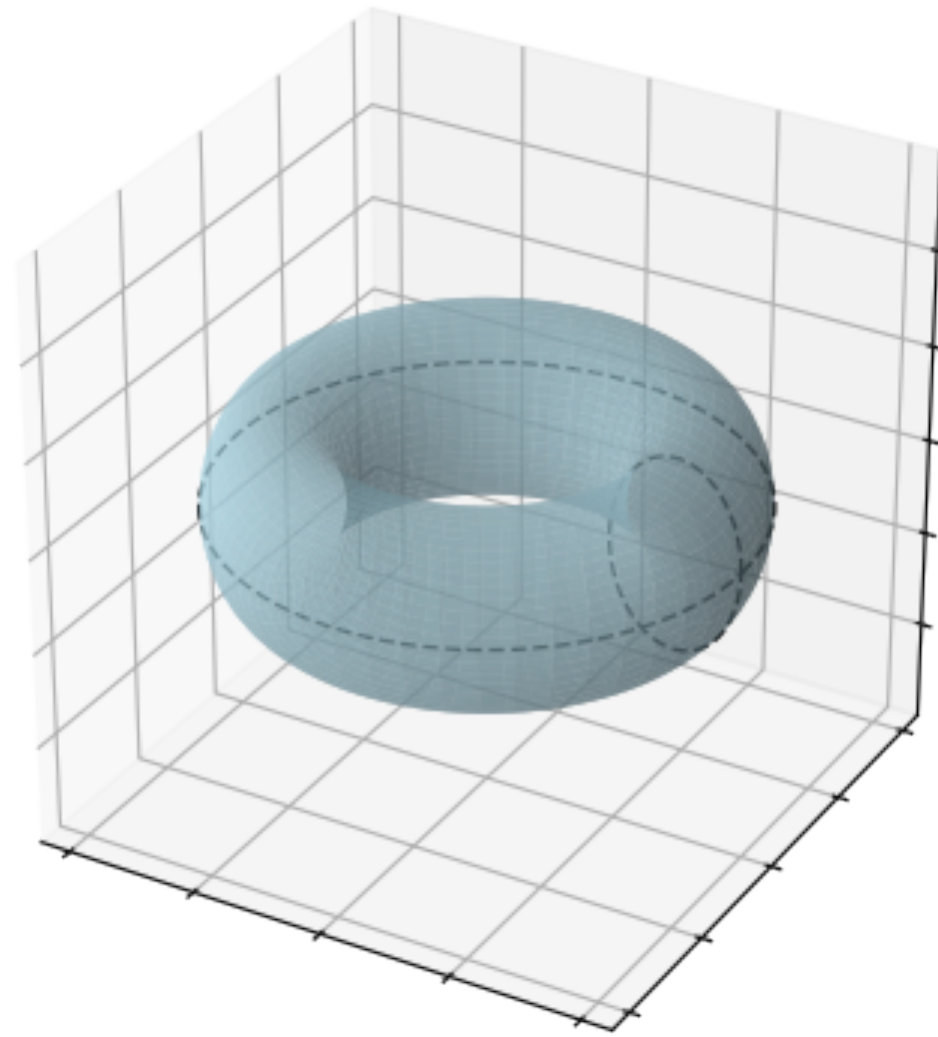
Percolation...

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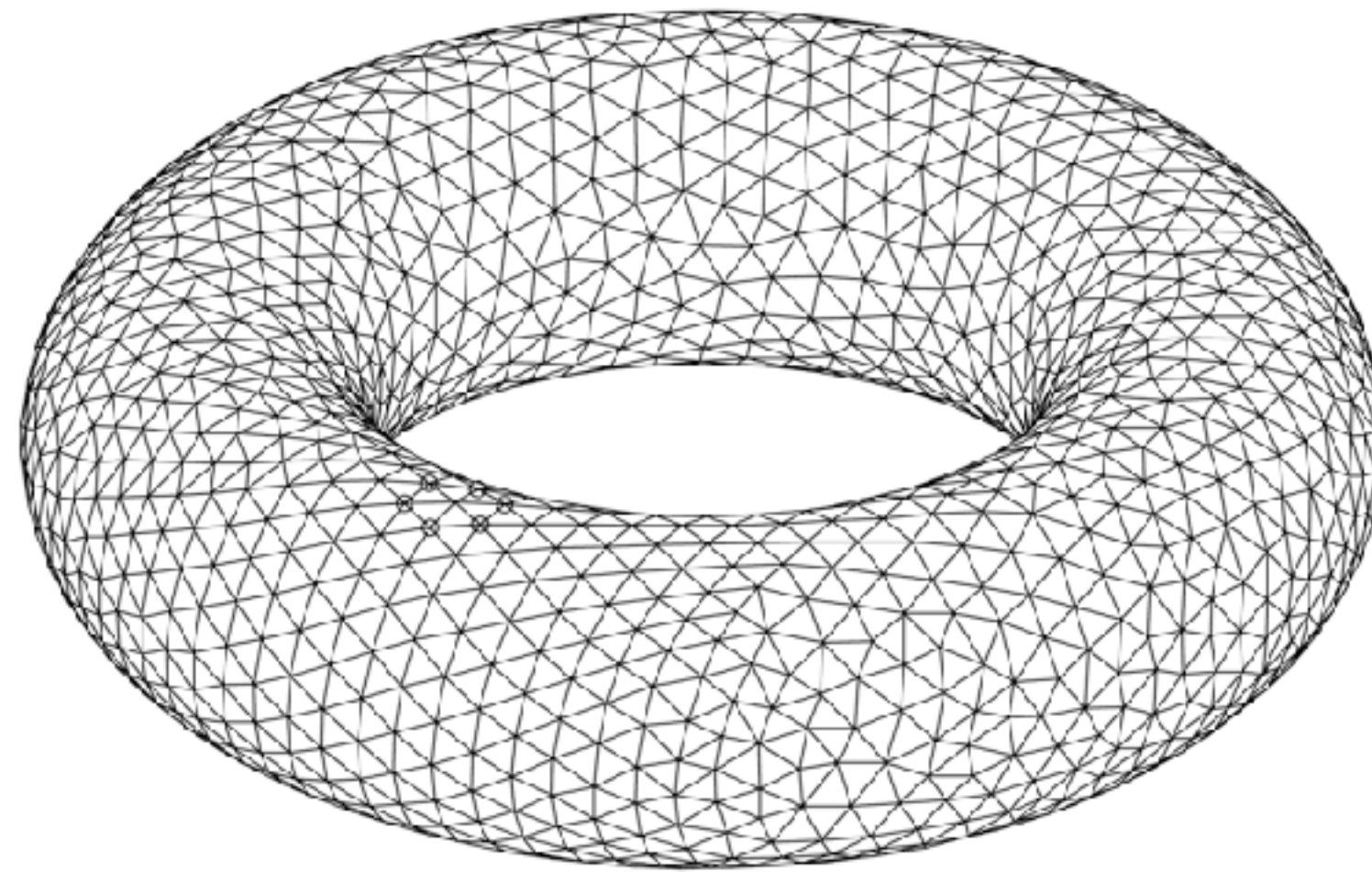


- Connectivity
- random fluctuations vs global structures

Topology

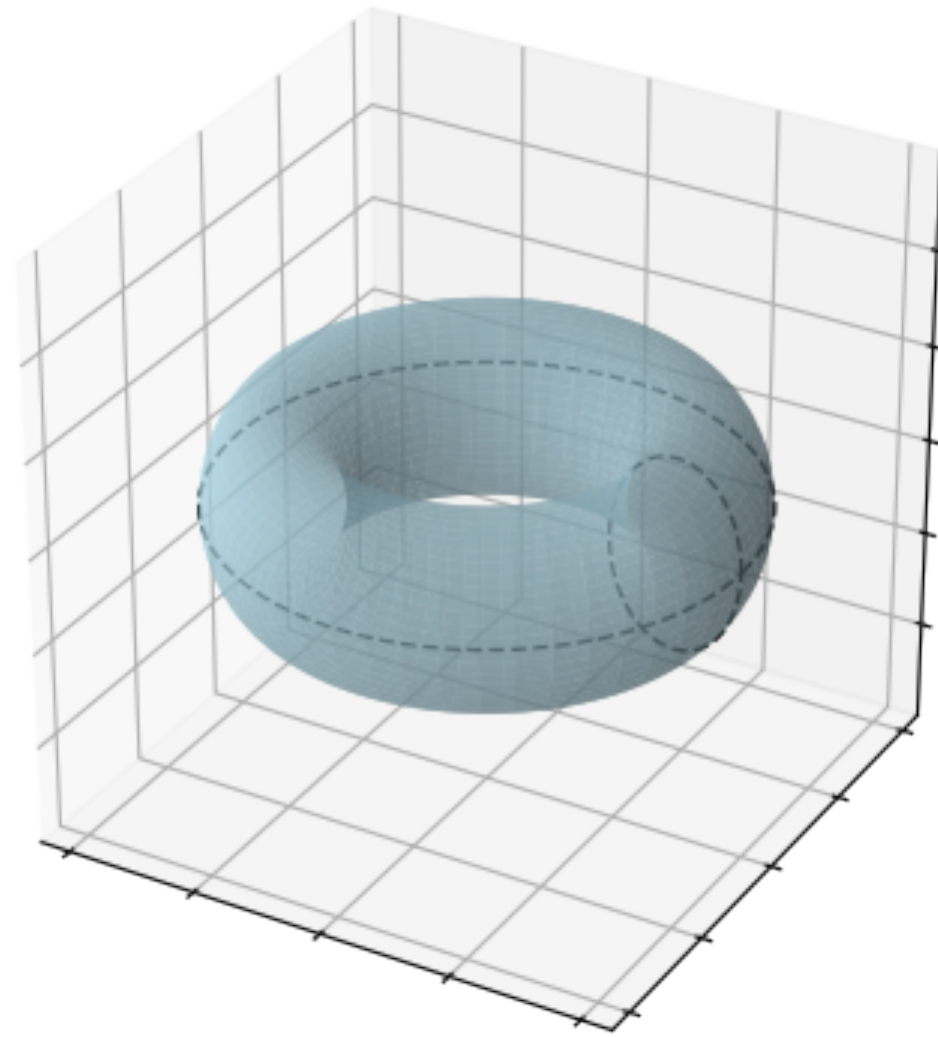


Andrey Yao

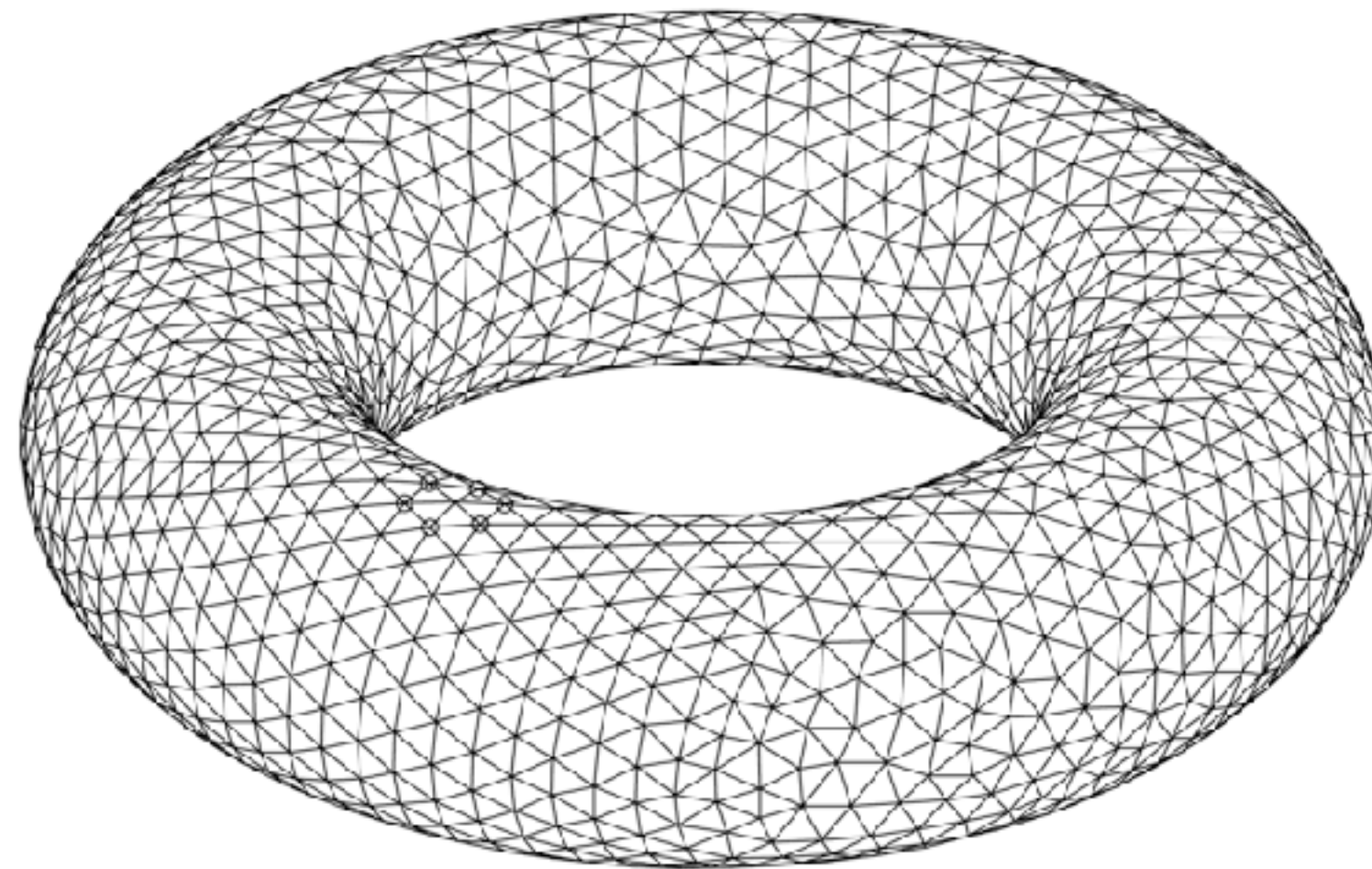


Maintenance Script (?)
<https://handwiki.org/wiki/File:Torus-triang.png>

Topology



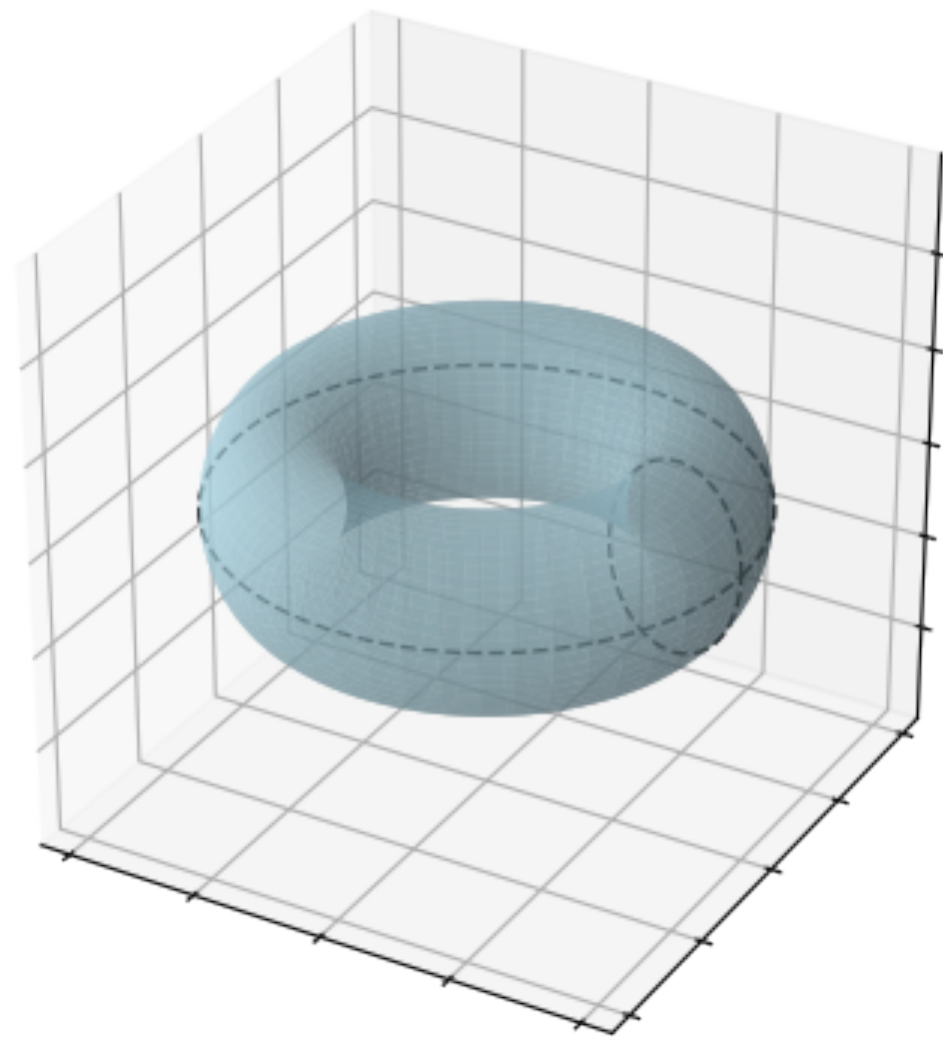
Andrey Yao



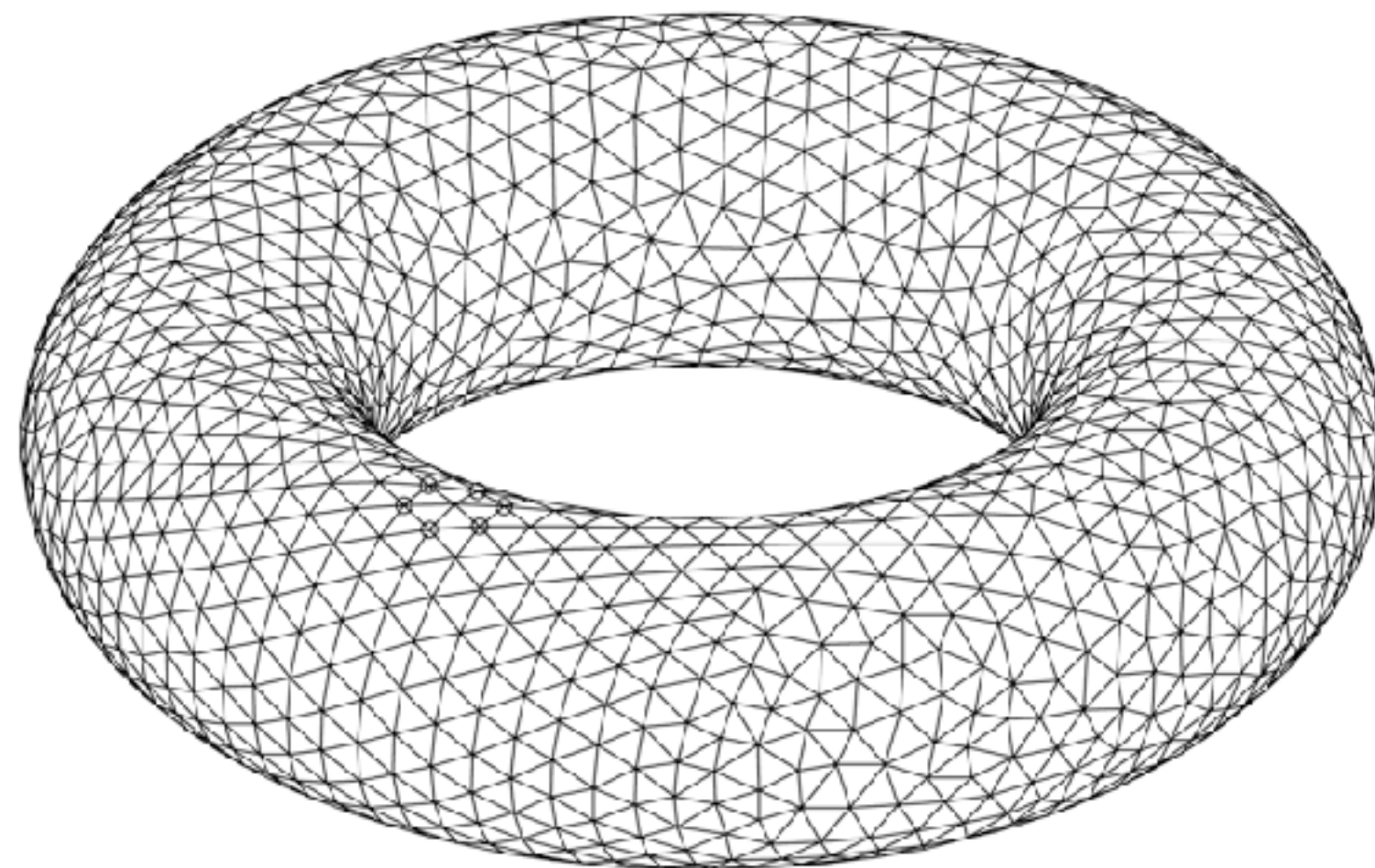
Maintenance Script (?)
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New Models

Topology



Andrey Yao

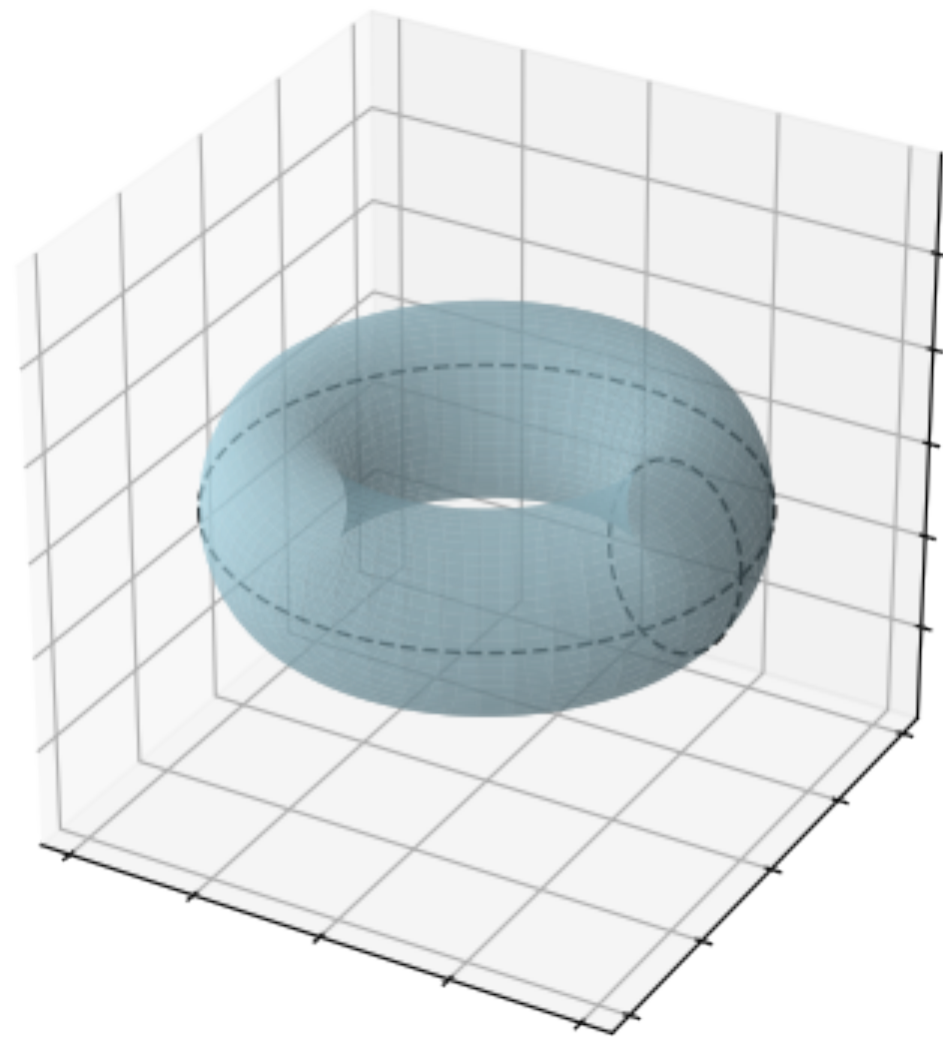


Maintenance Script (?)
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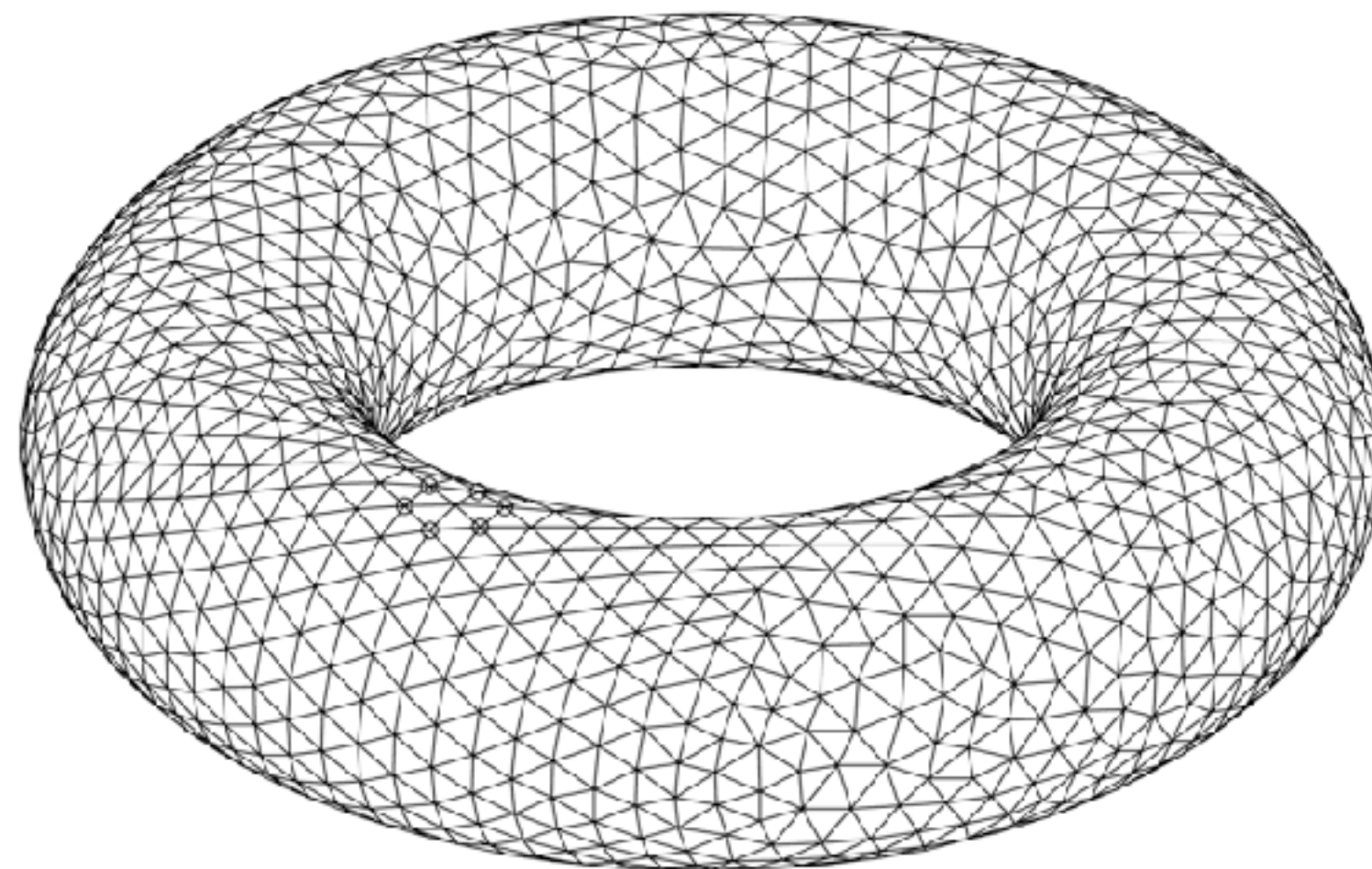
New Models

New Questions

Topology



Andrey Yao



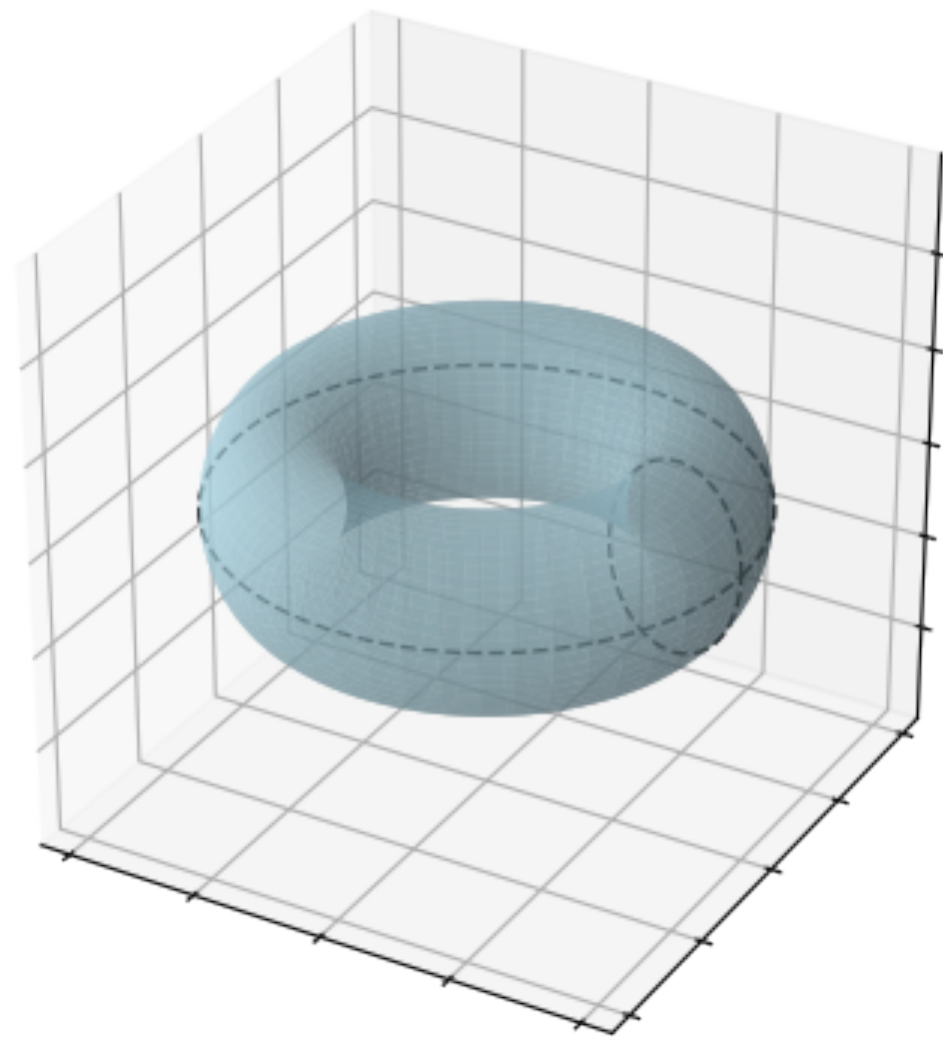
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New Models

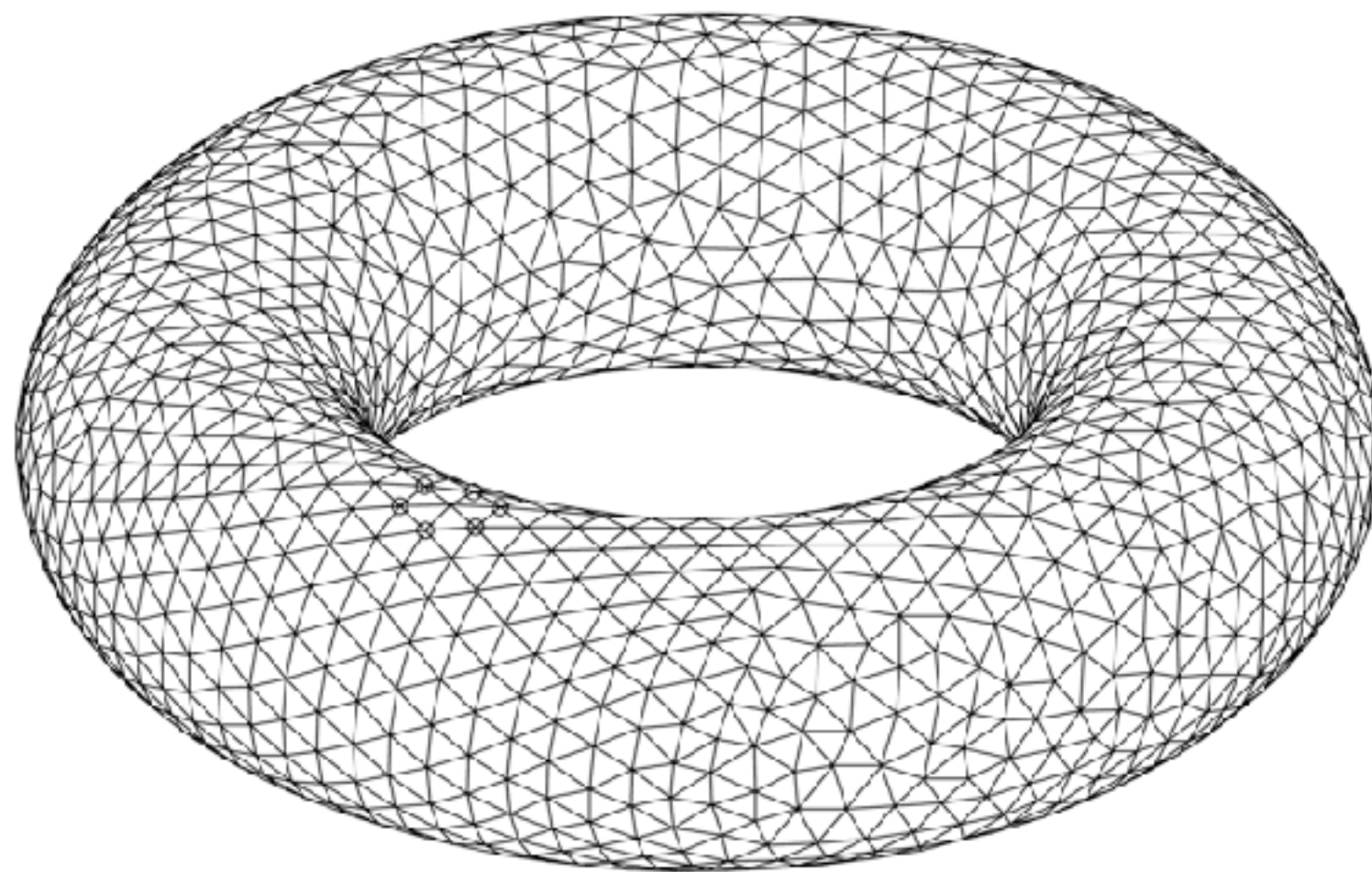
New Questions

New Techniques

Topology



Andrey Yao



Maintenance Script (?)
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New Models

New Questions

New Techniques

New Answers

Agenda

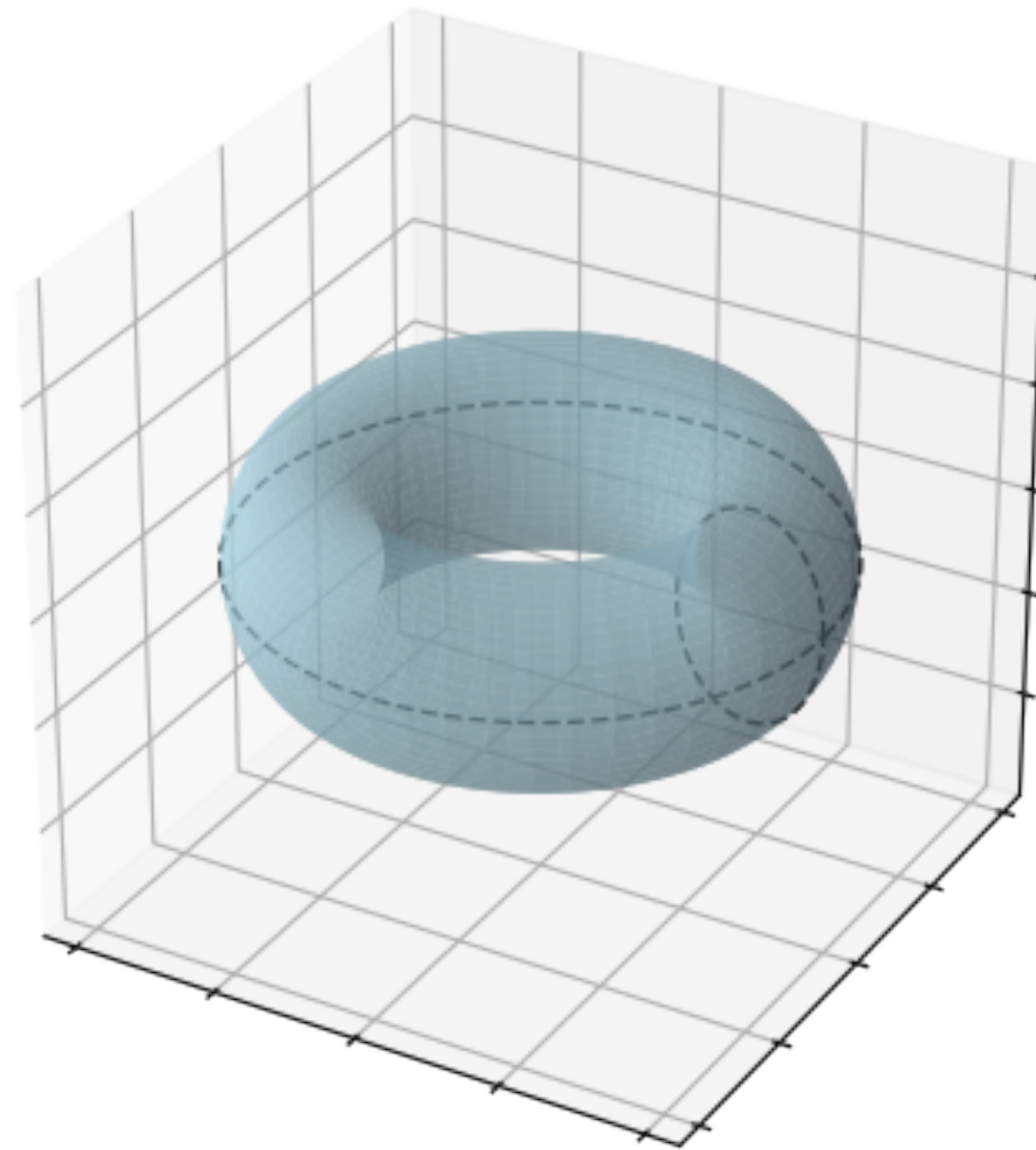


preferential attachment

Agenda

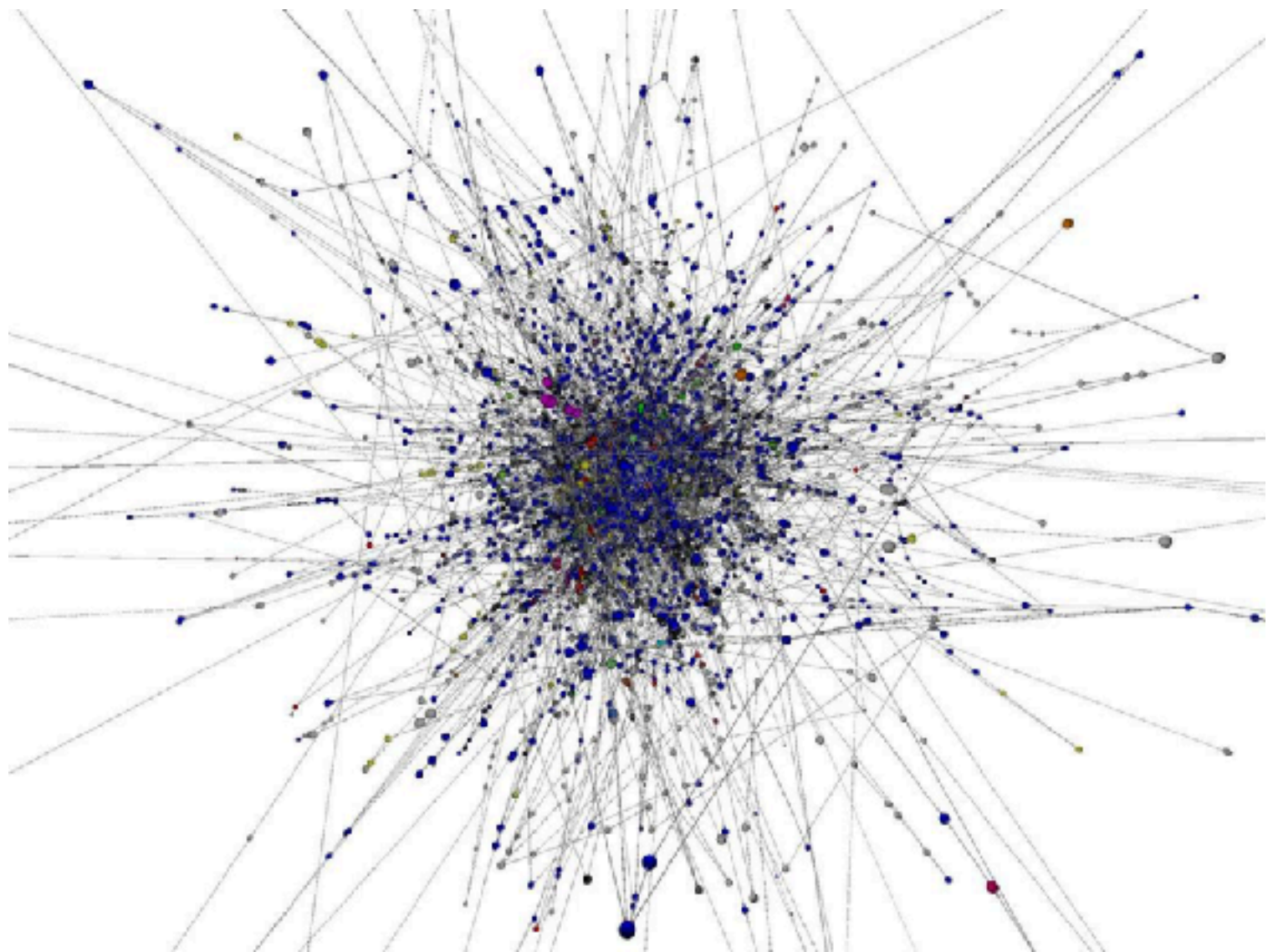


preferential attachment

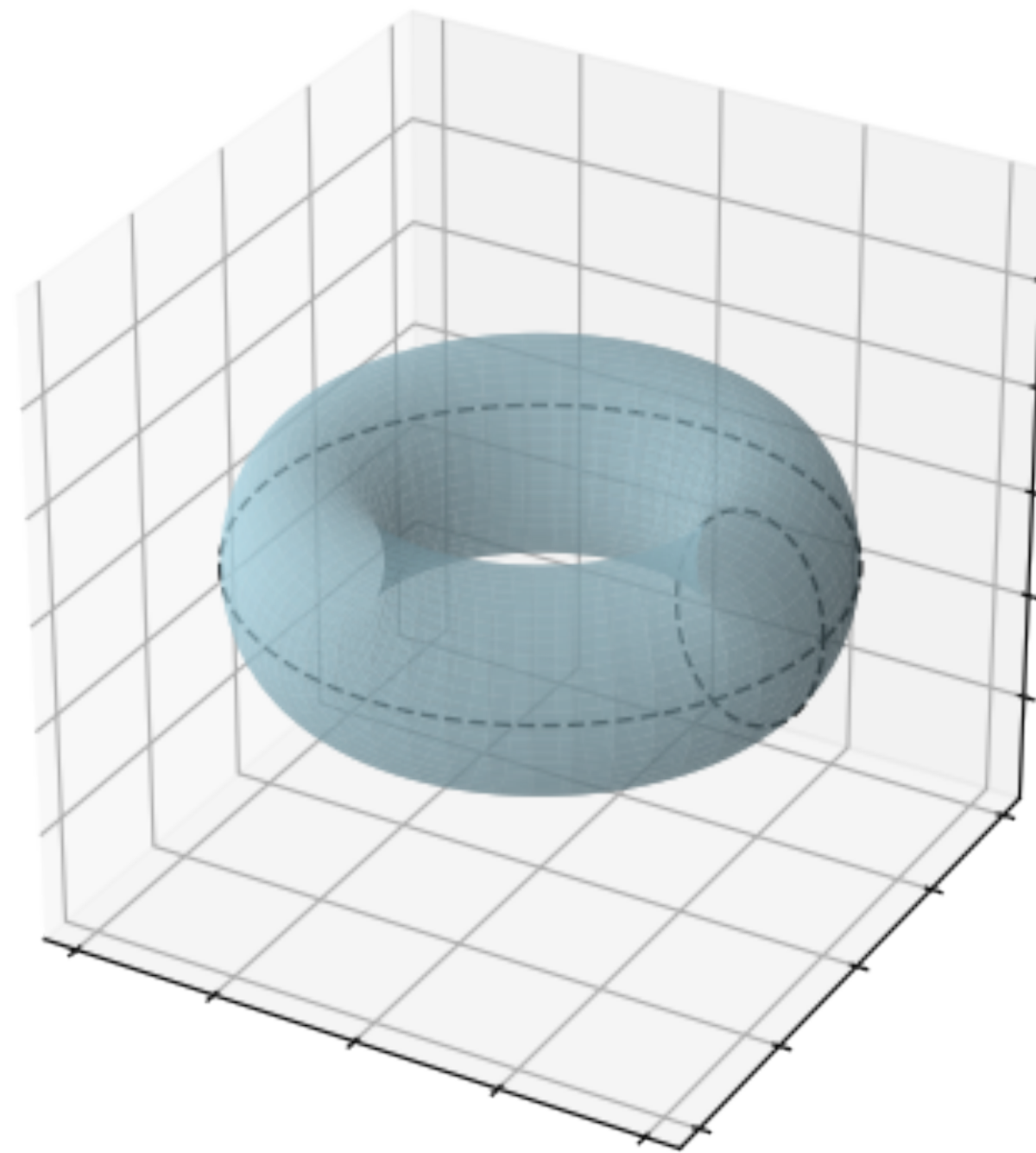


topology

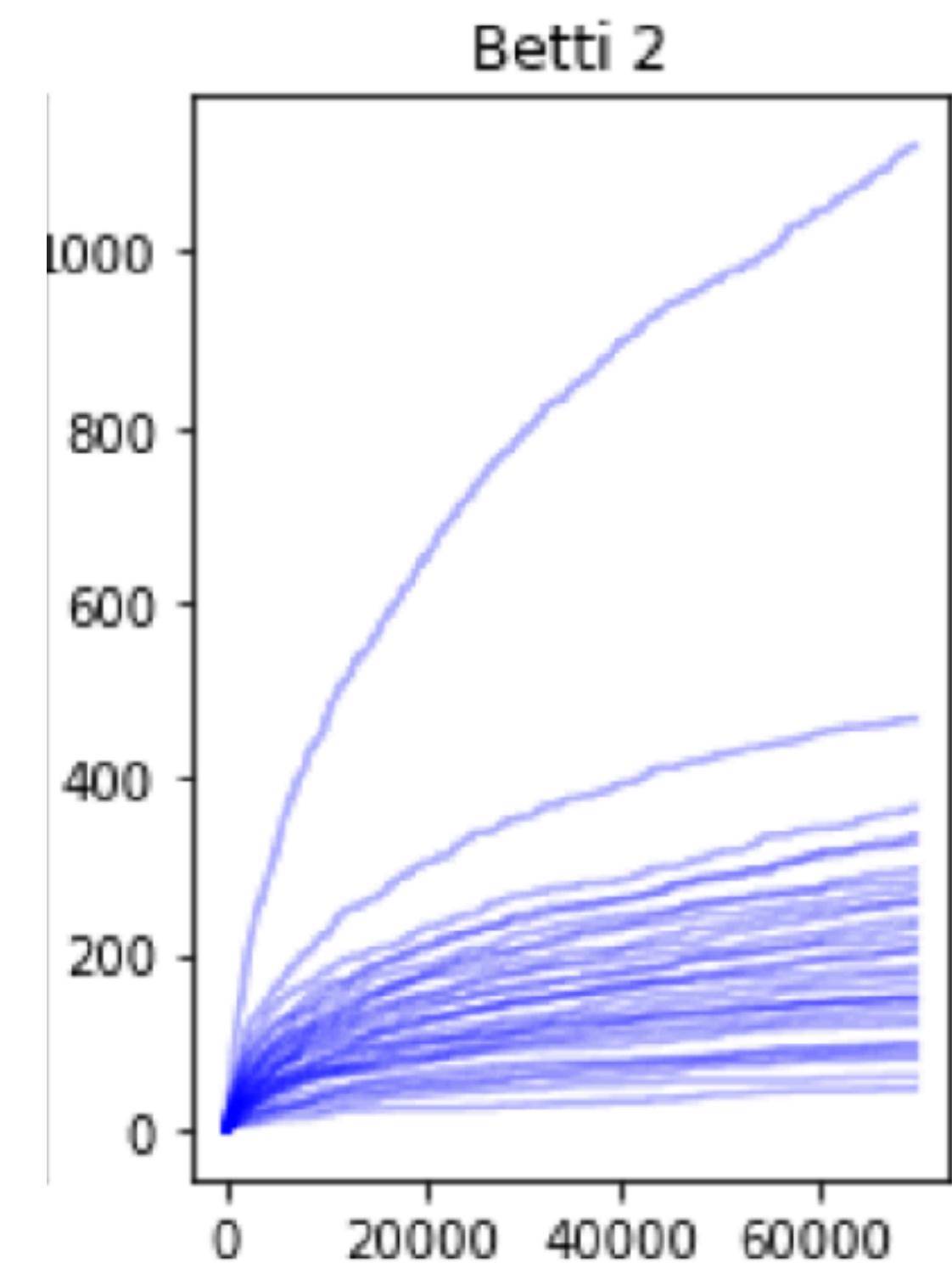
Agenda



preferential attachment



topology



our result

I. Preferential Attachment

.....

So, preferential attachment...



(Stephen Coast
<https://www.fractalus.com/steve/stuff/ipmap/>)

So, preferential attachment...

- Highly connected hubs



(Stephen Coast
<https://www.fractalus.com/steve/stuff/ipmap/>)

So, preferential attachment...

- Highly connected hubs
- Dense core of hubs?



(Stephen Coast
<https://www.fractalus.com/steve/stuff/ipmap/>)

Preferential Attachment

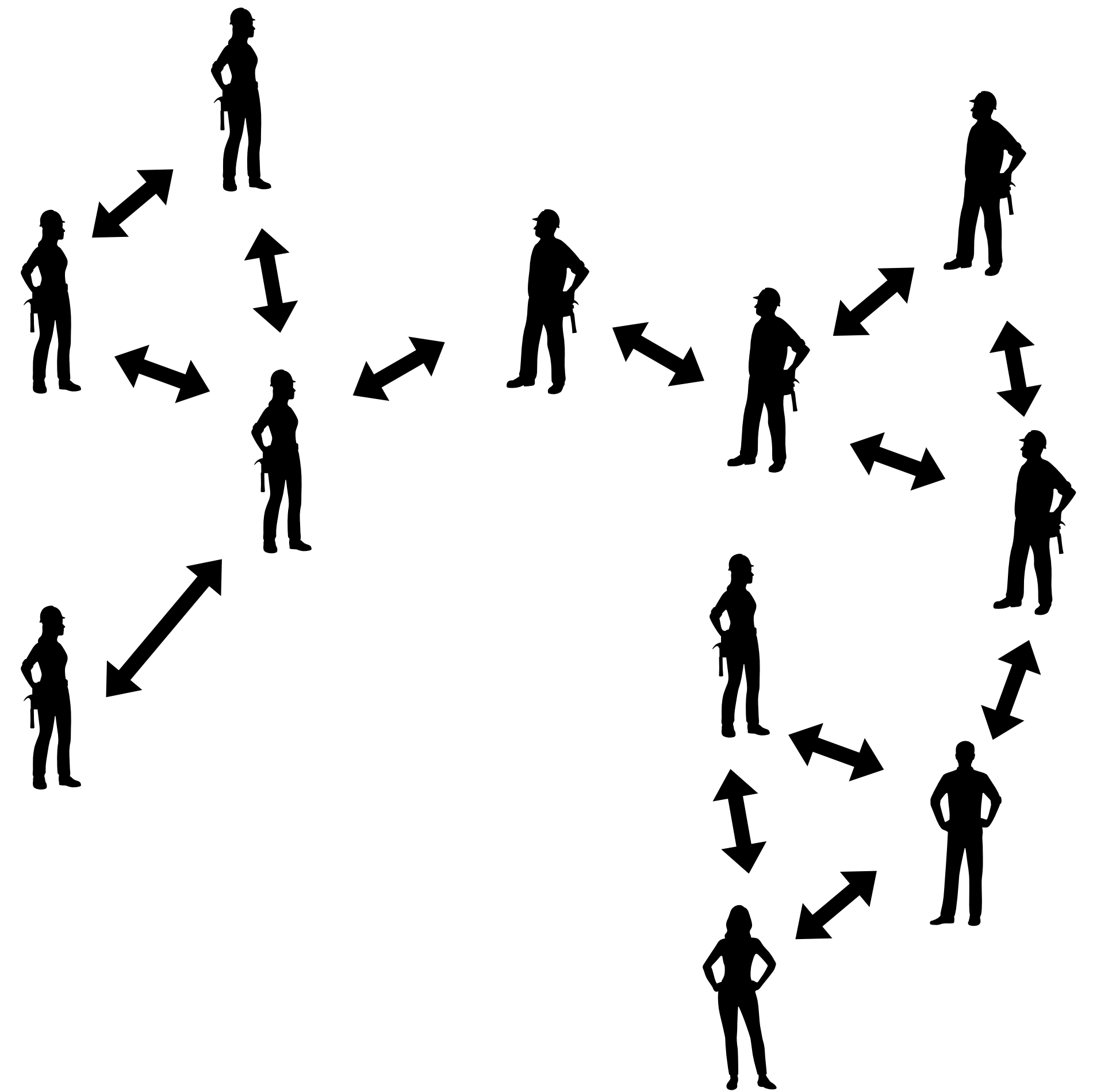
[Albert and Barabasi 1999]



(Stephen Coast
<https://www.fractalus.com/steve/stuff/ipmap/>)

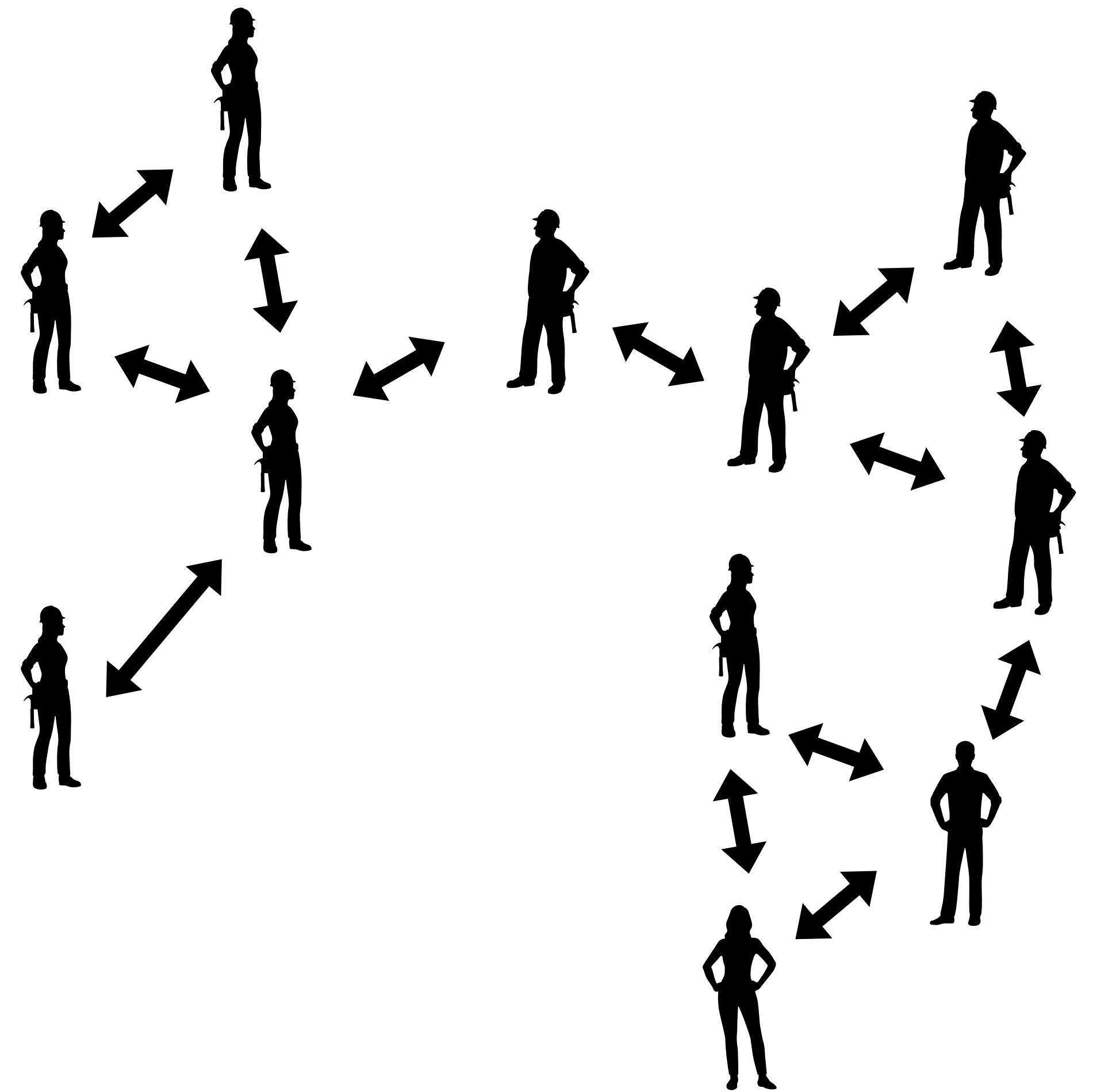
Preferential Attachment

[Albert and Barabasi 1999]



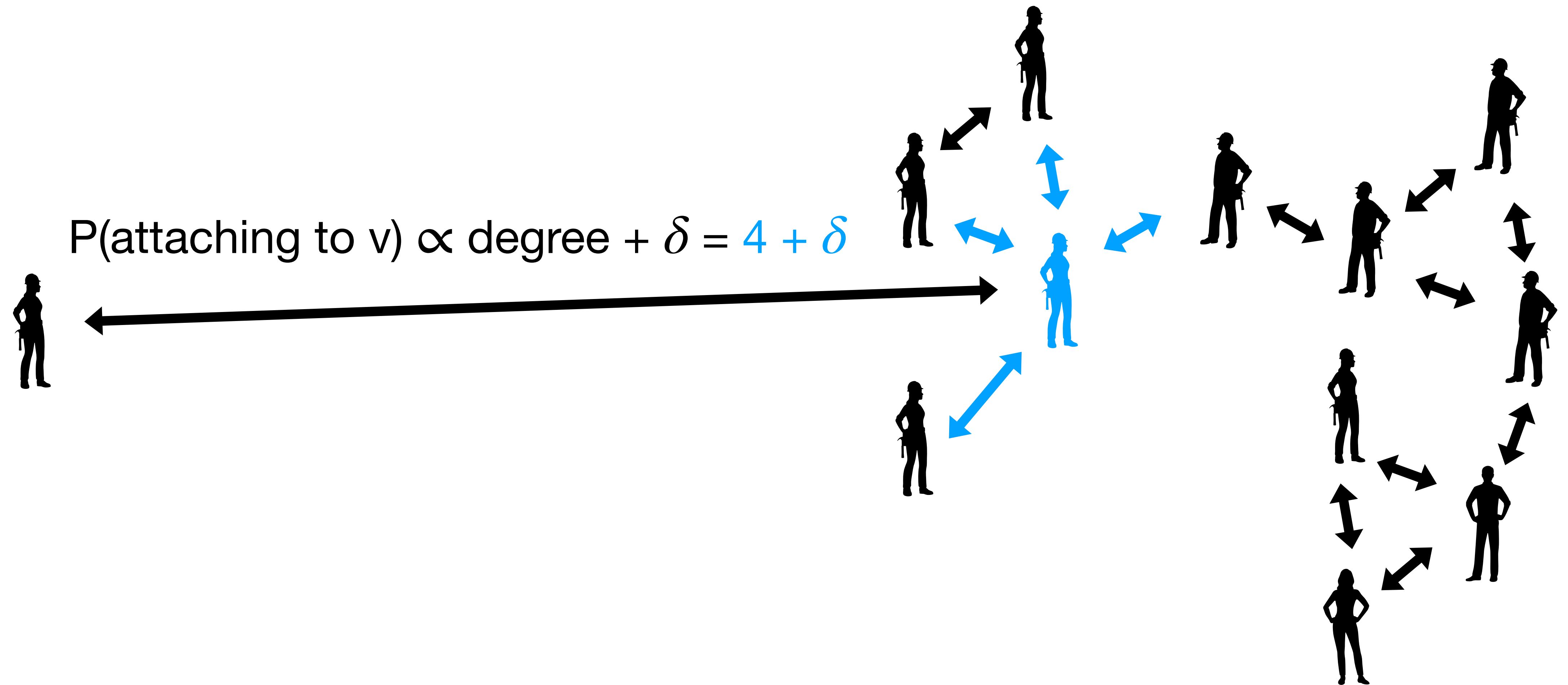
Preferential Attachment

[Albert and Barabasi 1999]



Preferential Attachment

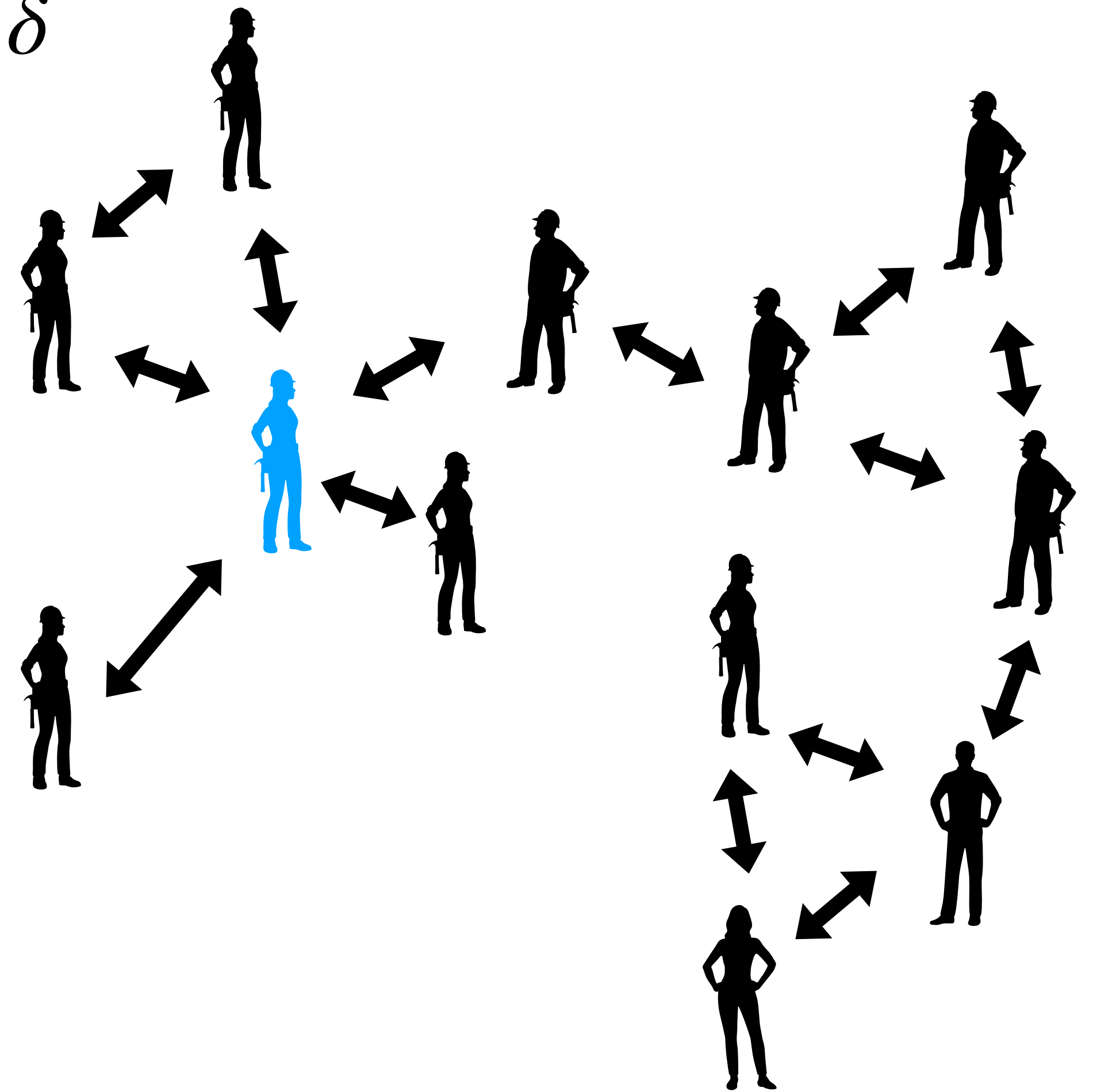
[Albert and Barabasi 1999]



Preferential Attachment

[Albert and Barabasi 1999]

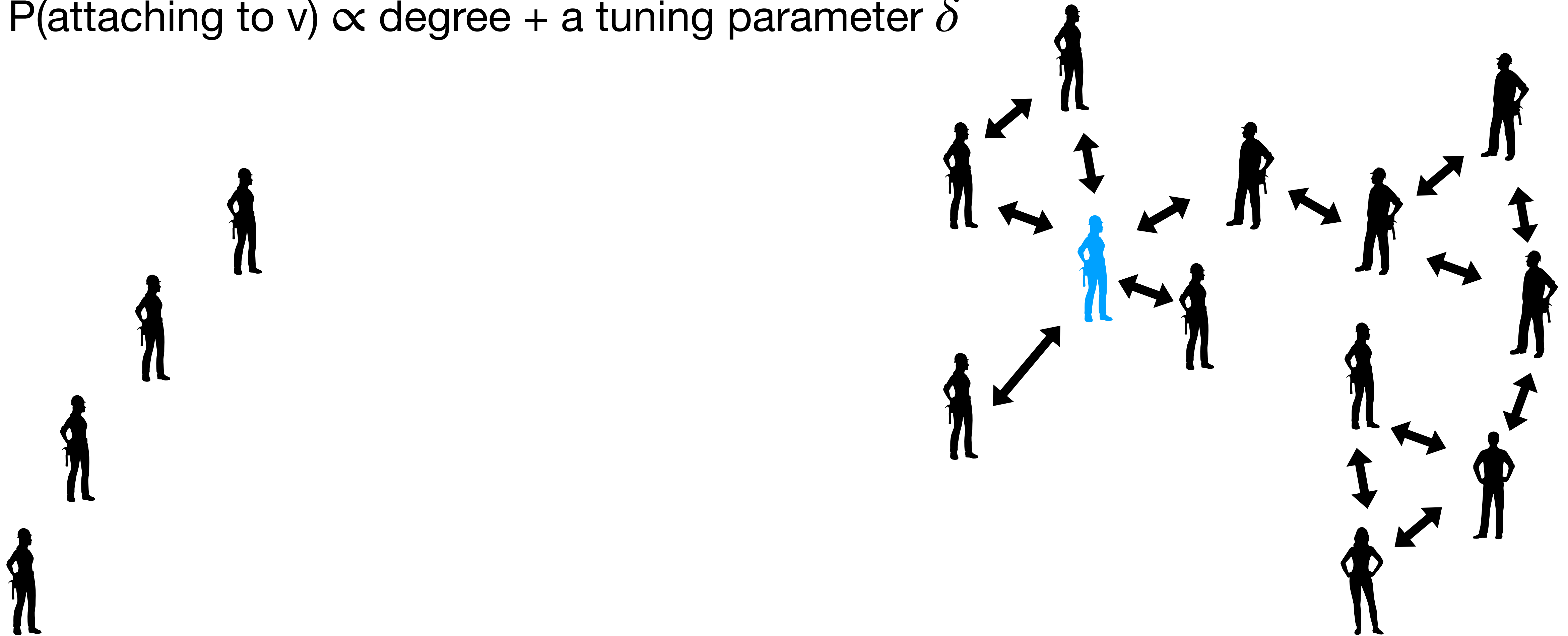
$P(\text{attaching to } v) \propto \text{degree} + \text{a tuning parameter } \delta$



Preferential Attachment

[Albert and Barabasi 1999]

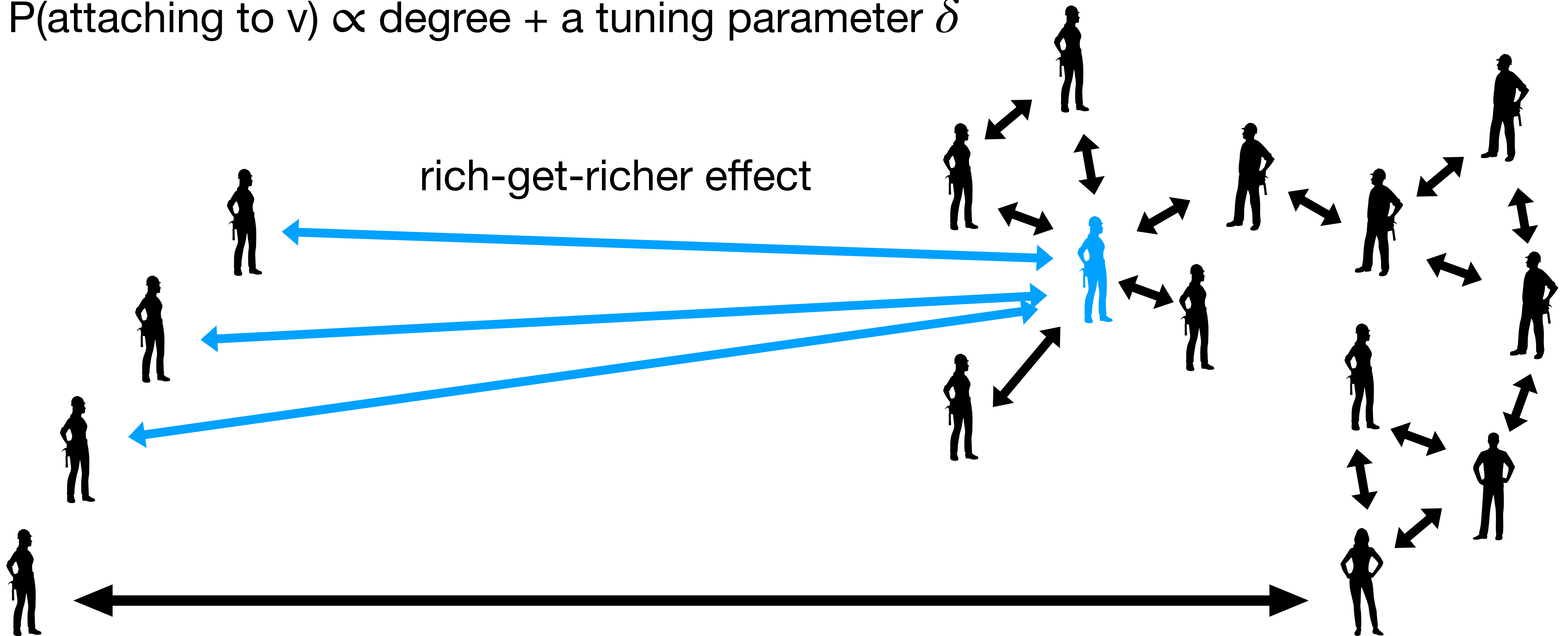
$P(\text{attaching to } v) \propto \text{degree} + \text{a tuning parameter } \delta$



Preferential Attachment

[Albert and Barabasi 1999]

$P(\text{attaching to } v) \propto \text{degree} + \text{a tuning parameter } \delta$



What do we know?

What do we know?

- Scale-freeness and Degree distribution

[Barabasi and Albert 1999; Dorogovtsev, Mendes and Samukhin 2000; Krapivsky, Redner and Leyvraz 2000]

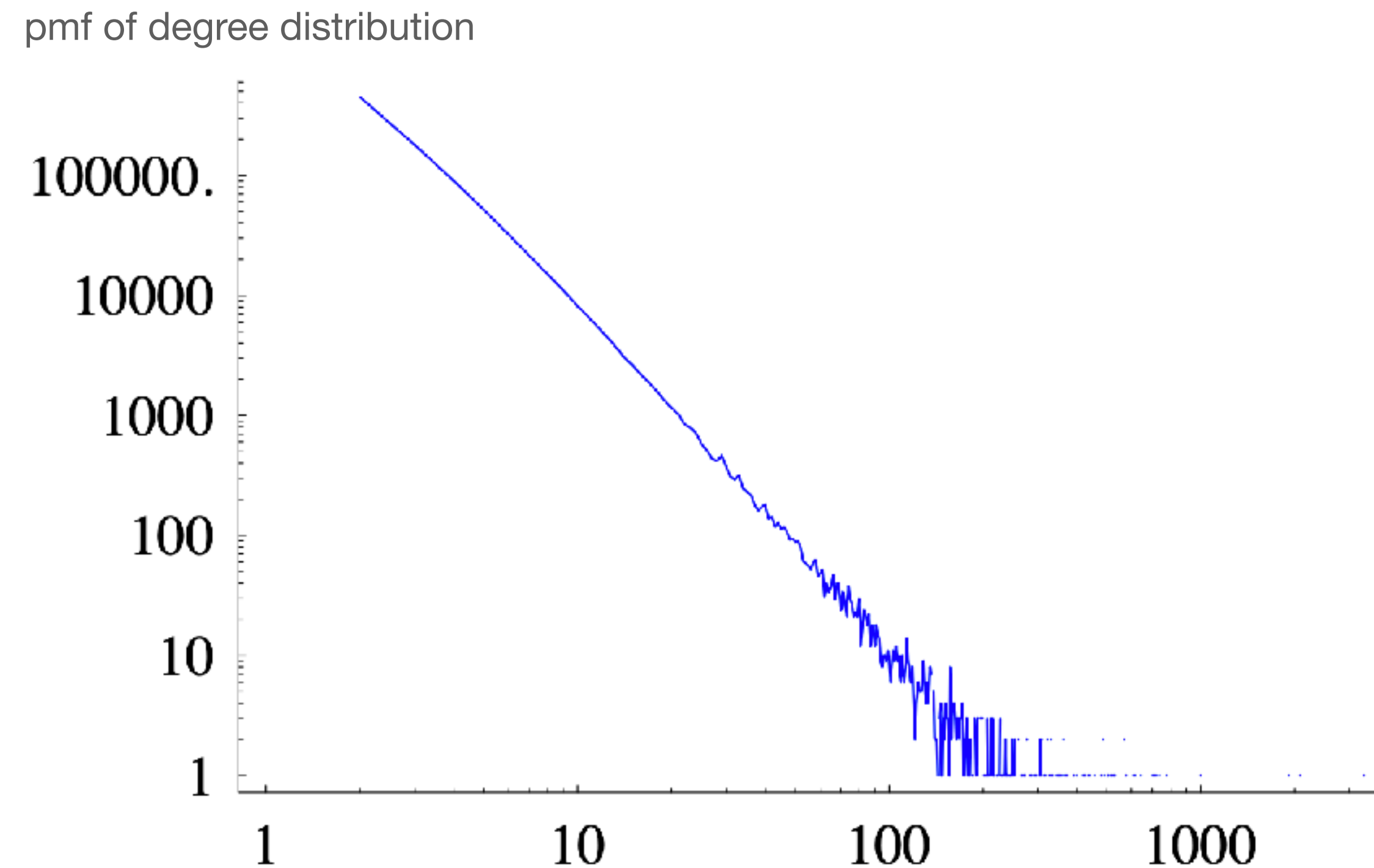


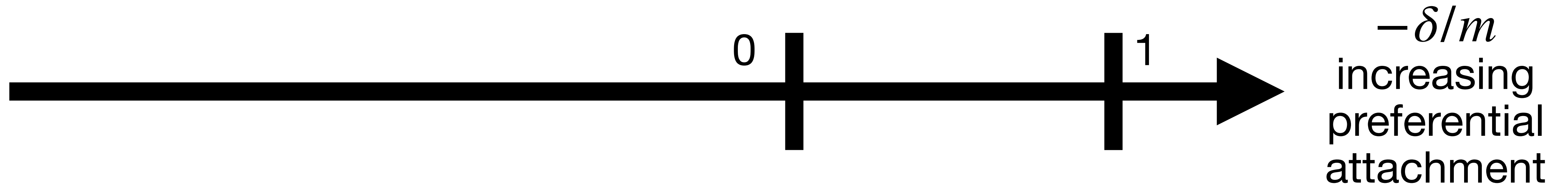
Fig 8.3 of R. Hofstad (2013).
Random Graphs and Complex Networks.
<https://doi.org/10.1017/9781316779422>

Phase transition

Recall

$P(\text{attaching to } v) \propto \text{degree} + \delta$

m = number of edges per new node

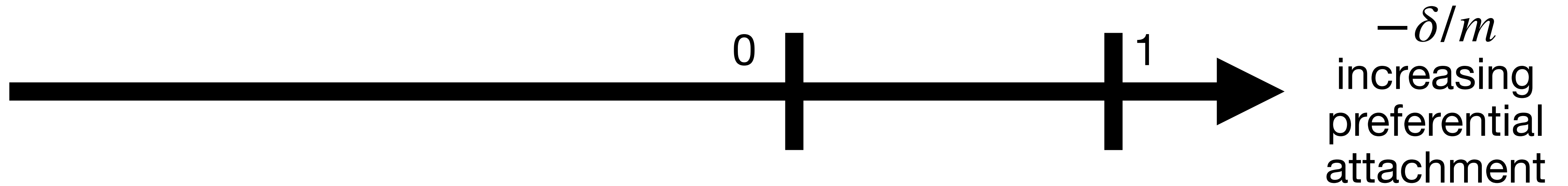


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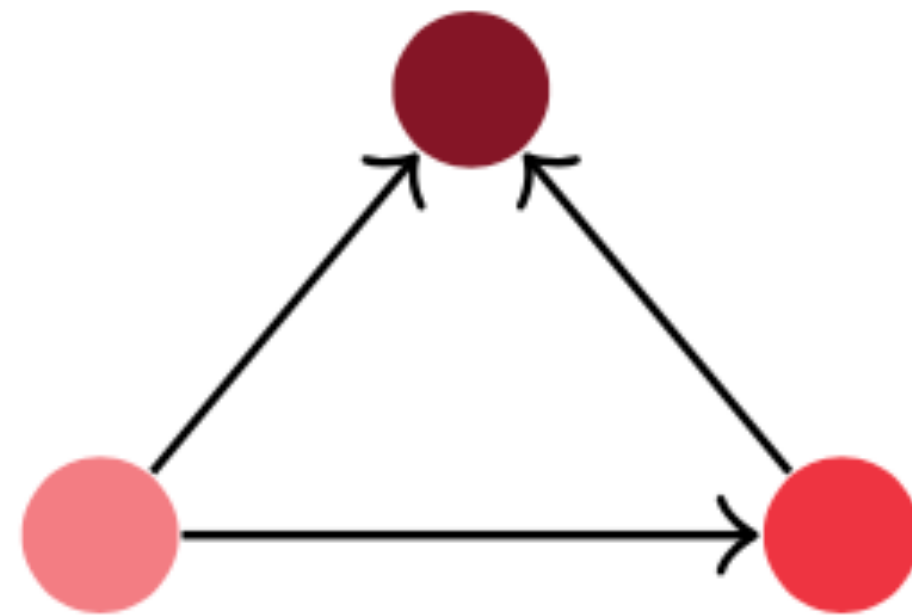
The limiting degree distribution has ...

finite variance

infinite variance

What do we know?

- triangle counts and clustering coefficient [Bollobas and Ridden 2002, Prokhorenkova et al 2013, Garavaglia and Stegehuis 2019]



(a) $t^{(3-\tau)/(\tau-1)} \log(t)$

$$\tau = 3 + \delta/m$$

Fig 2 of A. Garavaglia and C. Stegehuis (2019).
Subgraphs in Preferential Attachment Models.
<https://doi.org/10.1017/apr.2019.36>

What do we know?

- subgraph counts [Garavaglia and Stegehuis 2019]

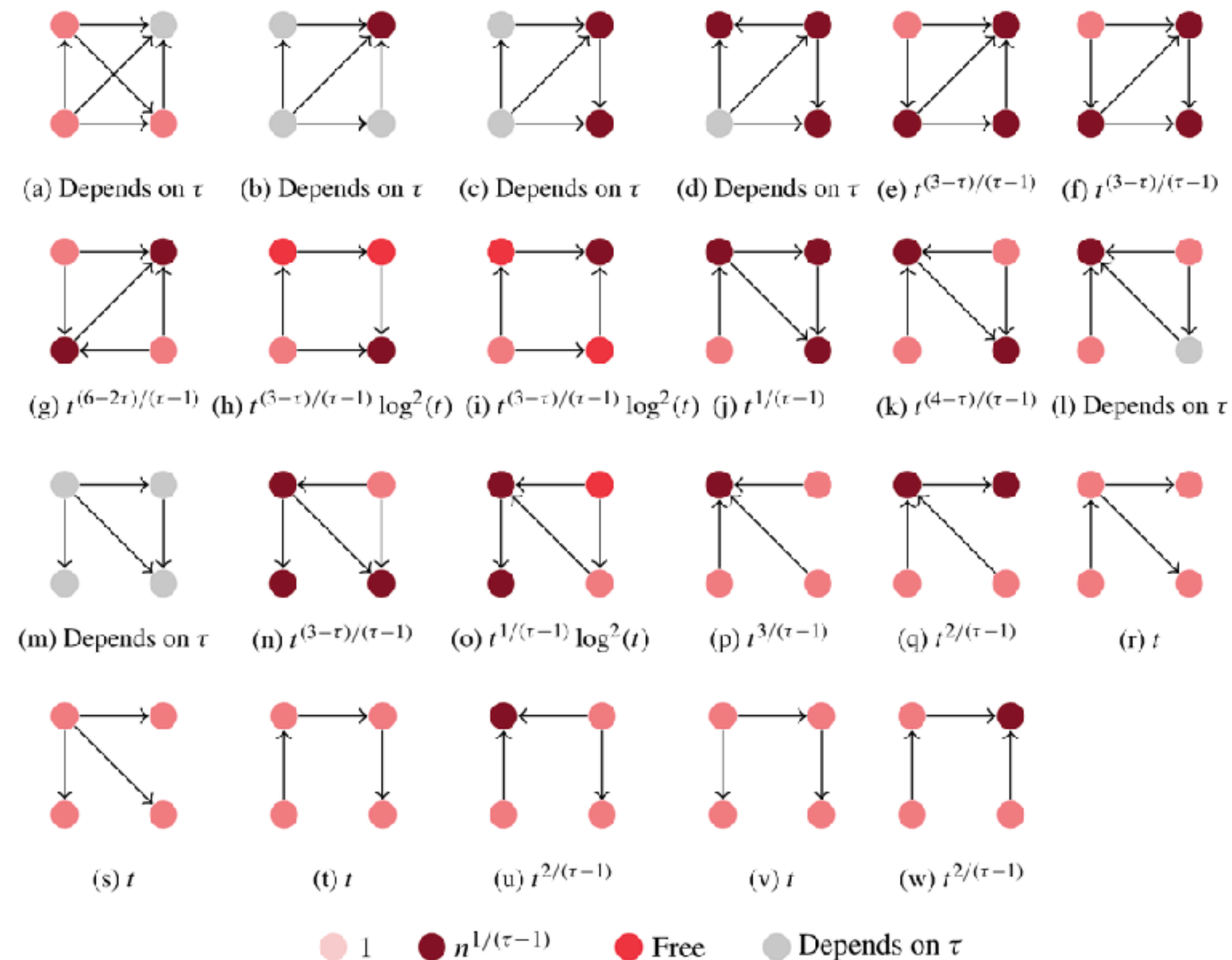
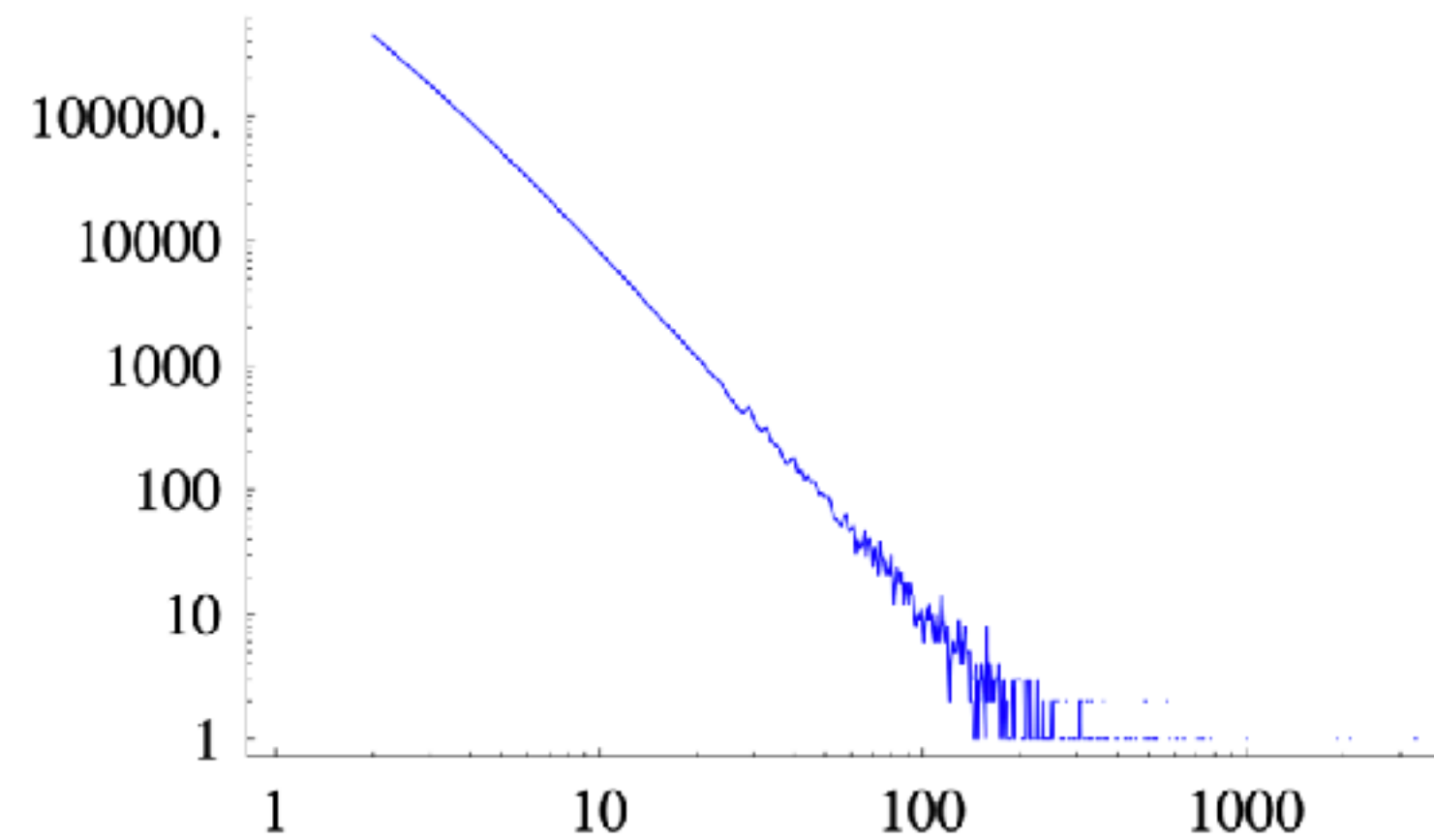
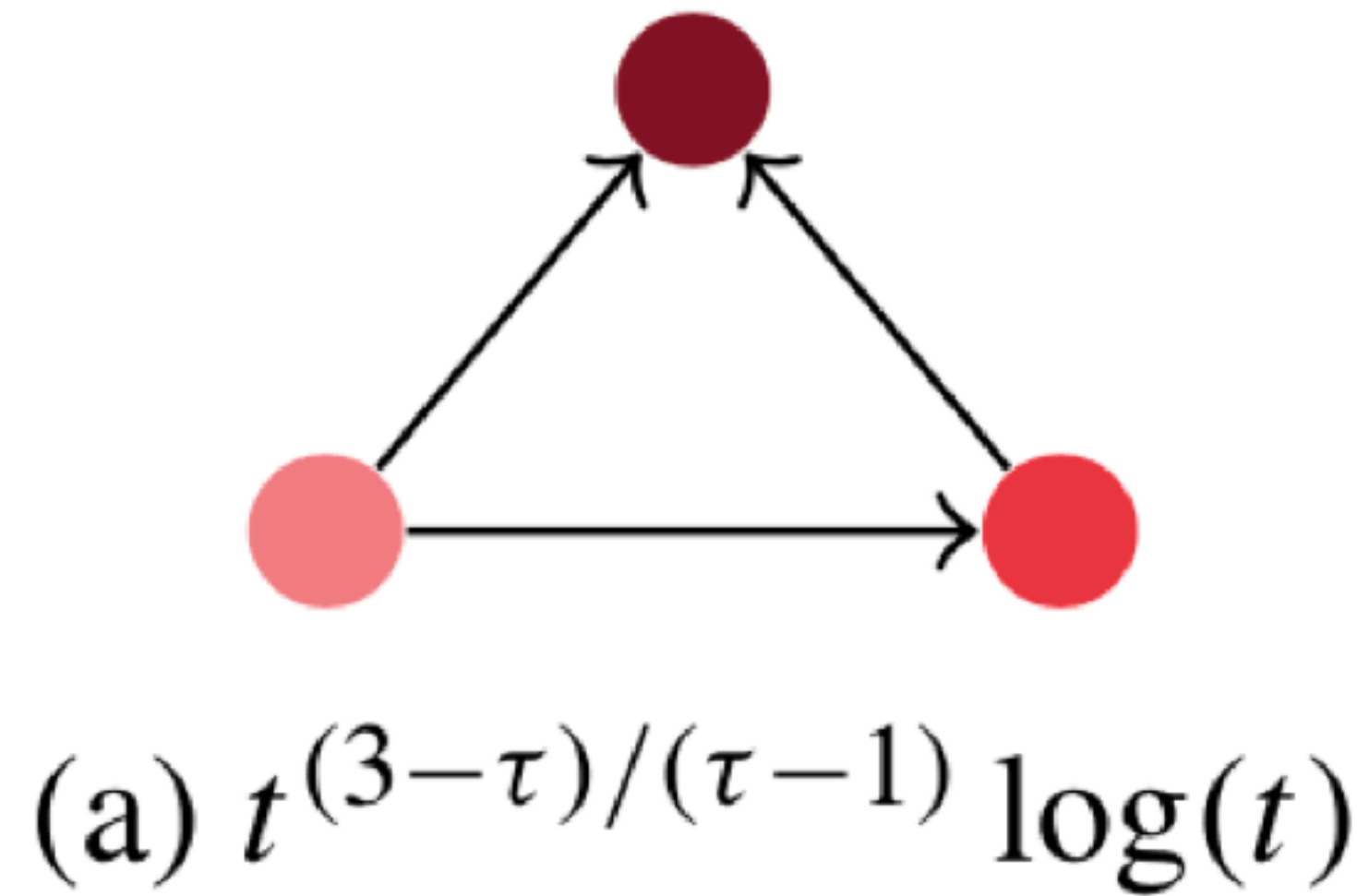


Fig 3 of A. Garavaglia and C. Stegehuis (2019).
Subgraphs in Preferential Attachment Models.
<https://doi.org/10.1017/apr.2019.36>

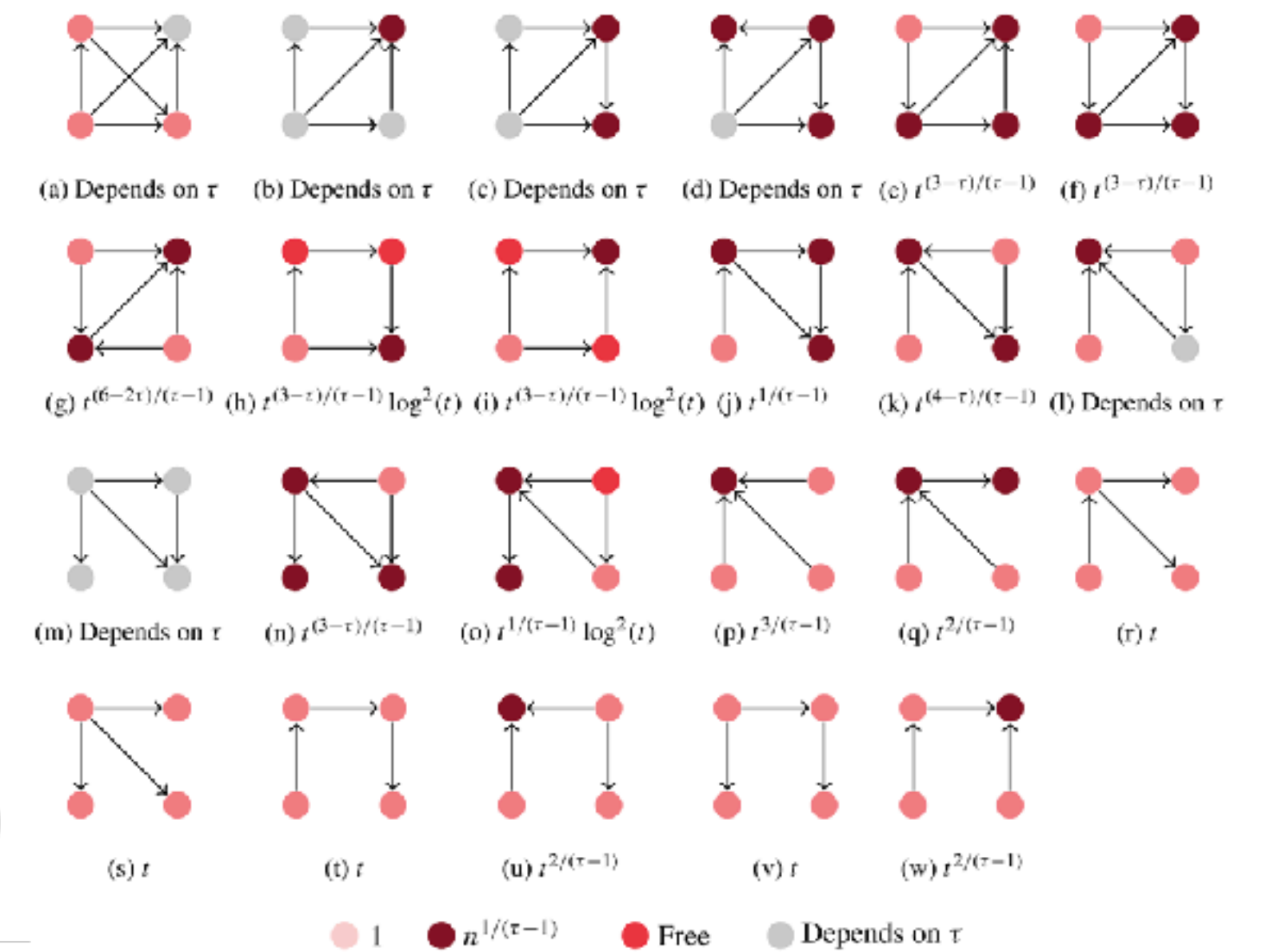
What do we know?



degree distribution

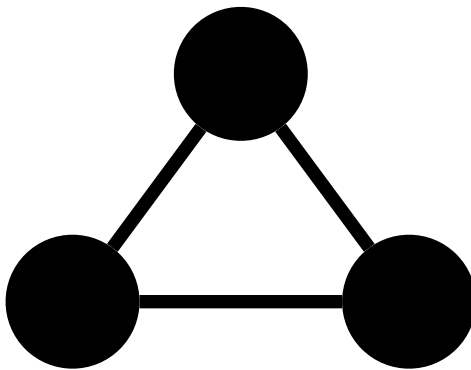
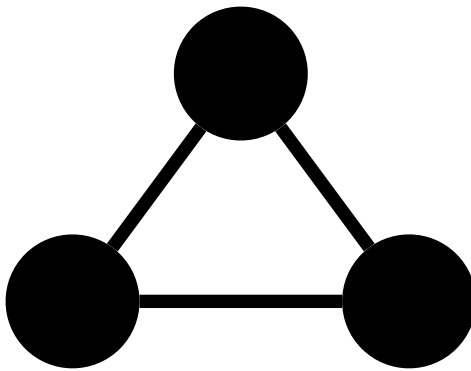
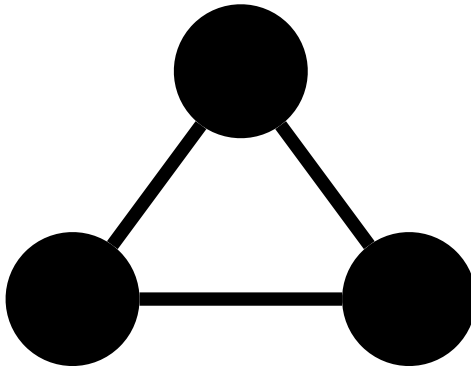
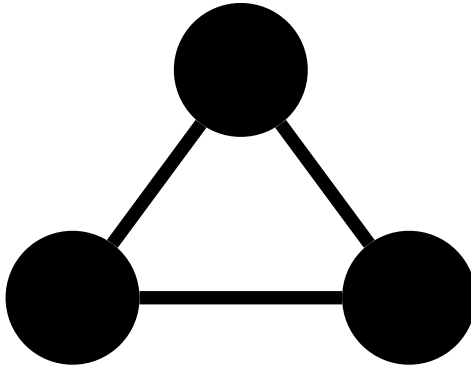


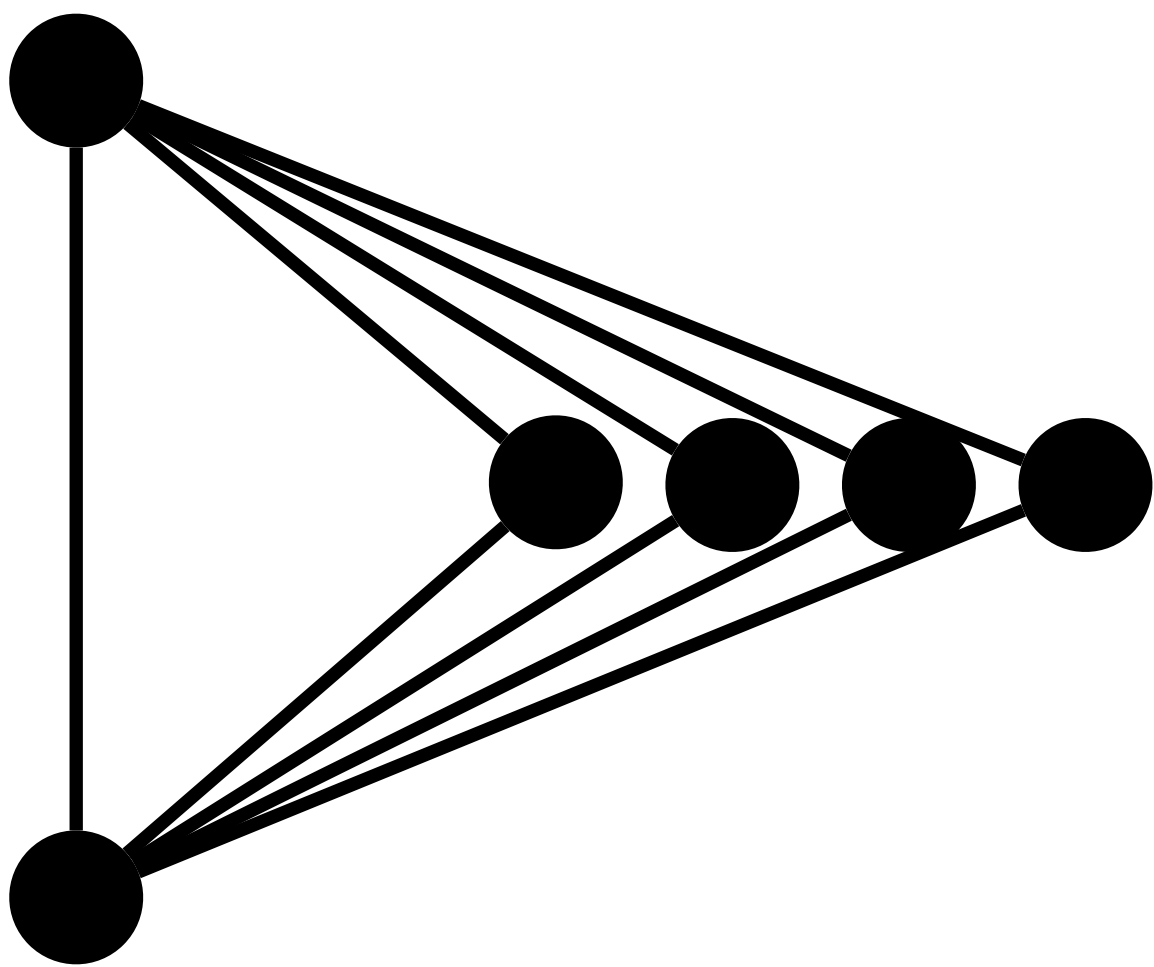
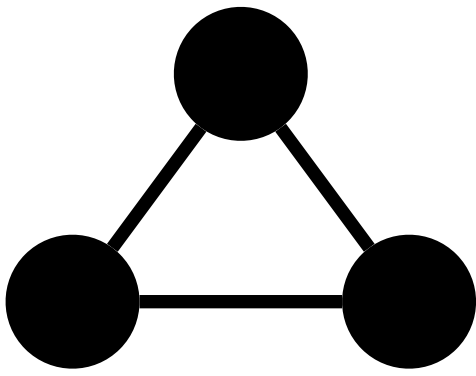
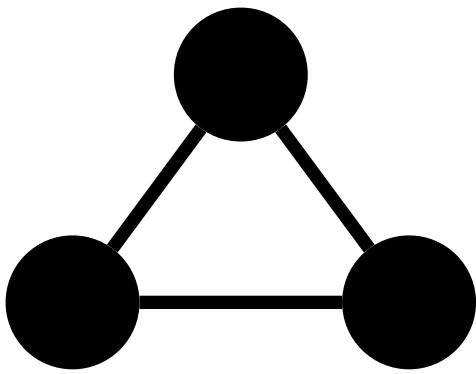
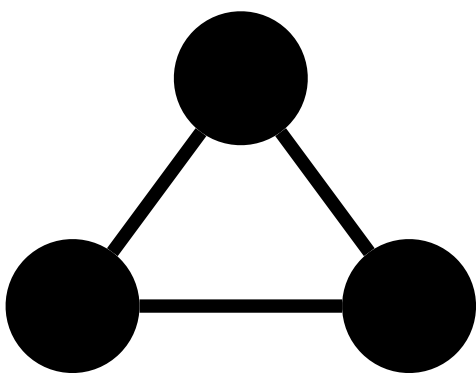
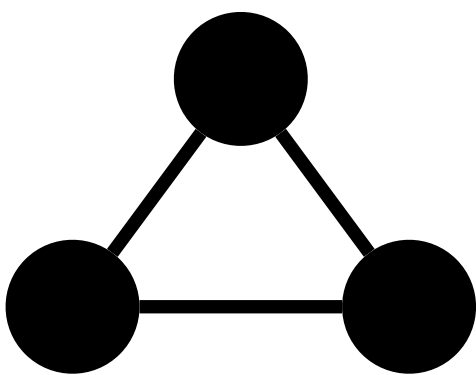
triangle counts



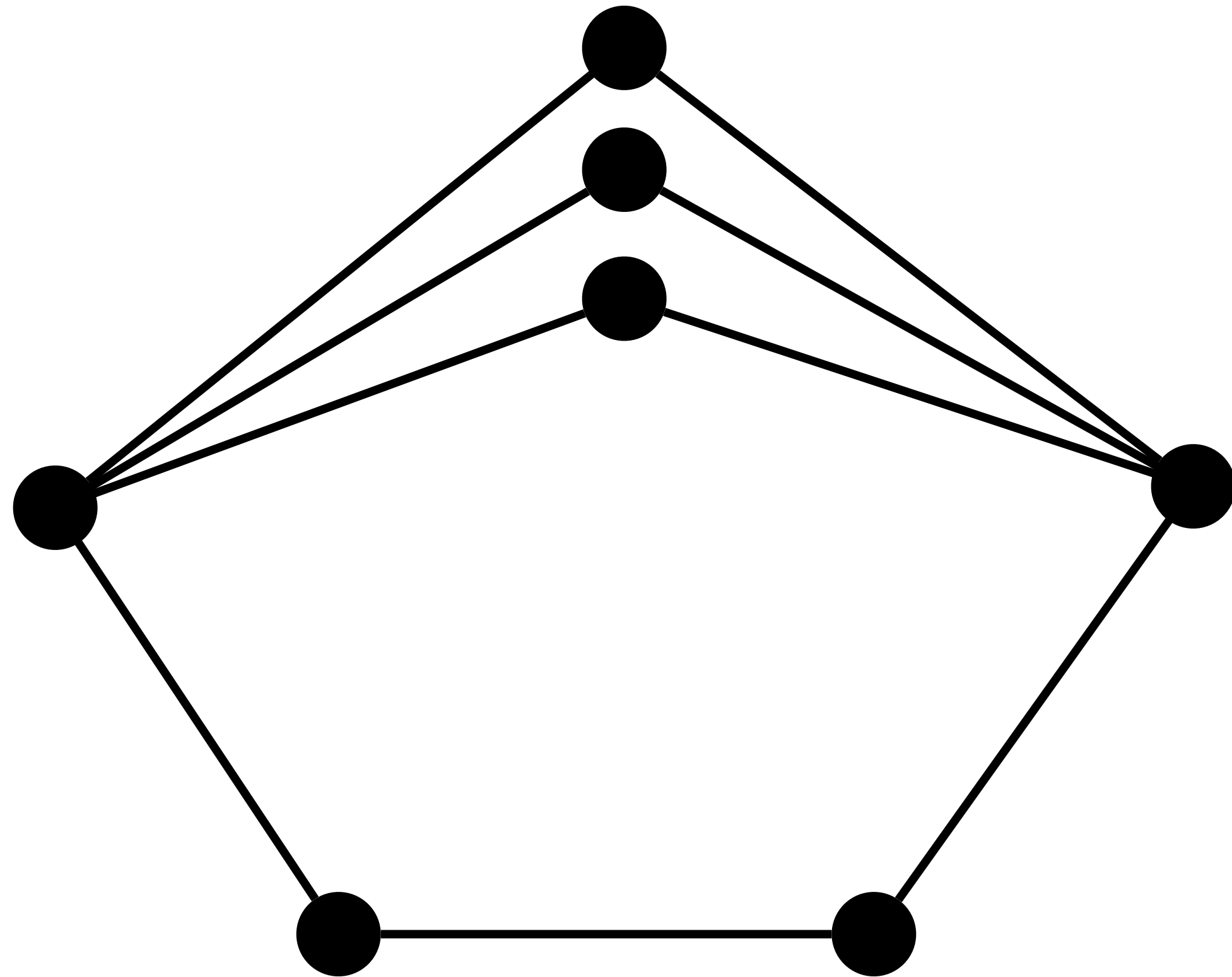
subgraph counts

**What should we count?
And how?**

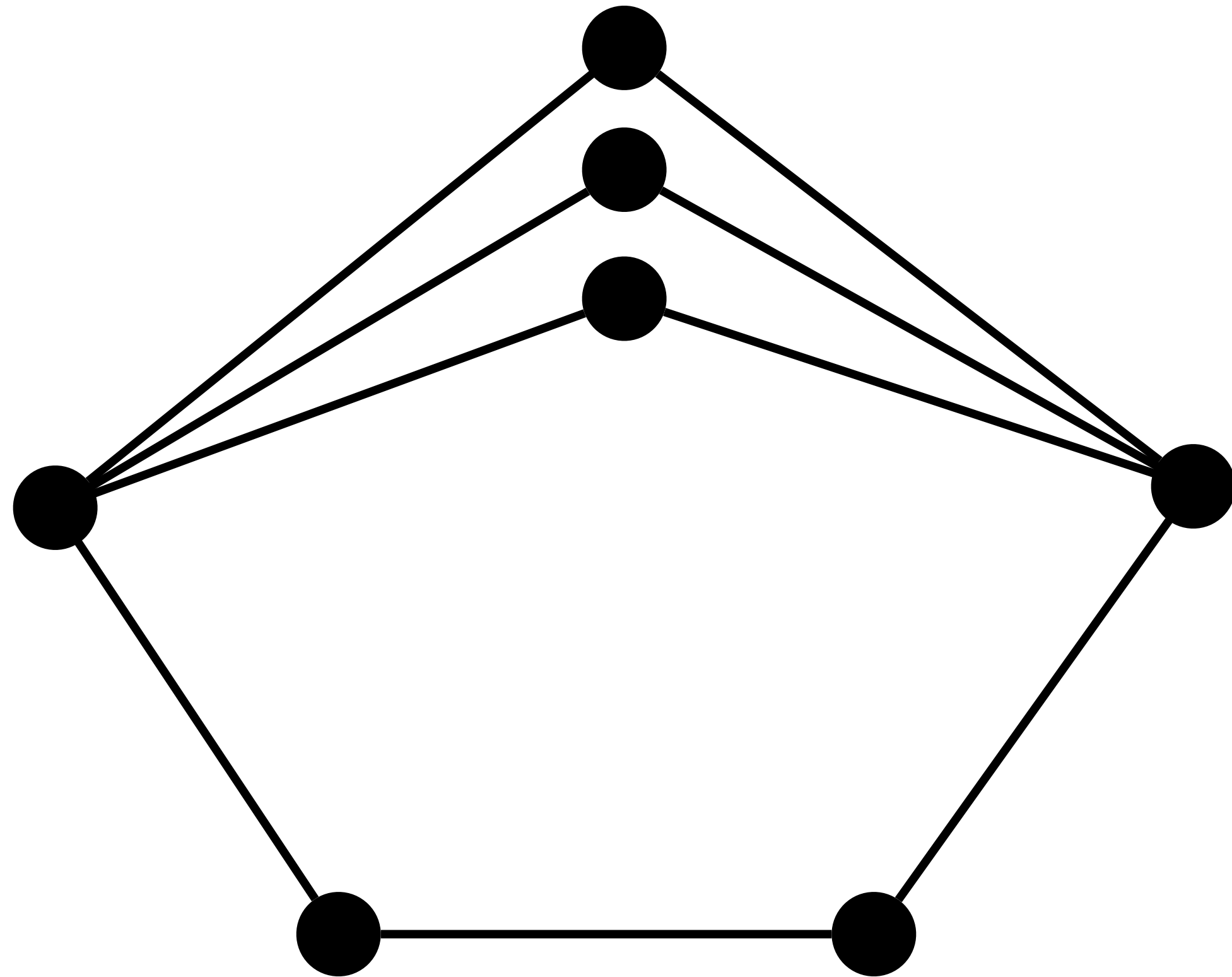




Paths from left to right?

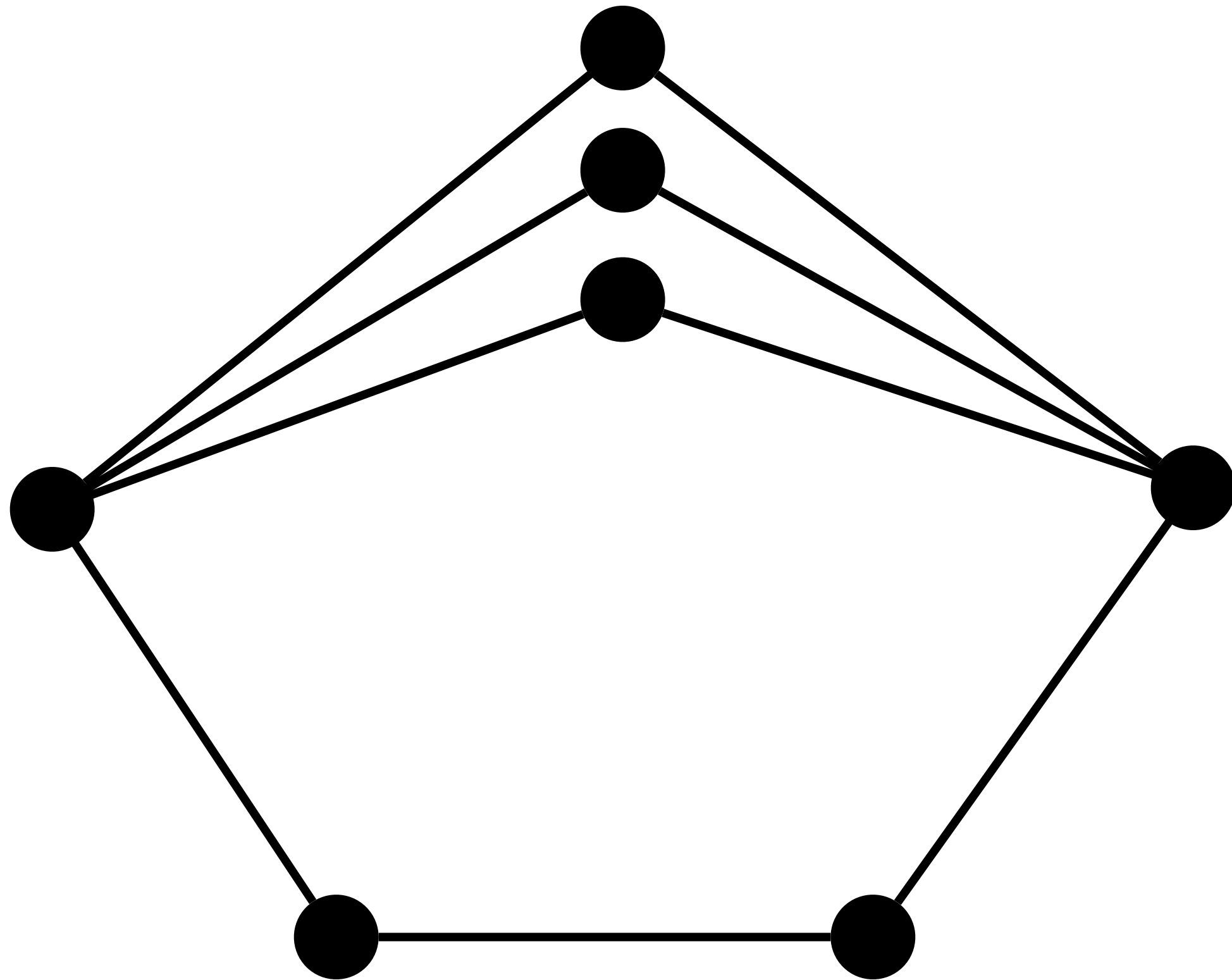


Paths from left to right?



- backtracking?

Paths from left to right?



- backtracking?
- concatenating with loops?

II. Into Topology

Counting everything in every dimension all at once

Betti numbers count repeated connections “in all dimensions”.

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“correct” way to count things

Betti numbers count repeated connections “in all dimensions”.



“correct” way to count things

homological algebra

Betti numbers count repeated connections “in all dimensions”.



“correct” way to count things

homological algebra

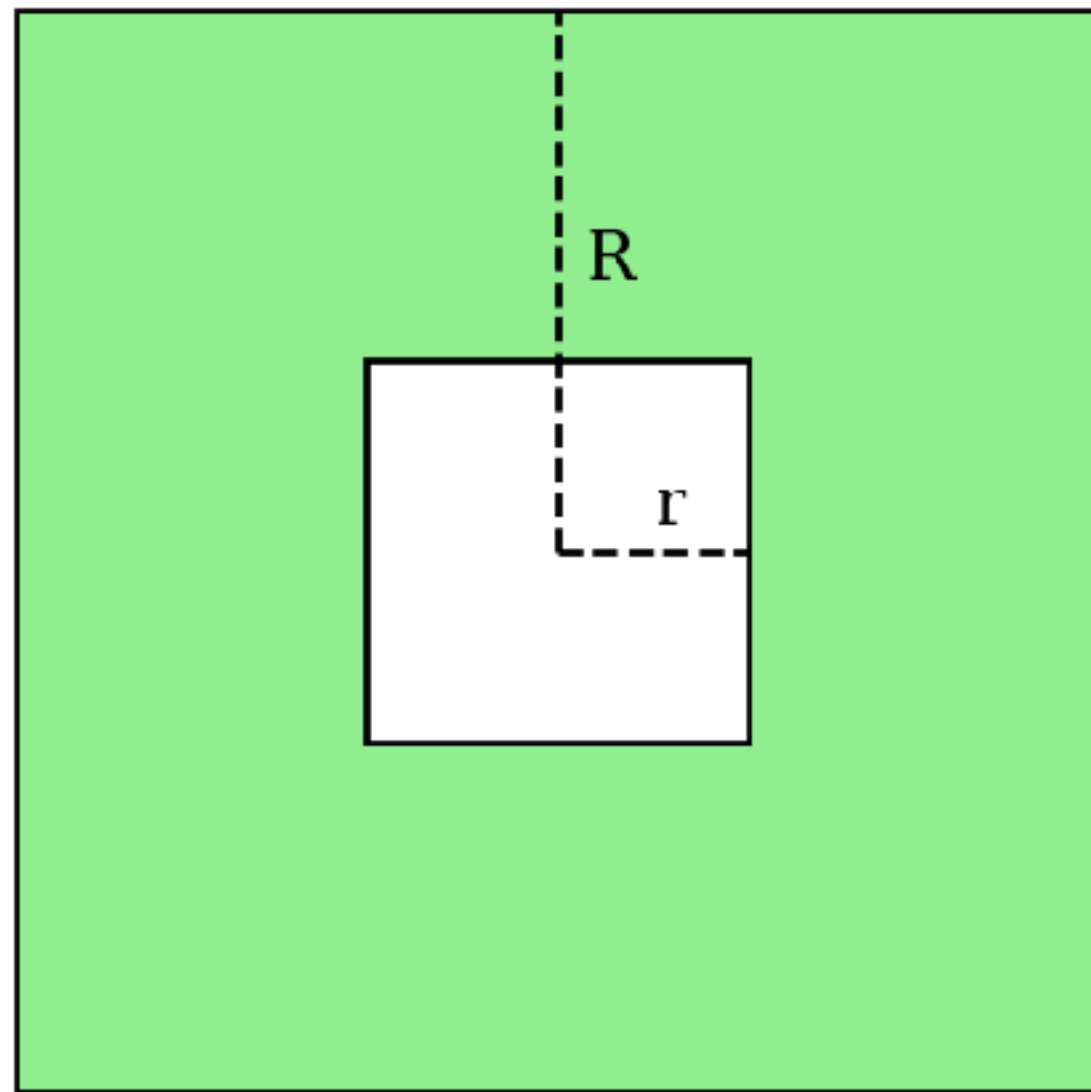


hard to write down

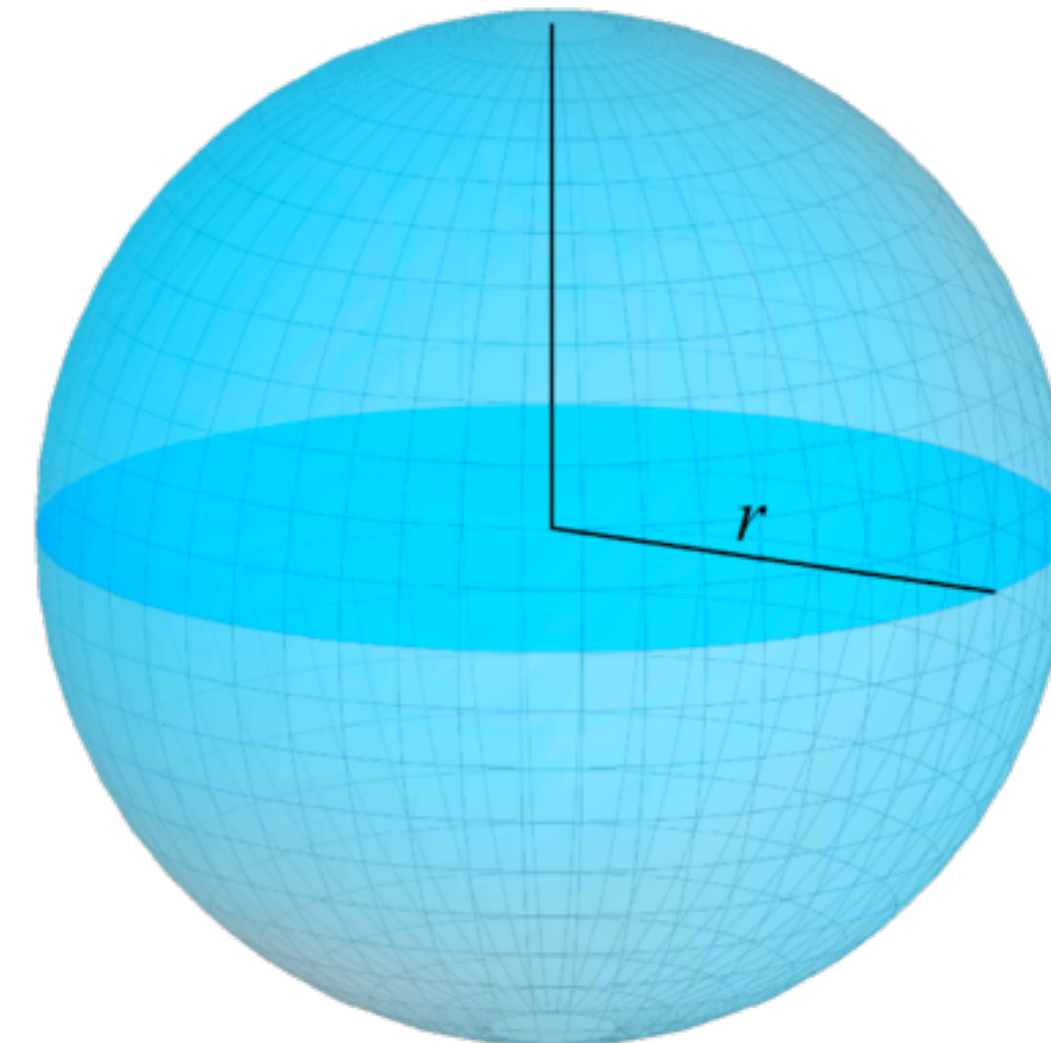
hard to do

Betti numbers β_k

Count of Holes



$$\beta_1 = 1 : 1 \text{ loop}$$

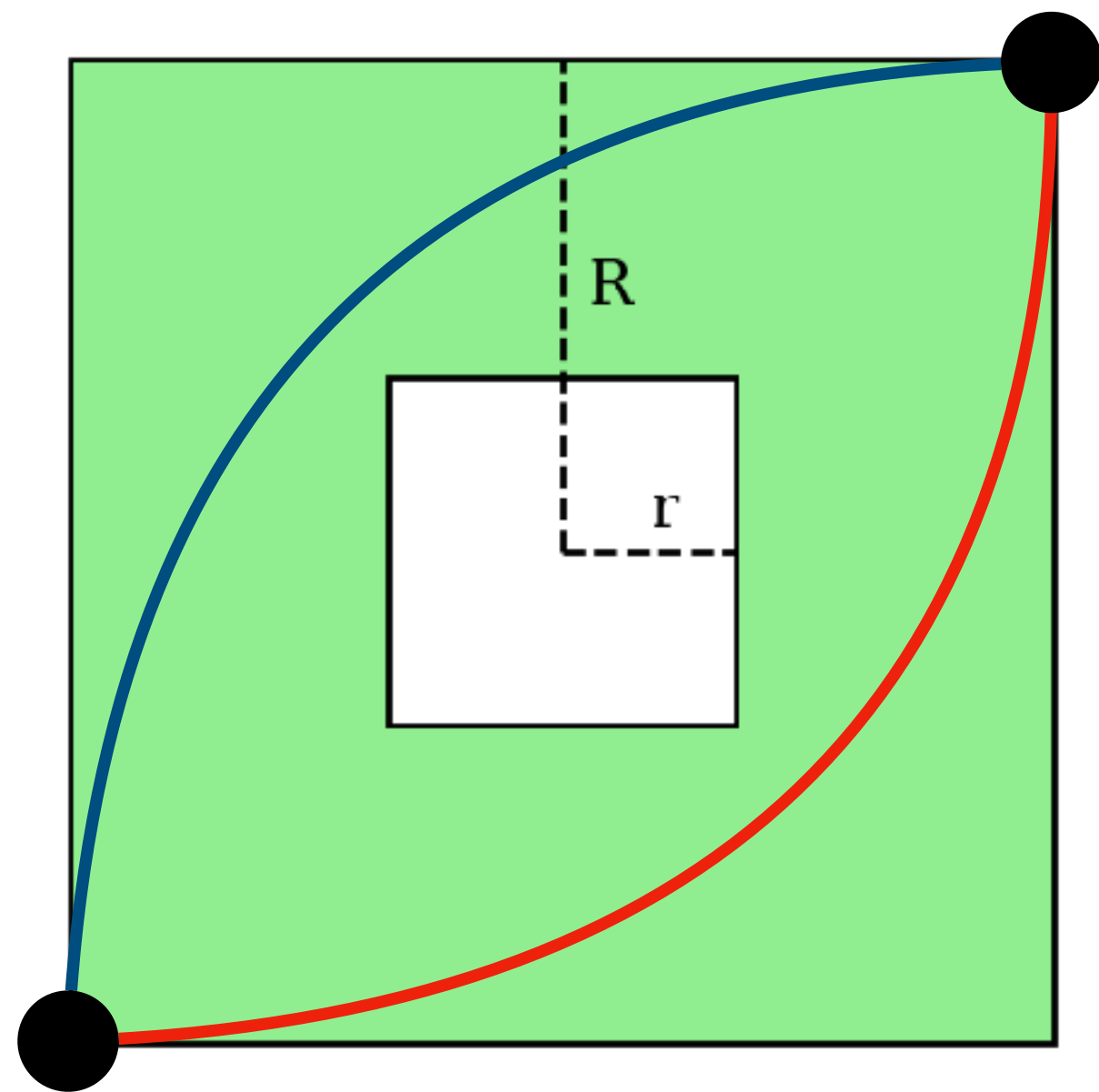


$$\beta_1 = 0 : 0 \text{ loop}$$

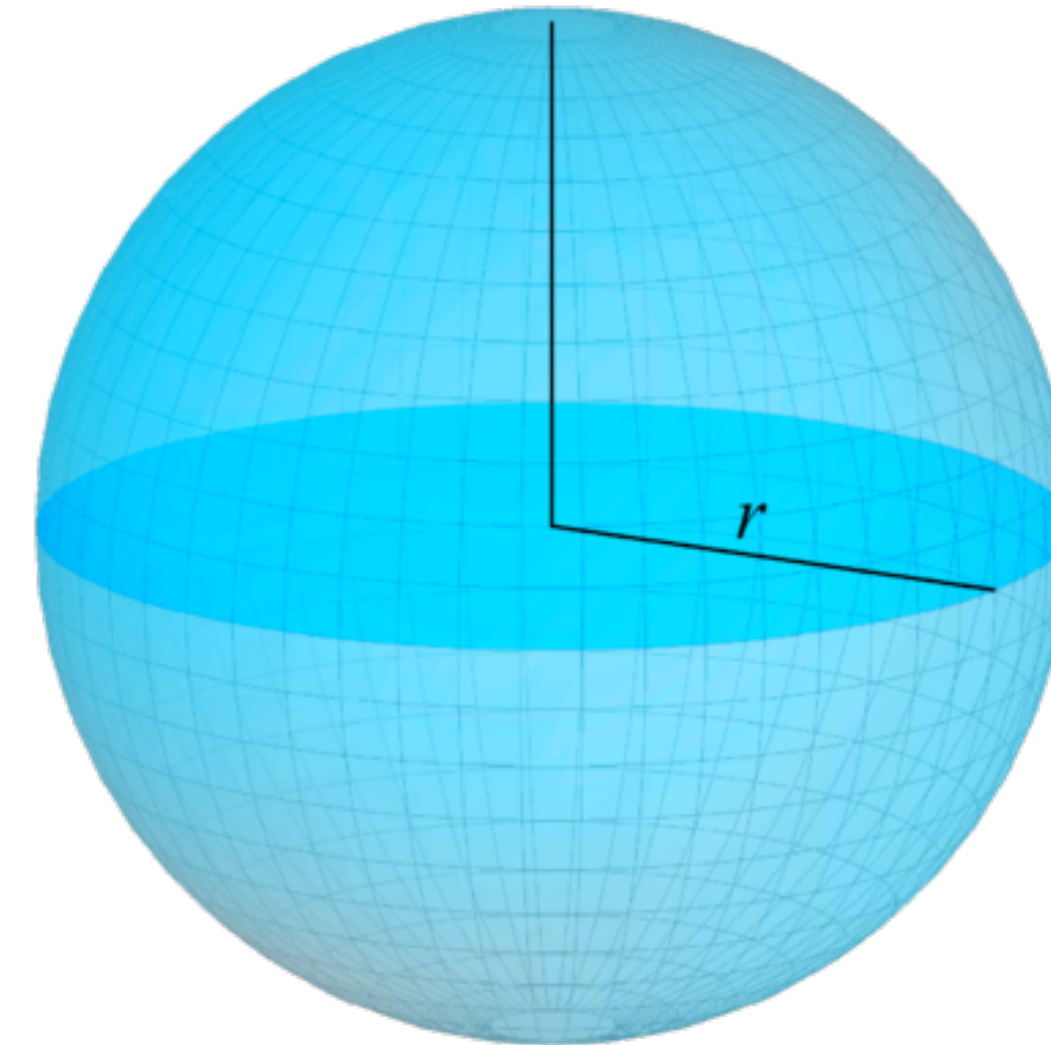
$$\beta_2 = 1 : 1 \text{ cavity}$$

Betti numbers β_k

Count of Repeated Connections



$$\beta_1 = 1 : 1 \text{ loop}$$

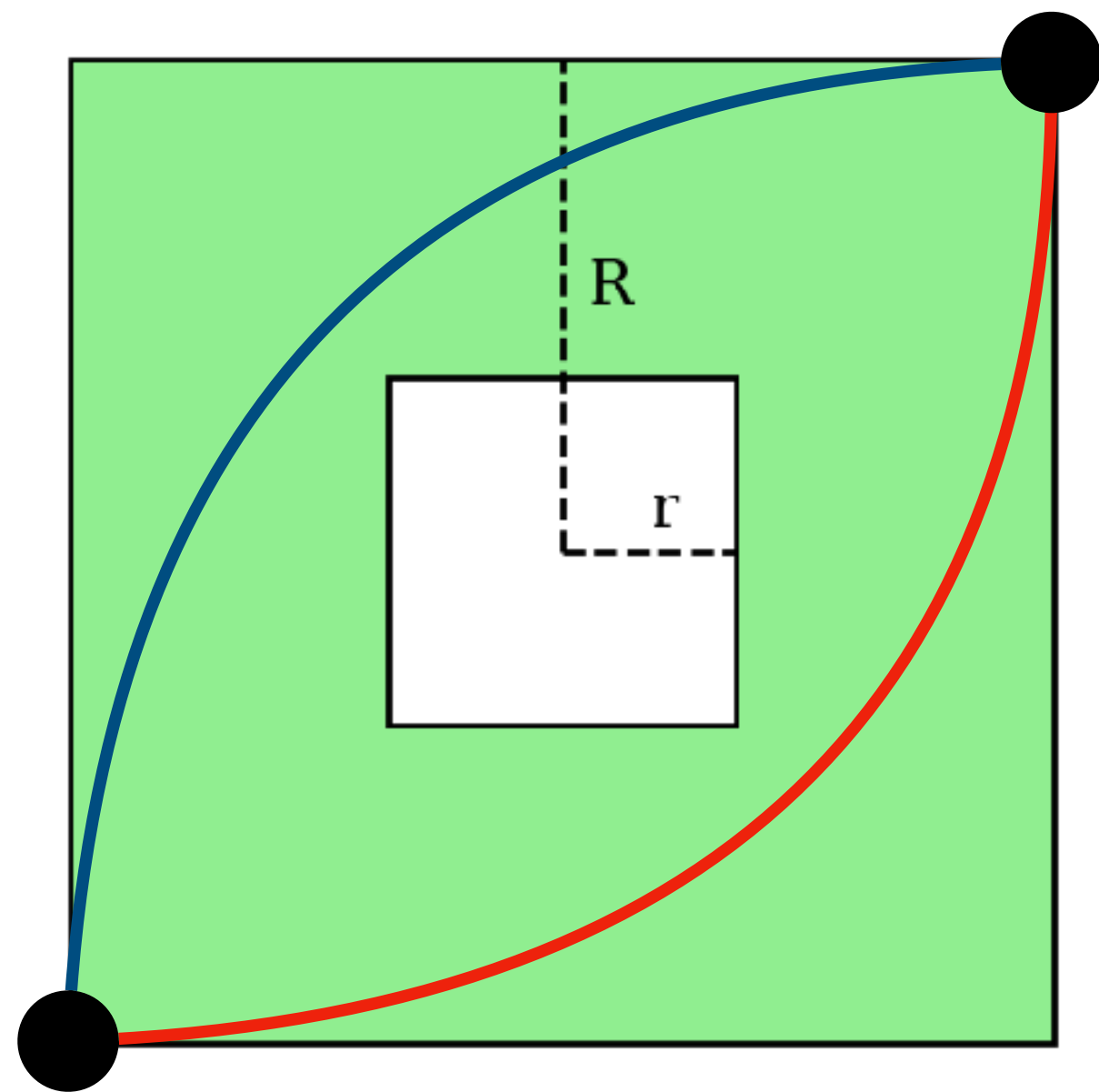


$$\beta_1 = 0 : 0 \text{ loop}$$

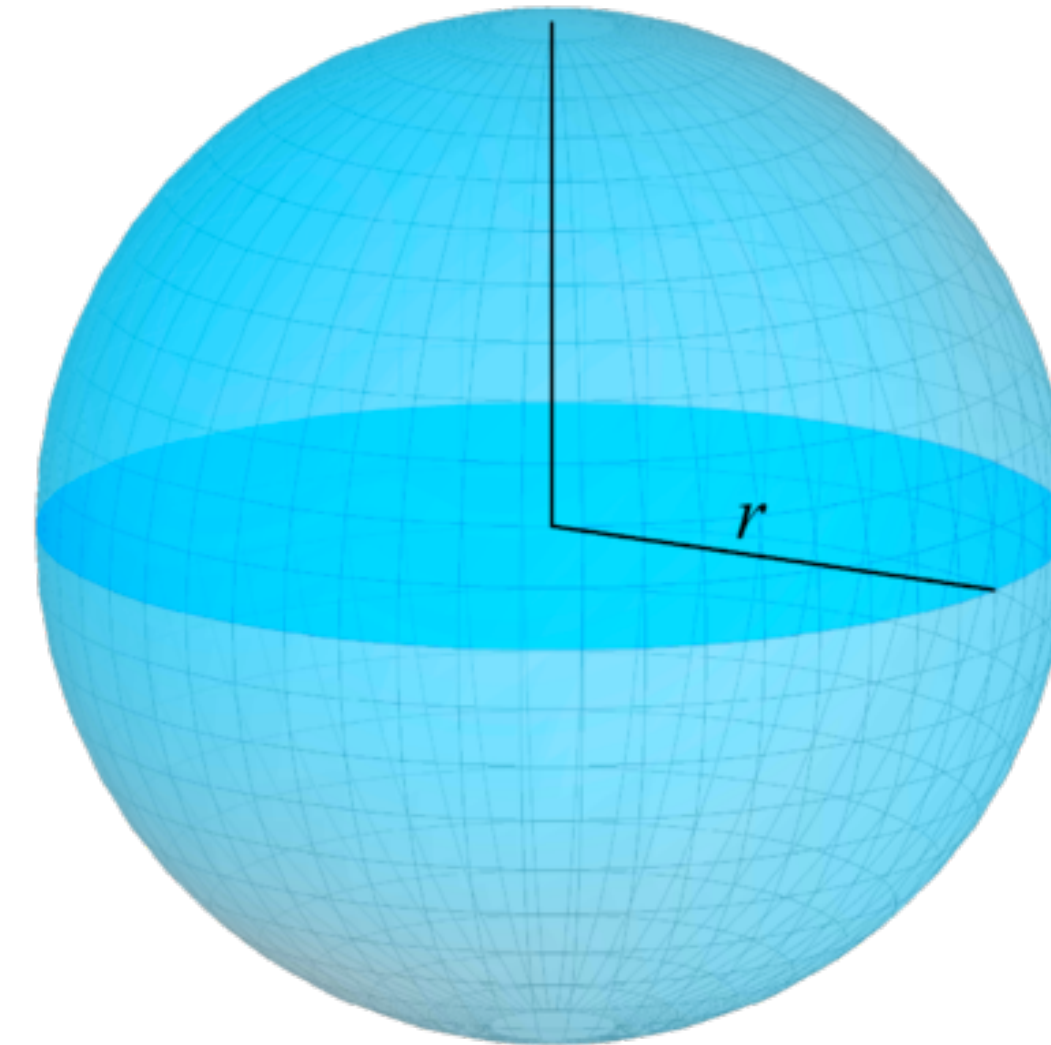
$$\beta_2 = 1 : 1 \text{ cavity}$$

Betti numbers β_k

Count of Repeated Connections



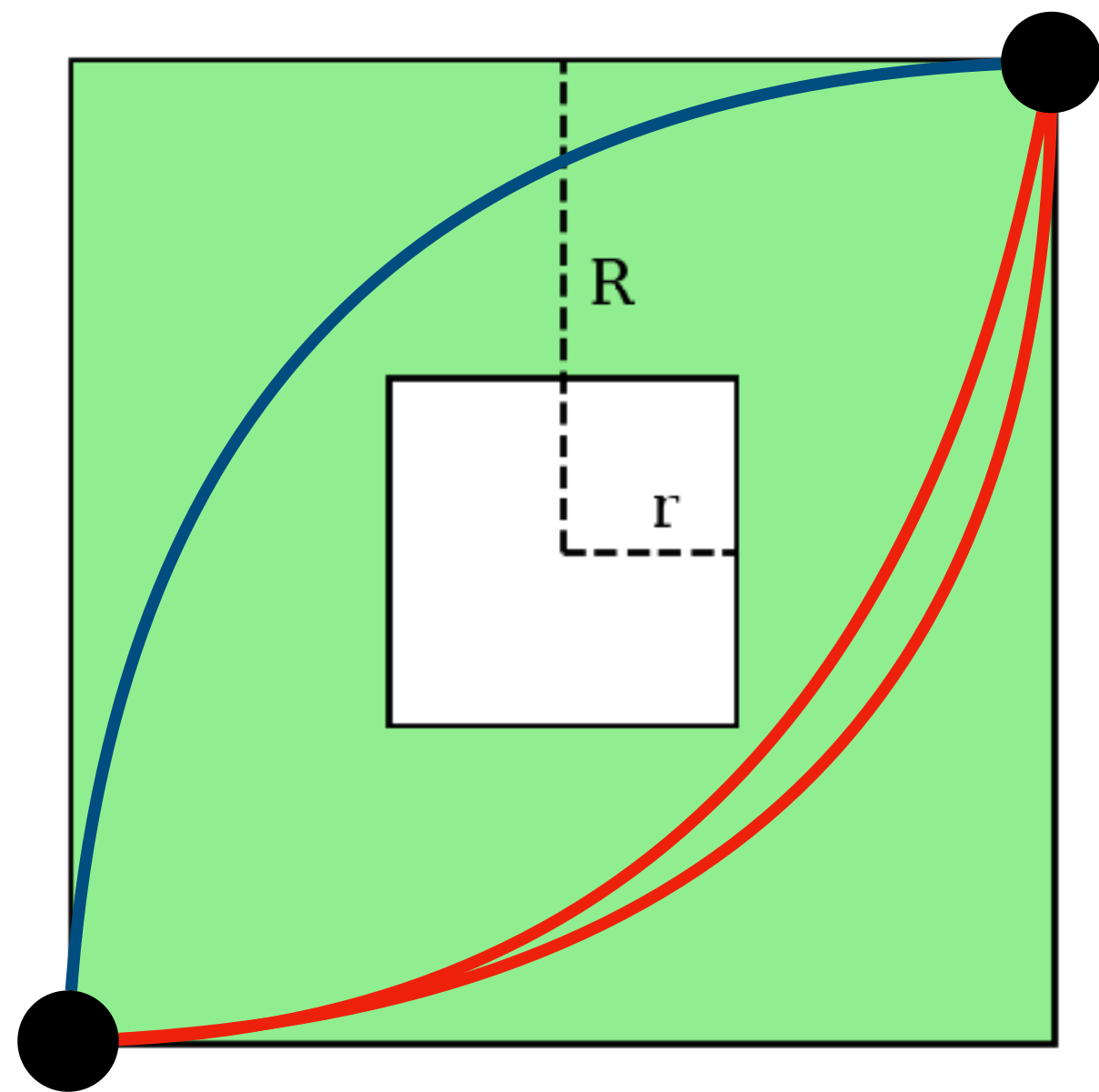
1 alternative path



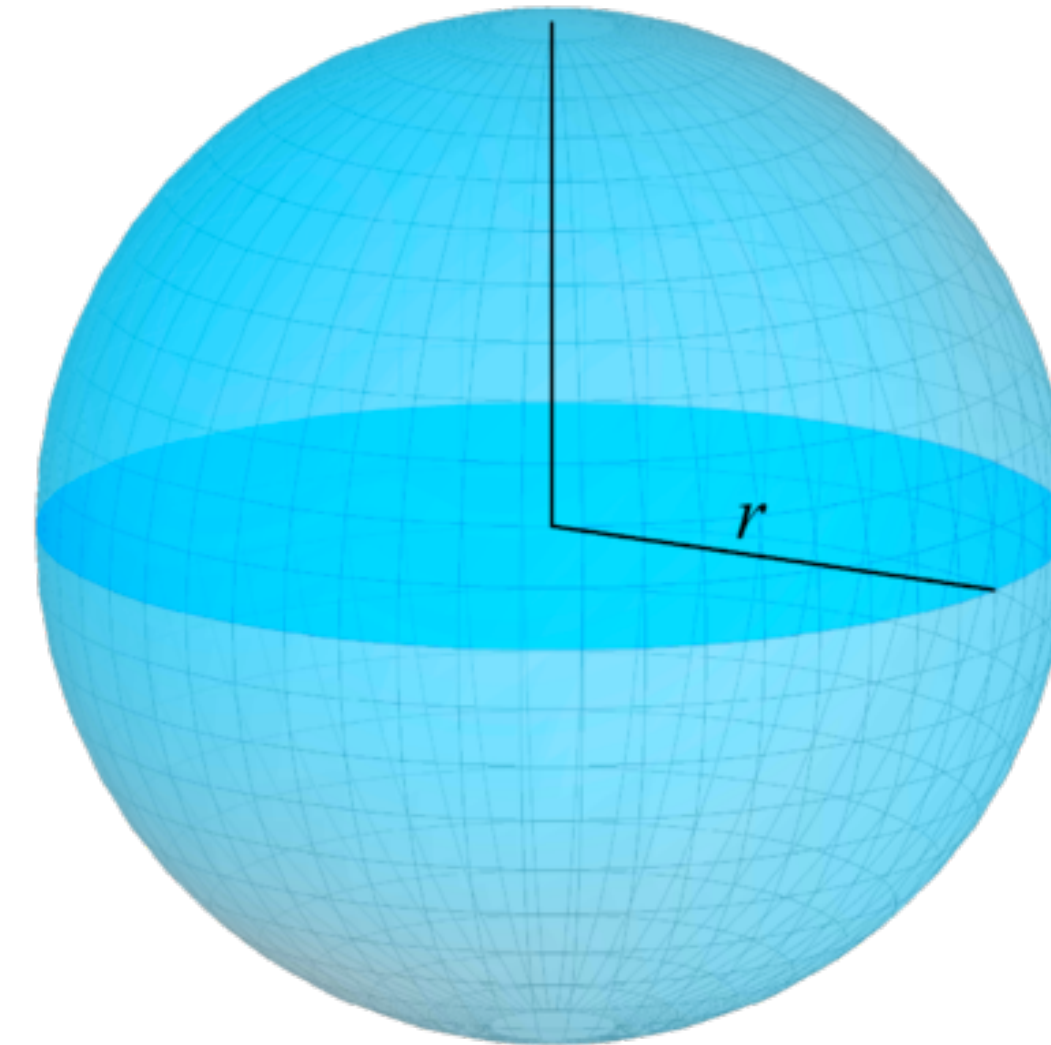
0 loop
1 cavity

Betti numbers β_k

Count of (Independent) Repeated Connections



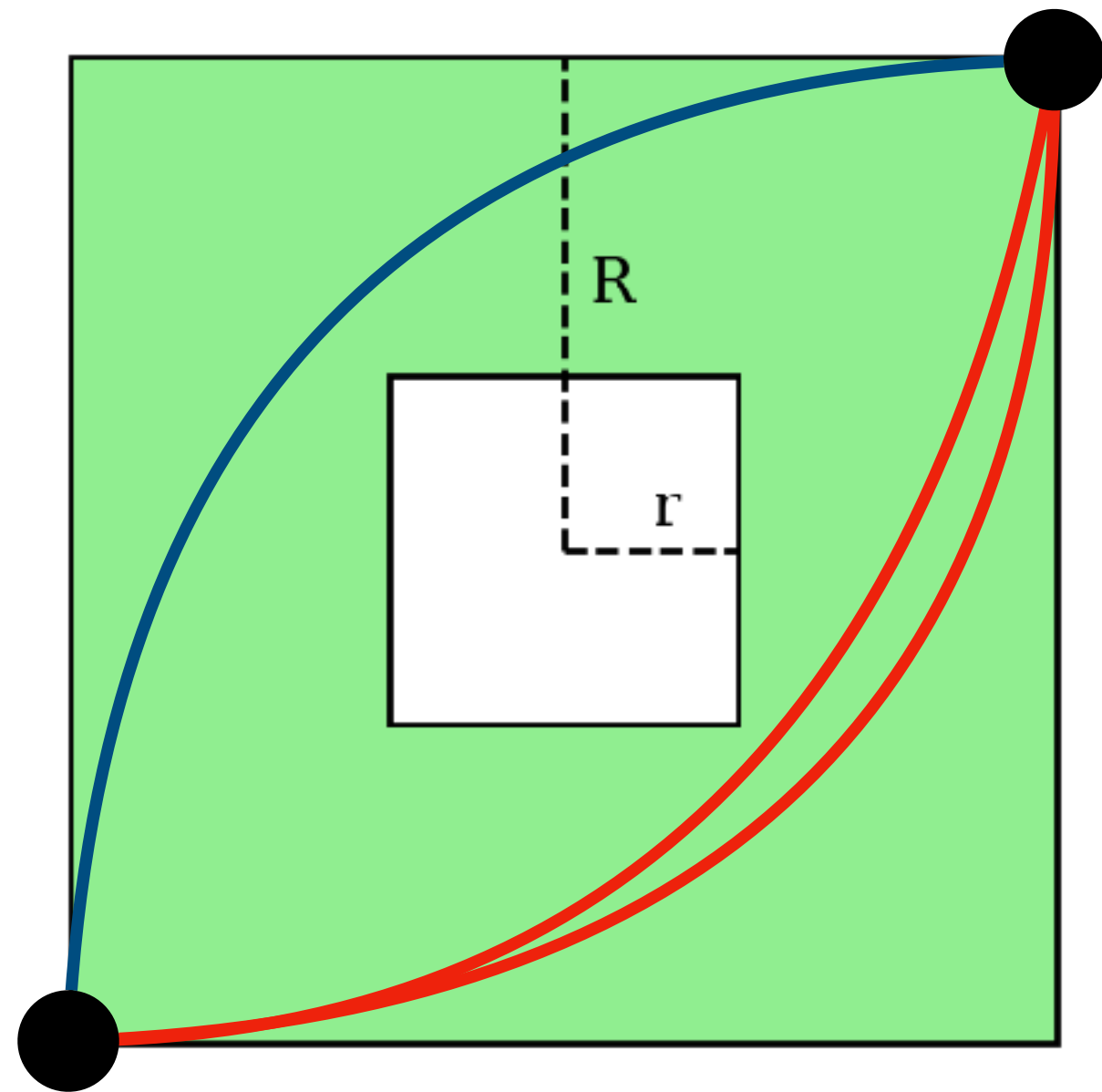
1 alternative path



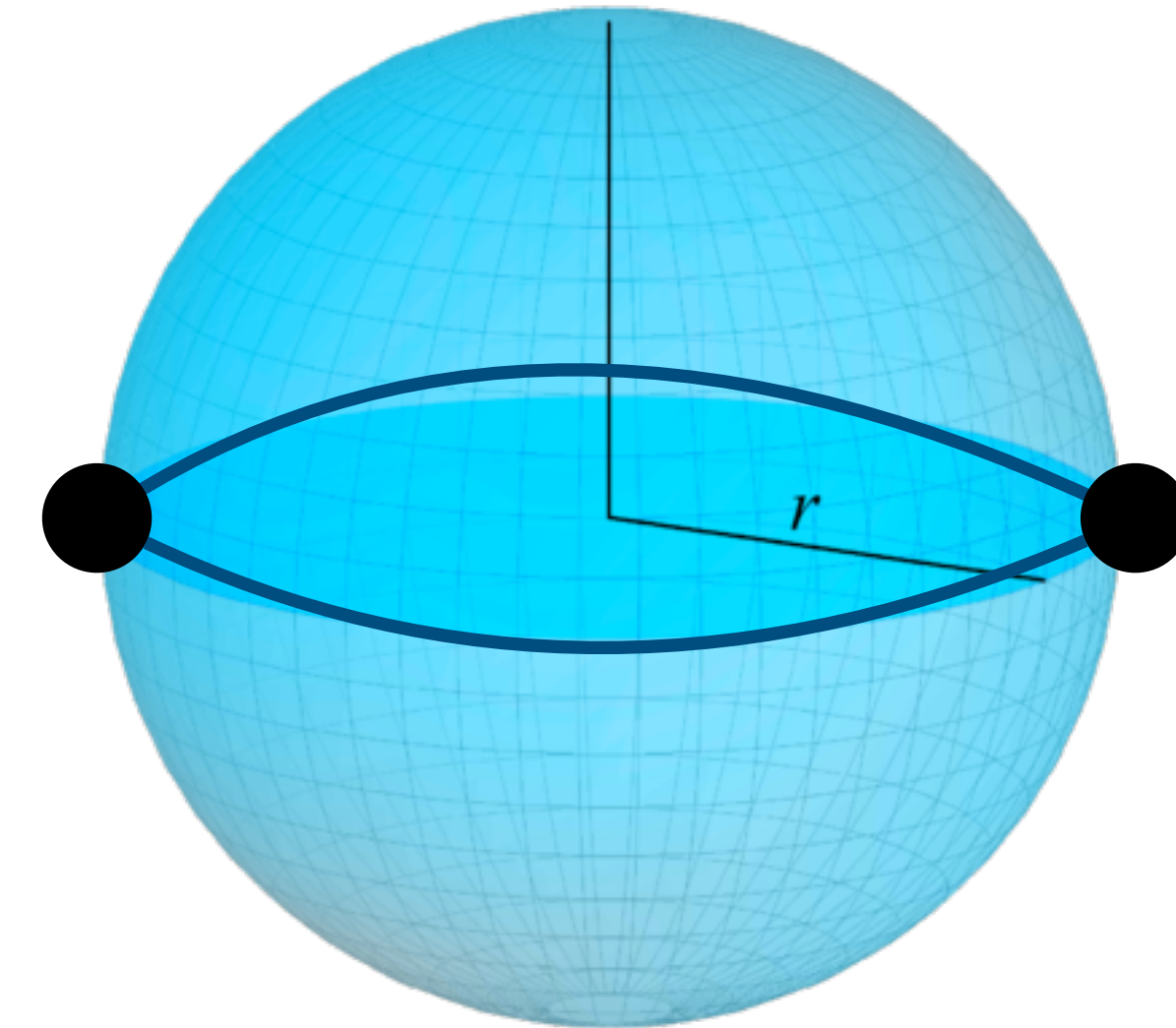
0 loop
1 cavity

Betti numbers β_k

Count of (Independent) Repeated Connections



1 alternative path

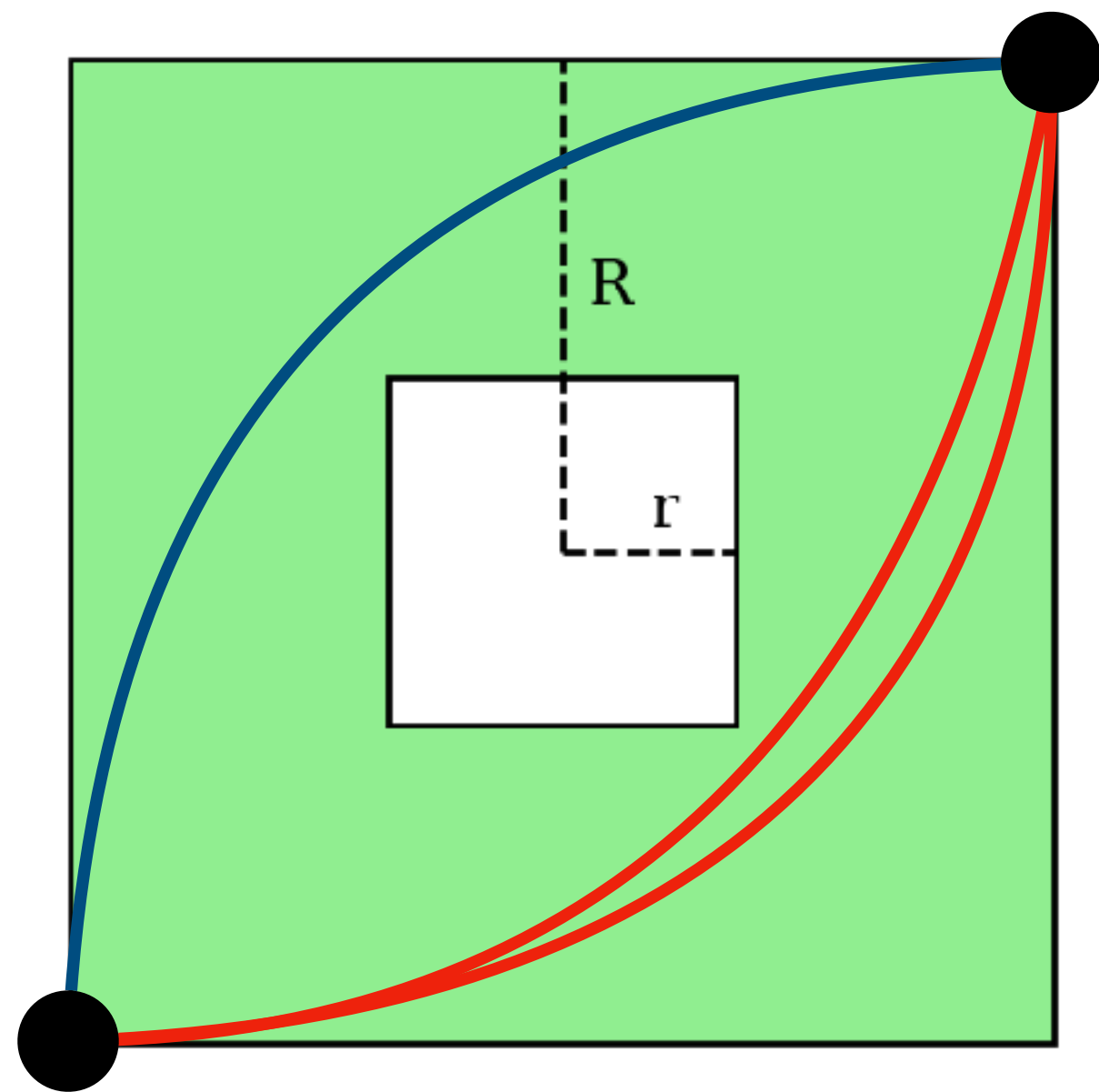


0 alternative path (slide through upper hemisphere)

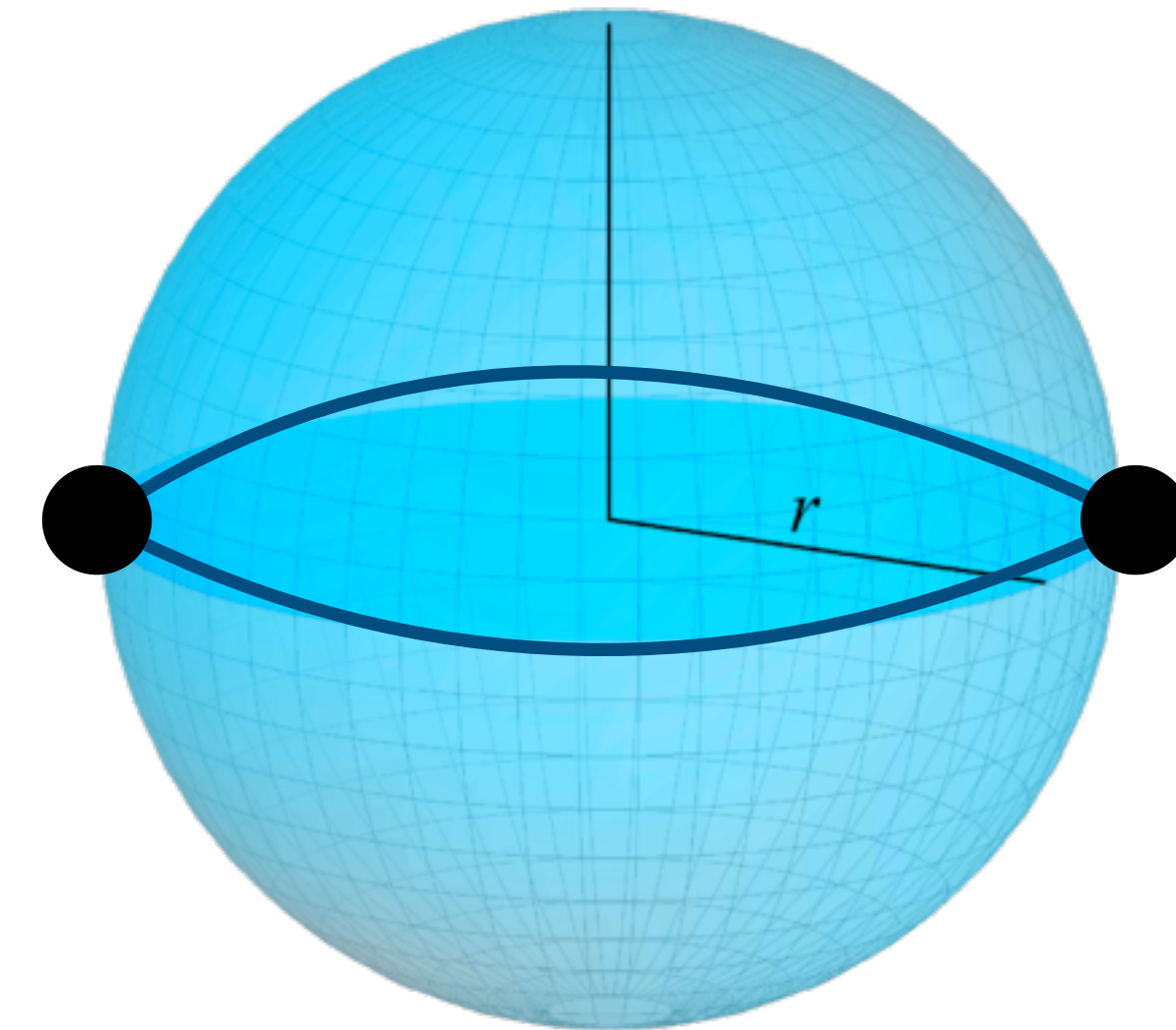
1 cavity

Betti numbers β_k

Count of (Independent) Repeated Connections



1 alternative path



0 alternative path (slide through upper hemisphere)

1 alternative way to slide a path

**Betti numbers count
repeated connections “in all dimensions”.**

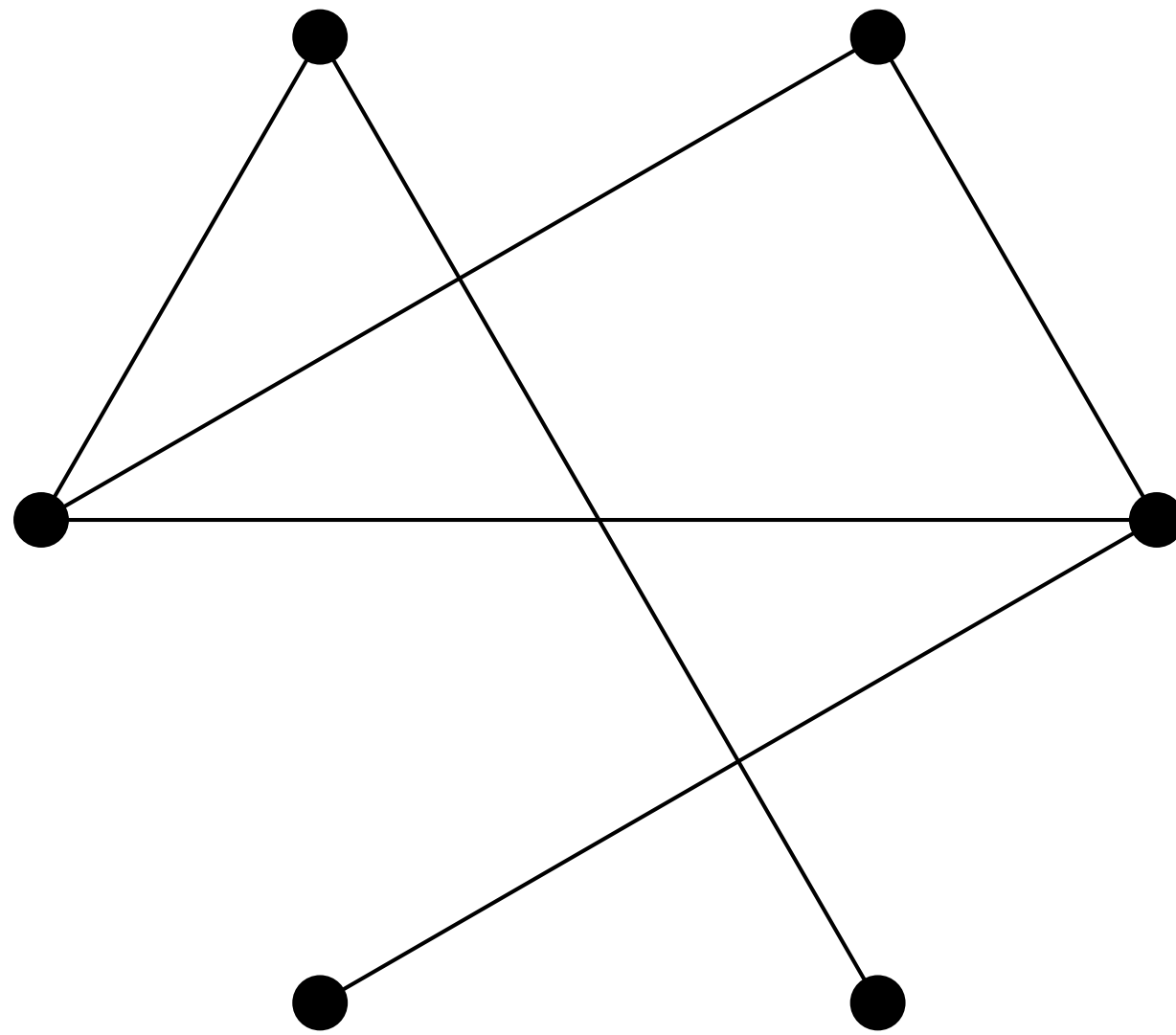
Interlude:

Random Walk in the Literature

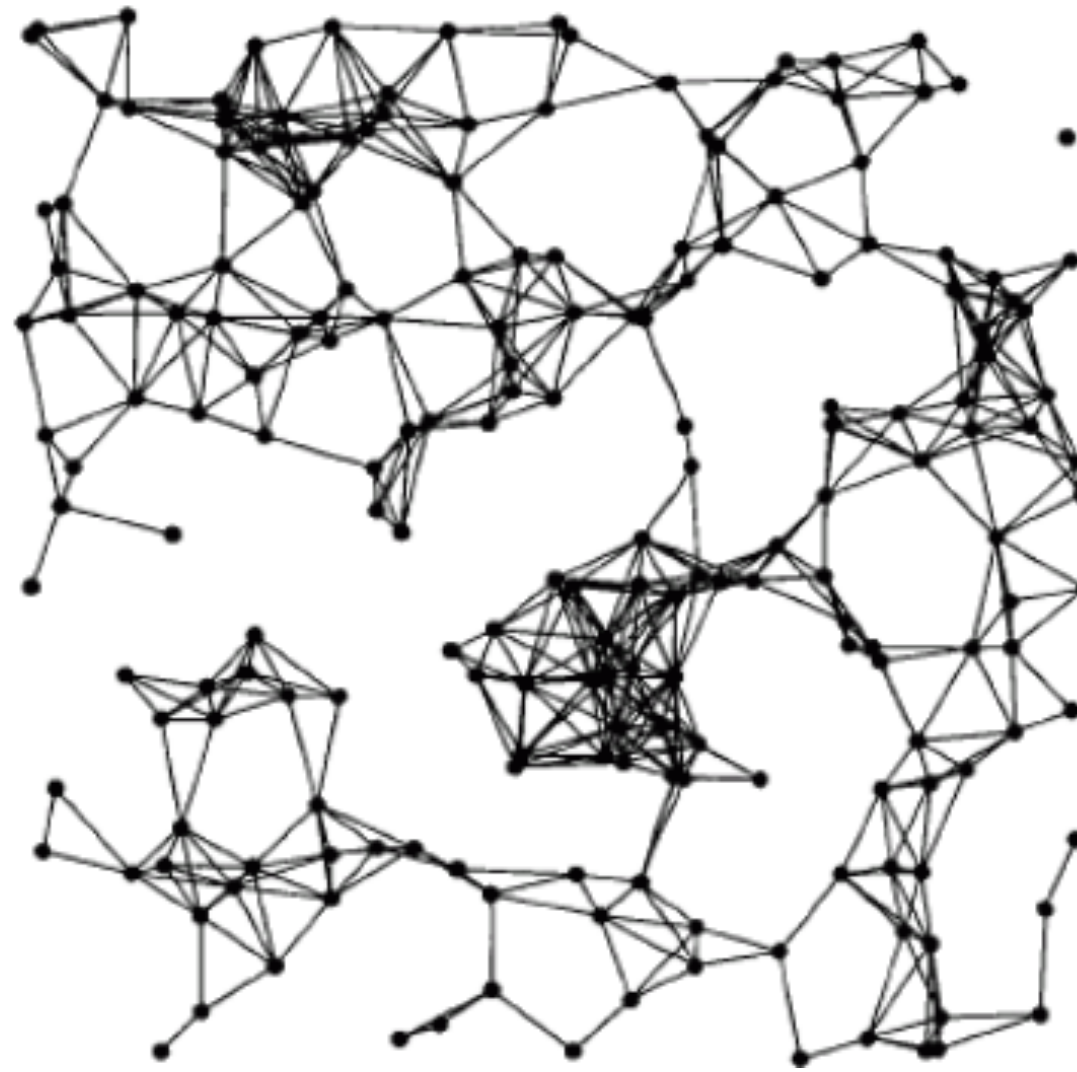
What Random Topologists Already Know

..

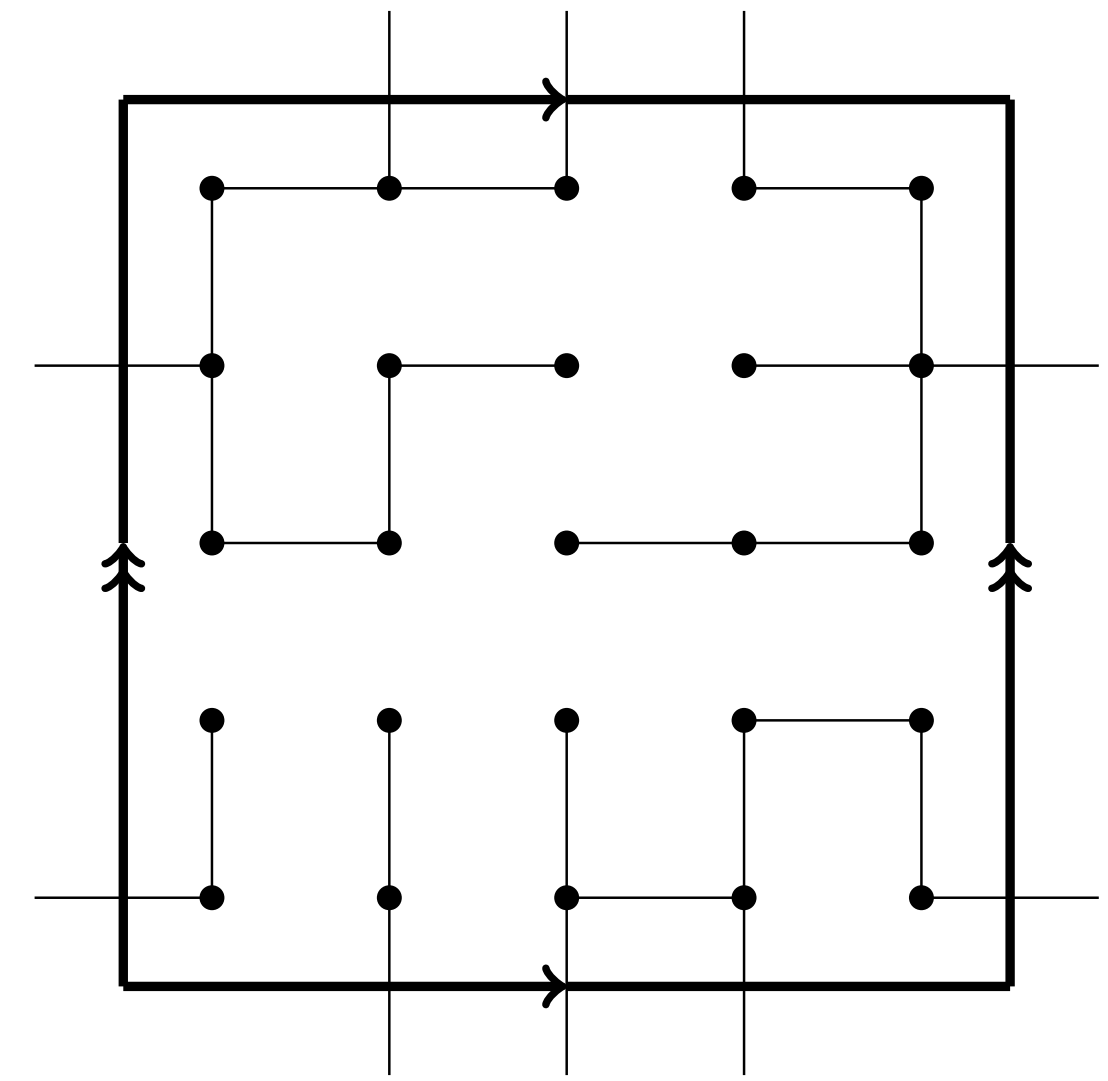
Tapas of Random Topology



Erdo-Renyi Complexes

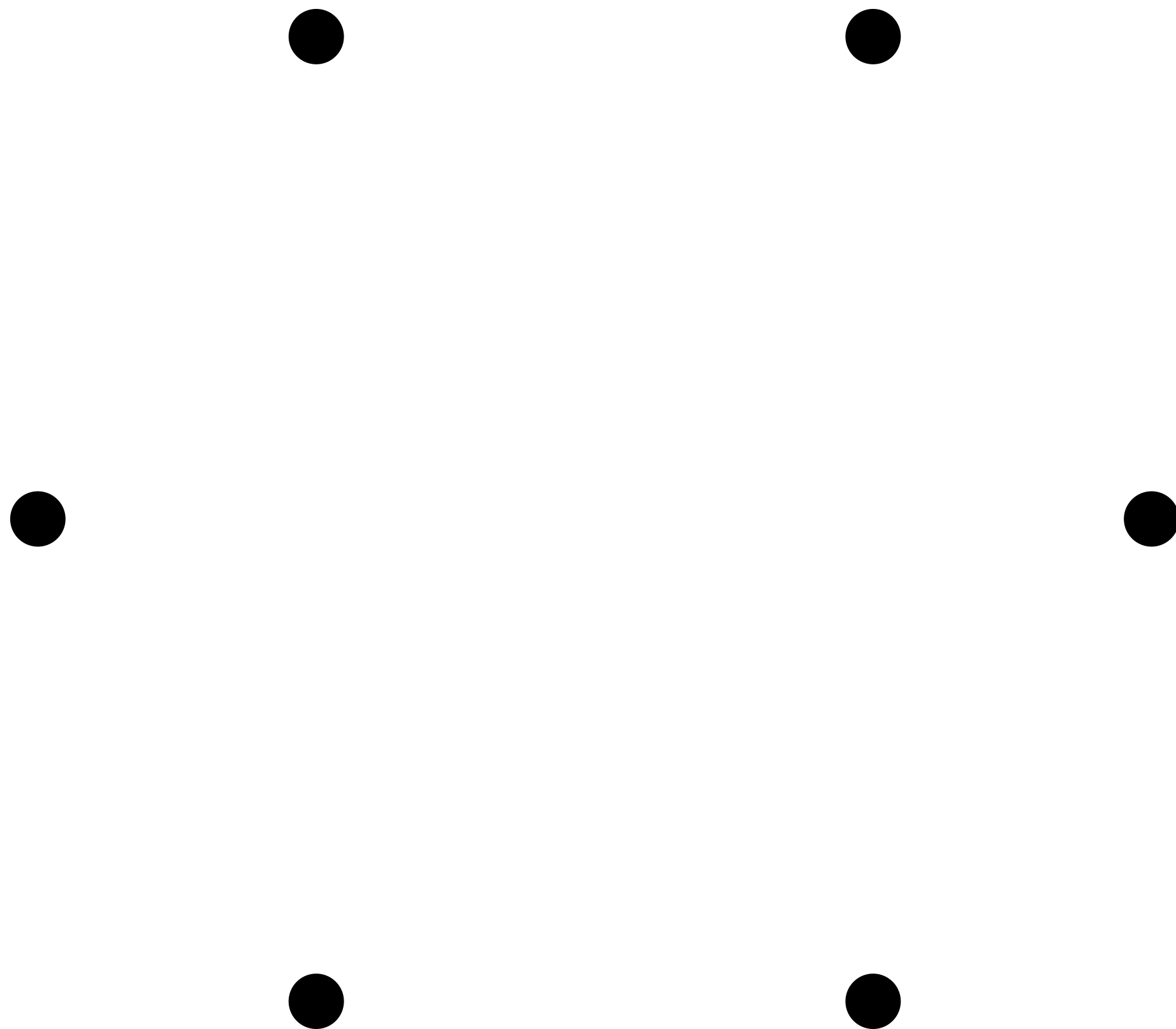


Geometric Complexes

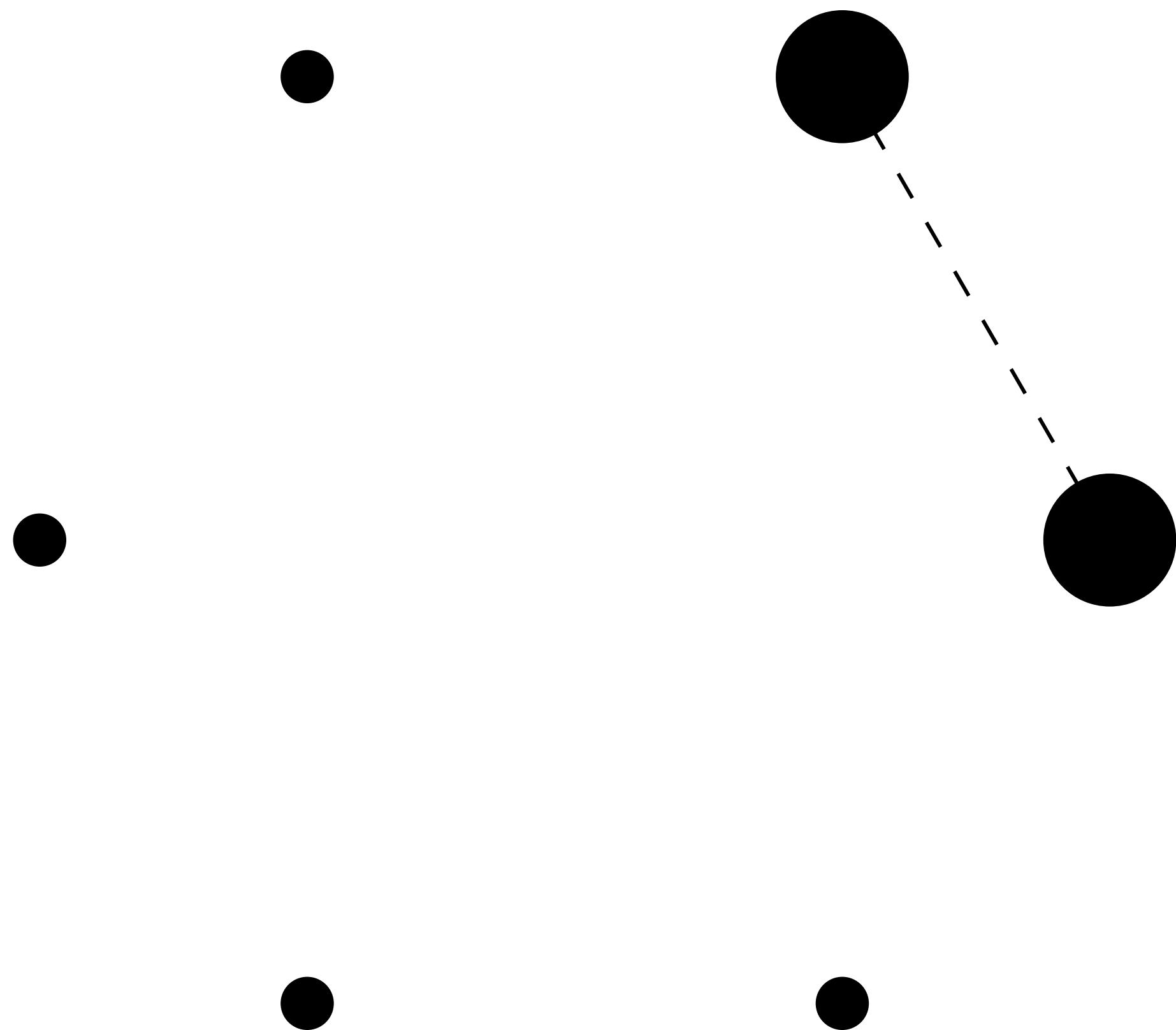


Topological Percolation

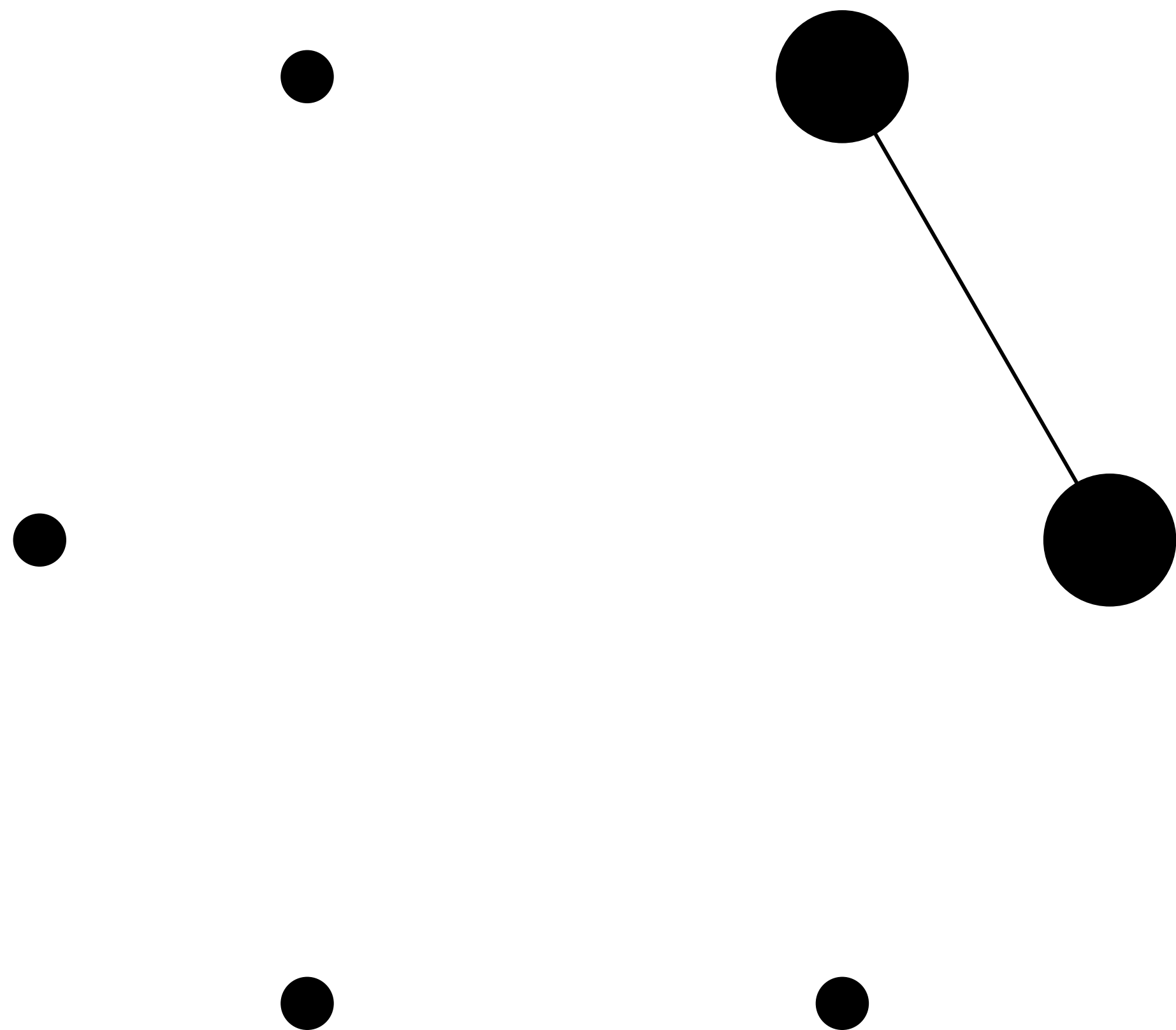
Erdos-Renyi graphs



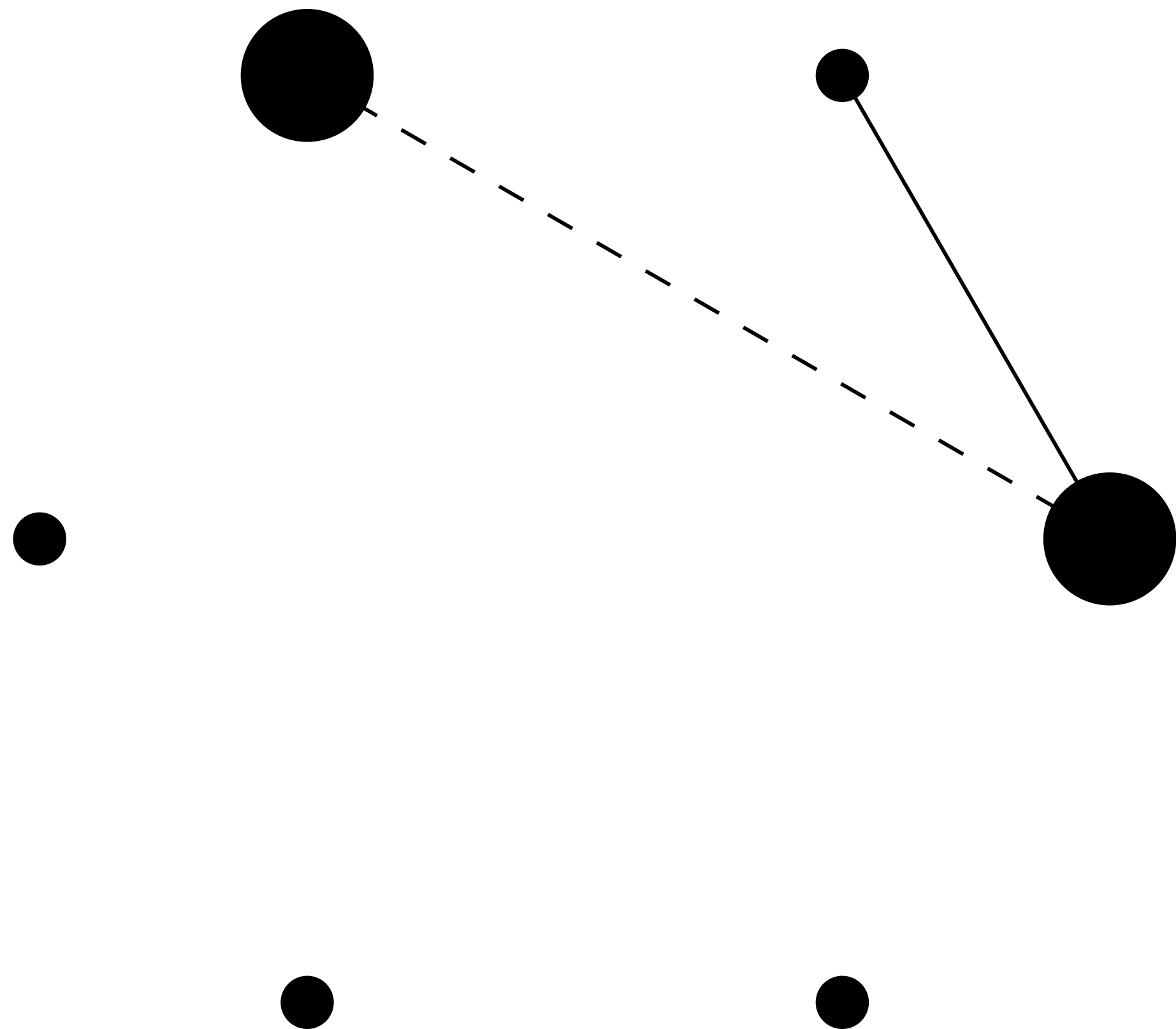
Erdos-Renyi graphs



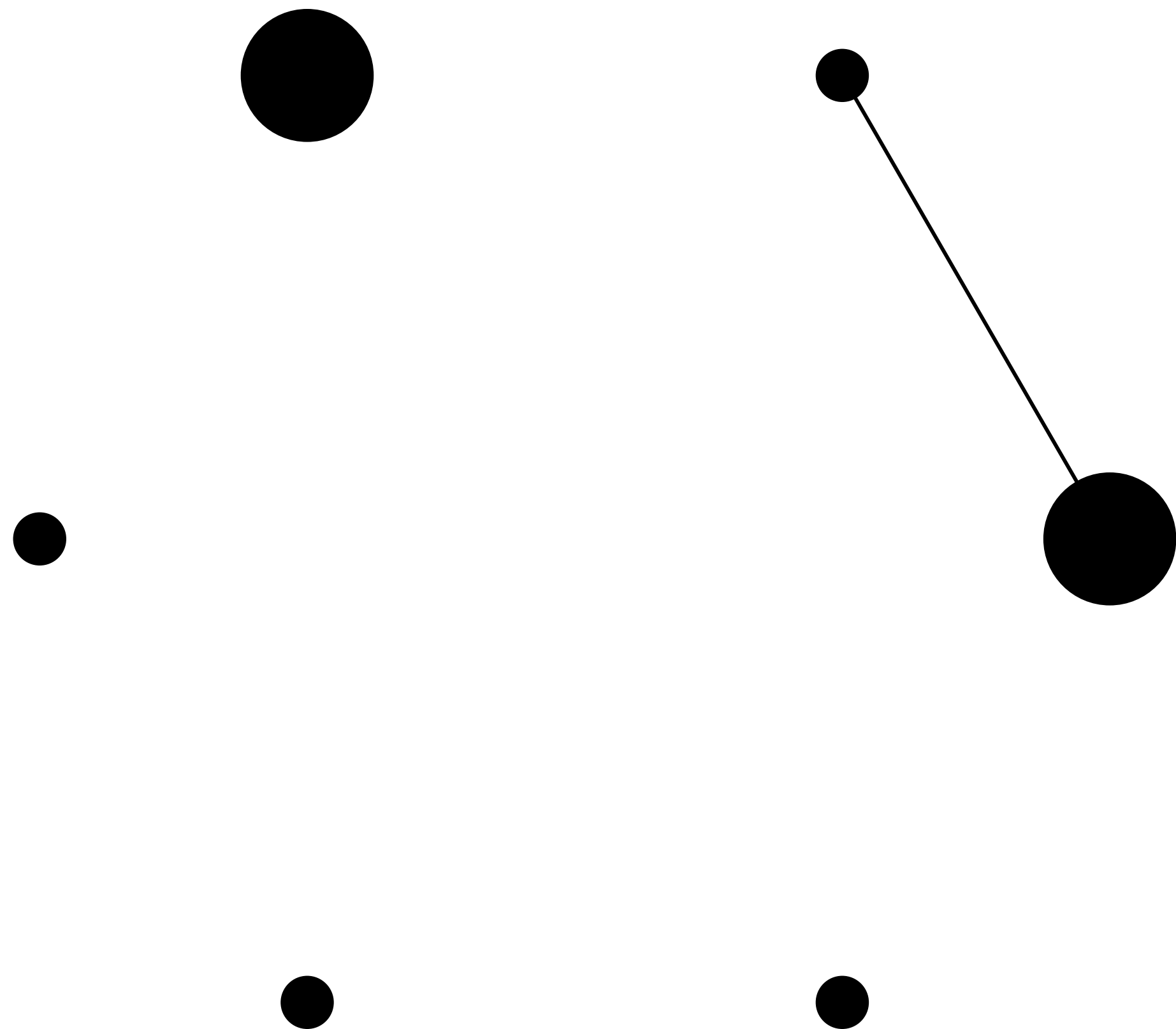
Erdos-Renyi graphs



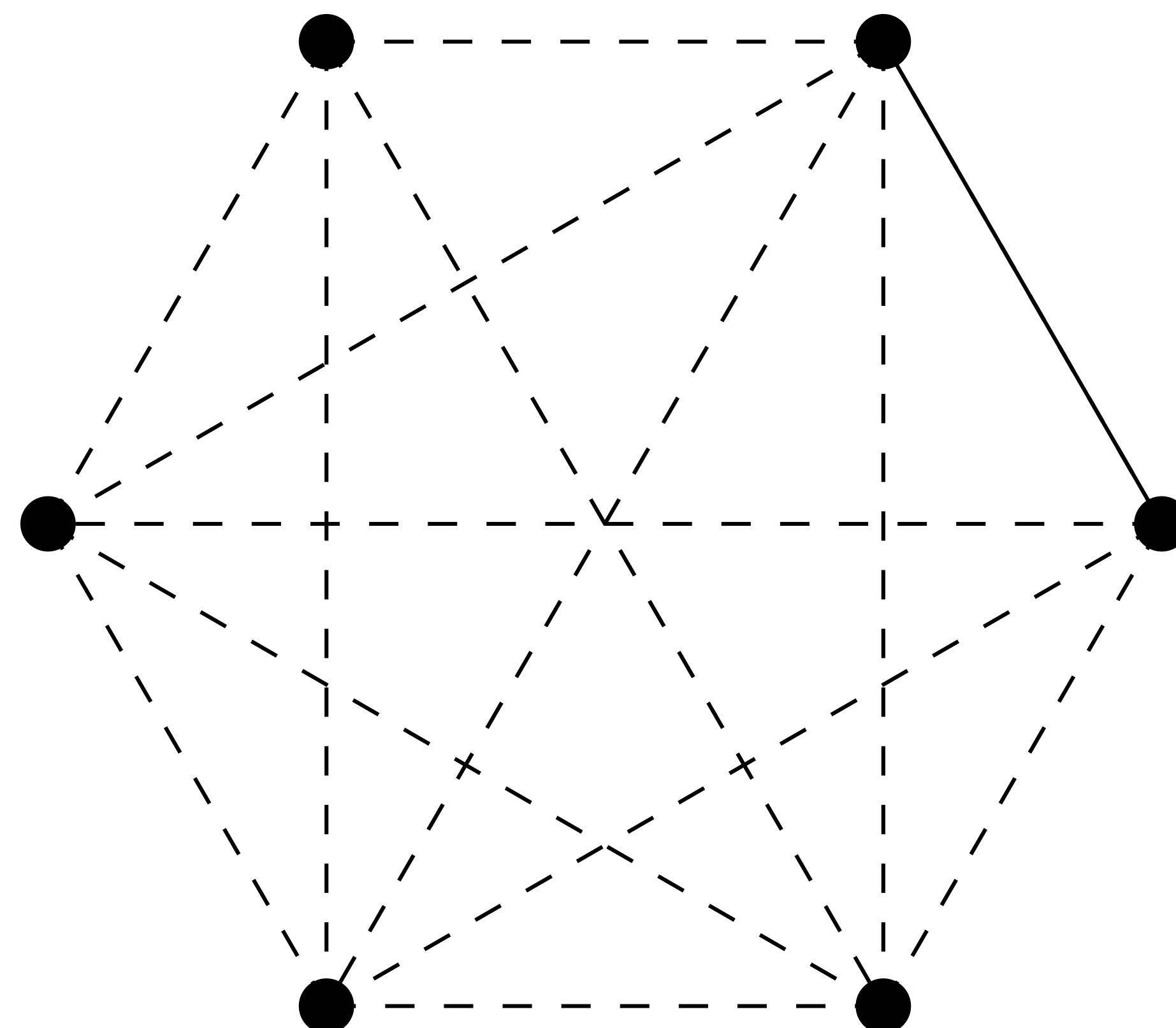
Erdos-Renyi graphs



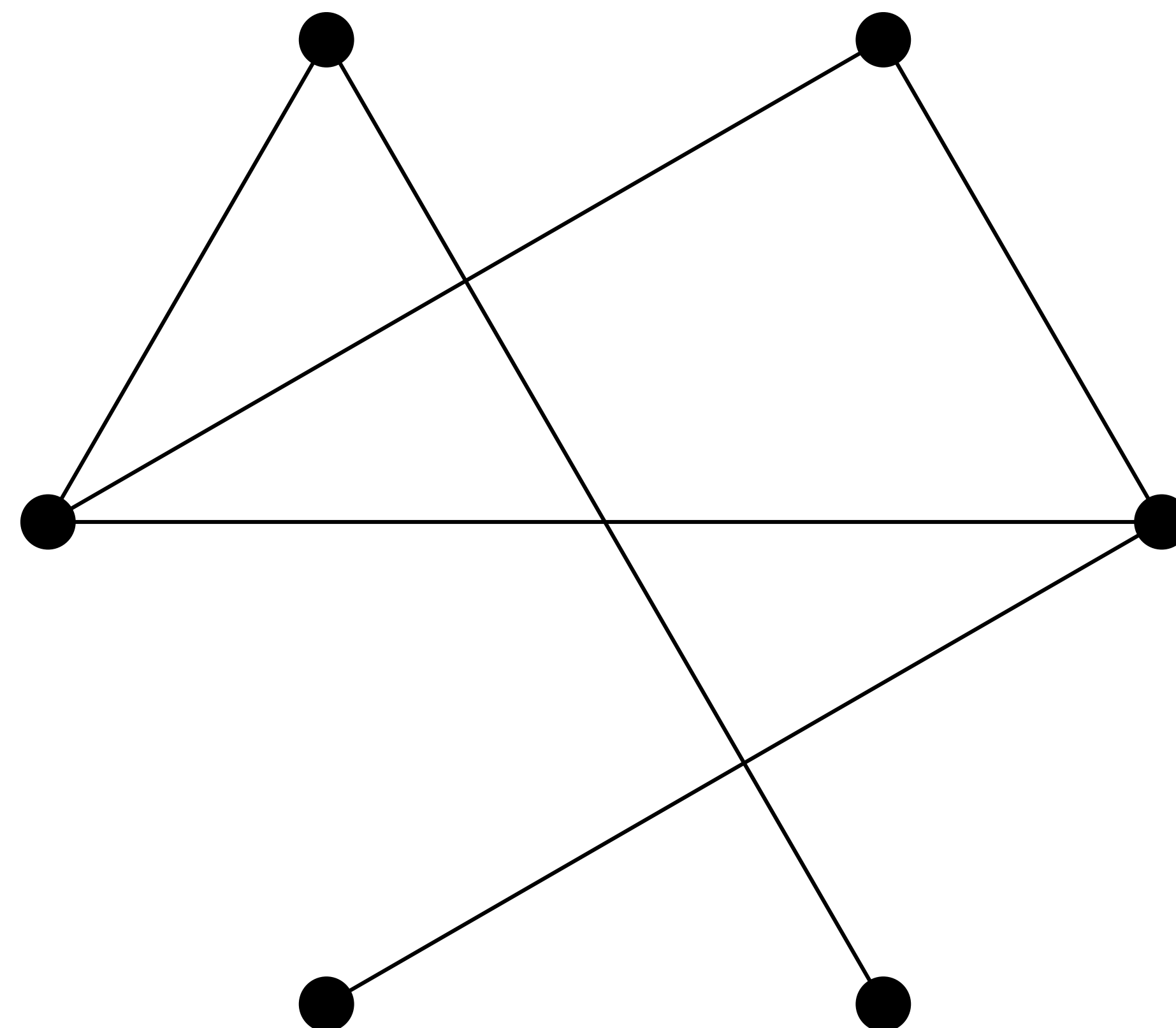
Erdos-Renyi graphs



Erdoes-Renyi graphs

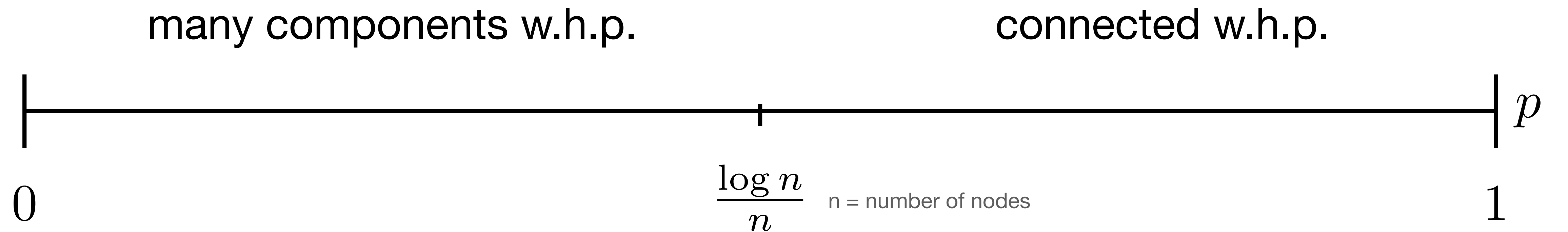


Erdos-Renyi graphs



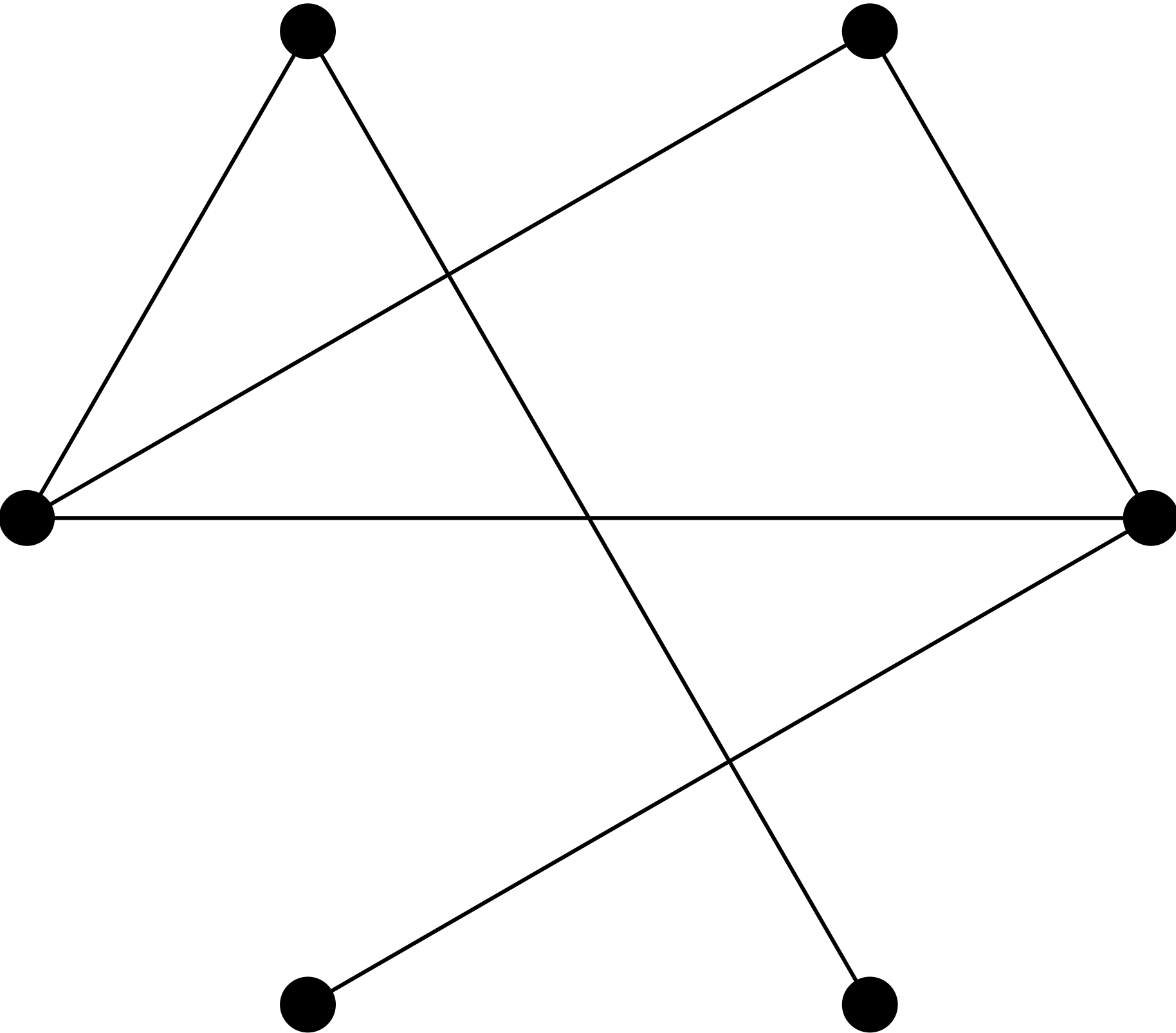
Phase Transition

[Erdos-Renyi 1960]

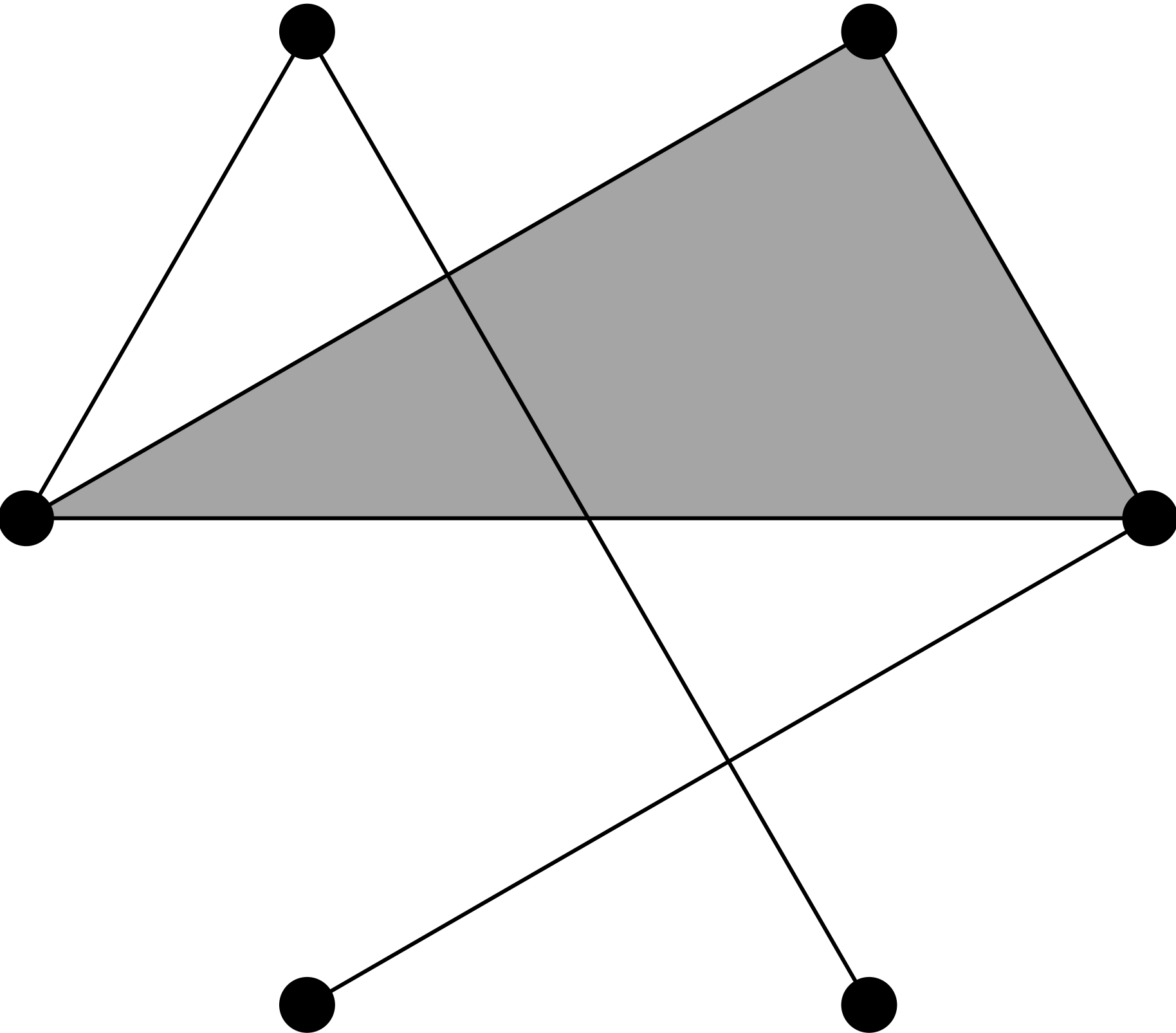


all log terms and constants forgone

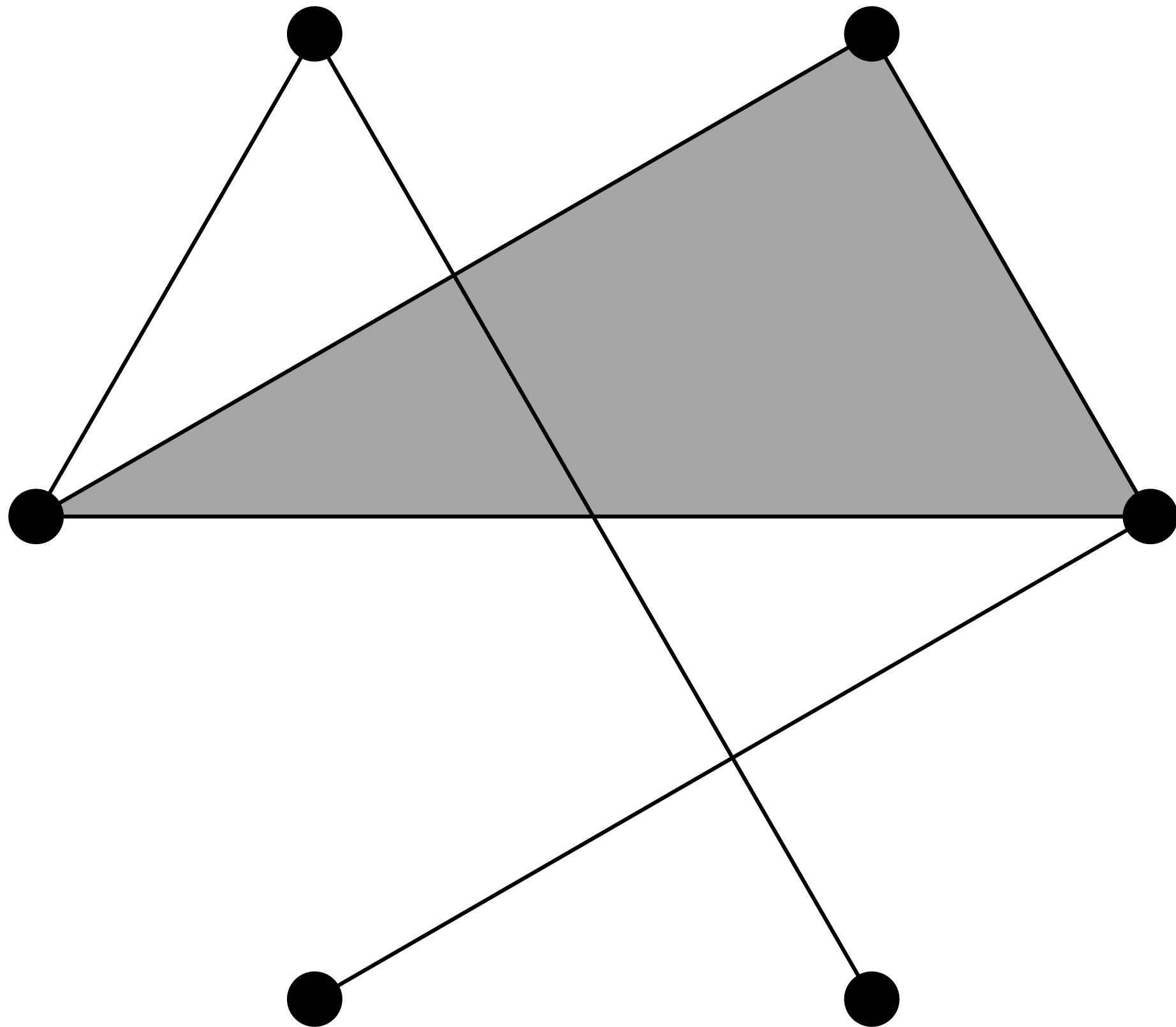
Erdos-Renyi Clique Complex



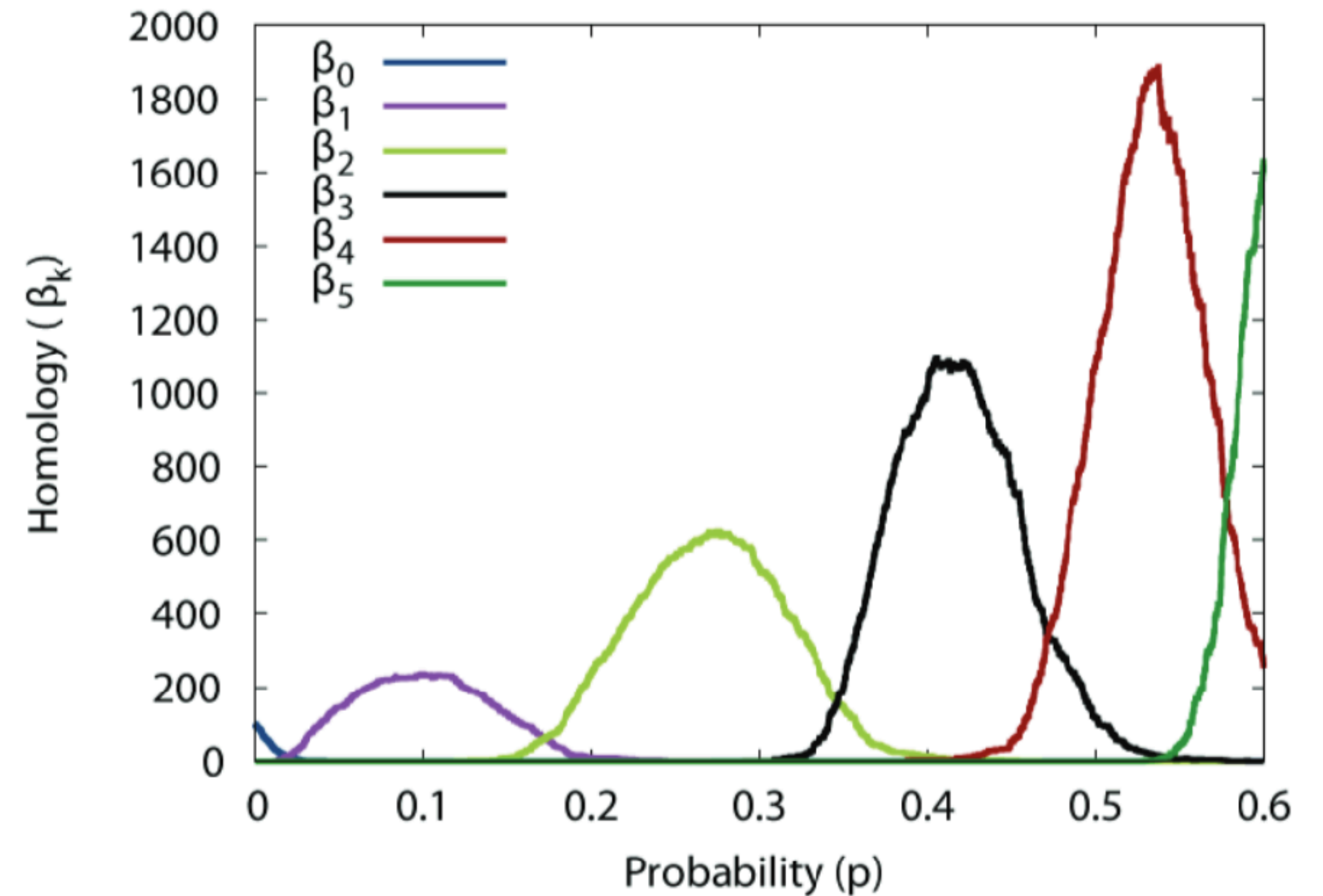
Erdos-Renyi Clique Complex



Betti Numbers



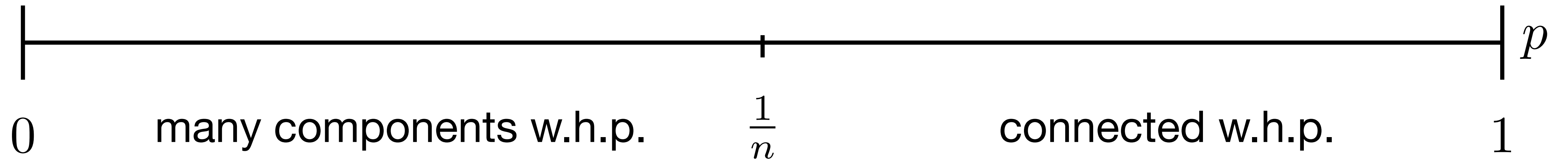
Erdős–Rényi random complex on $n=100$ vertices



computation and plotting done by Zomorodian

Phase Transition

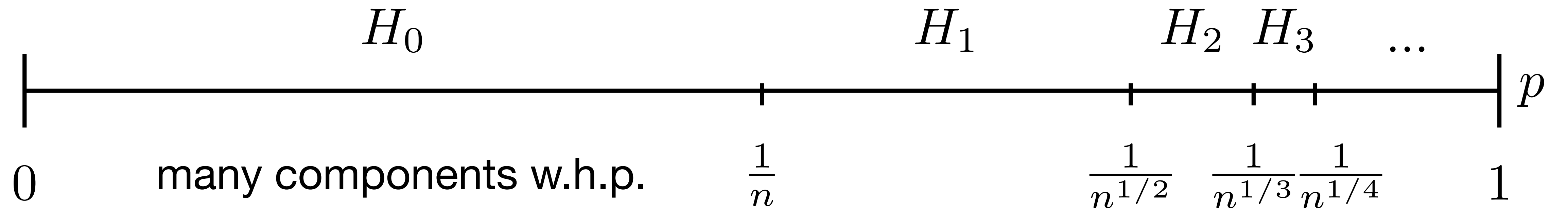
[Erdos-Renyi 1960]



n = number of nodes
all log terms and constants forgone

Phase Transition

[Kahle 2009, 2014]

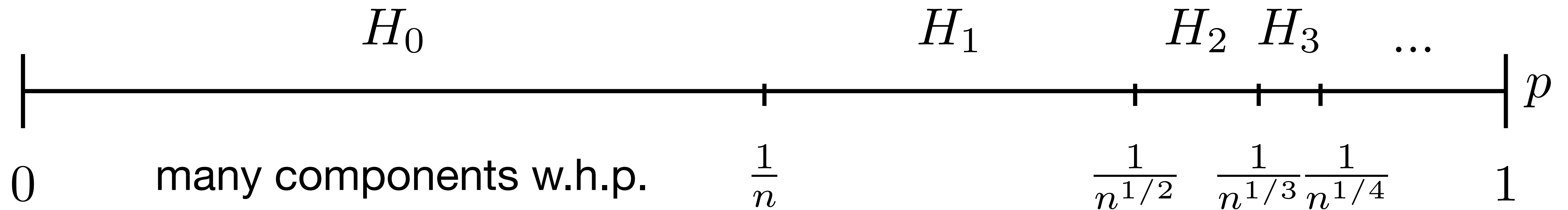
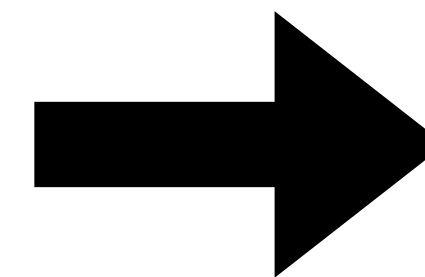


n = number of nodes
all log terms and constants forgone

Phase Transition

[Kahle 2009, 2014]

Holes get filled.



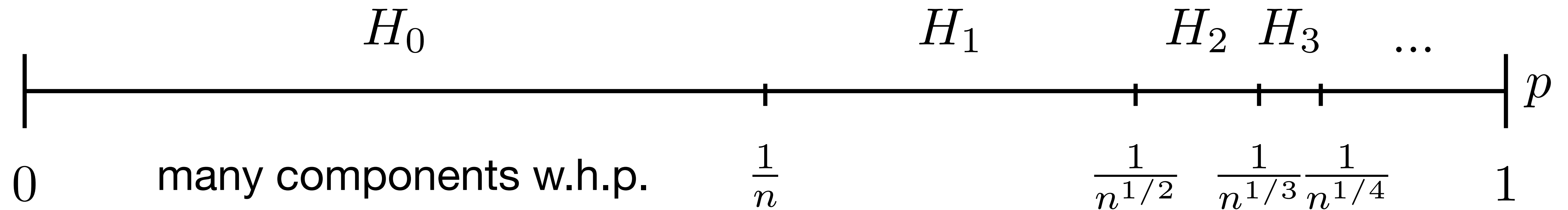
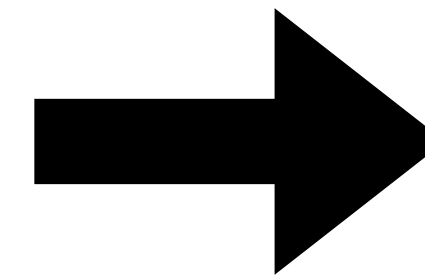
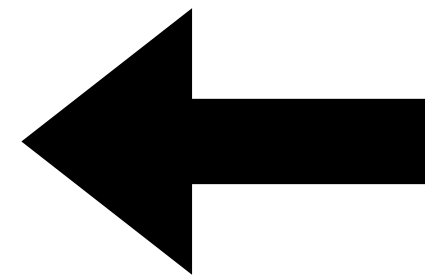
n = number of nodes
all log terms and constants forgone

Phase Transition

[Kahle 2009, 2014]

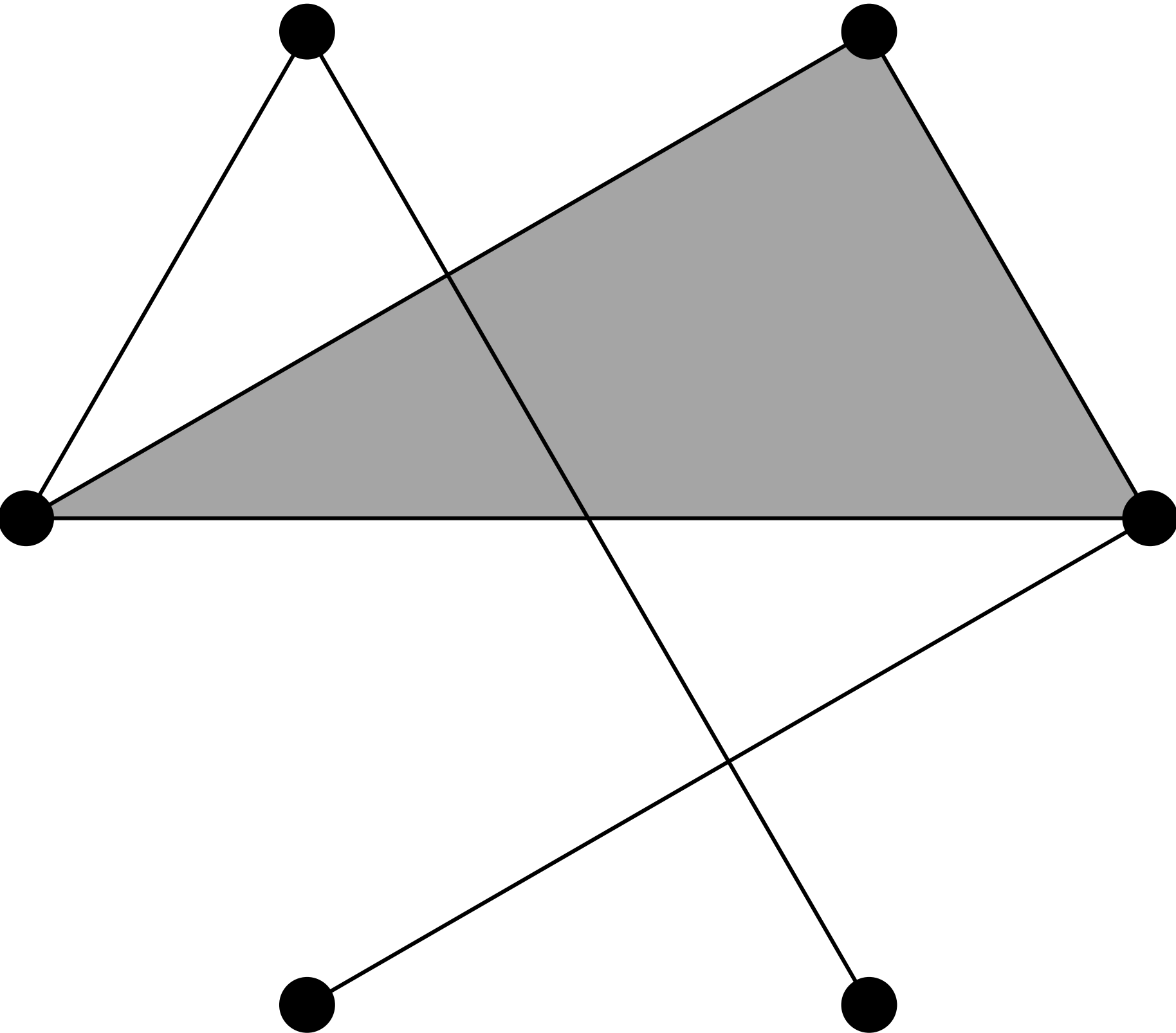
Holes can't form.

Holes get filled.



n = number of nodes
all log terms and constants forgone

Erdos-Renyi Clique Complex



Geometric Complexes



image credit: Penrose

Geometric Complexes

- Rips
- Cech



image credit: Penrose

Geometric Complexes

- Rips (clique)
- Cech



image credit: Penrose

Geometric Complexes

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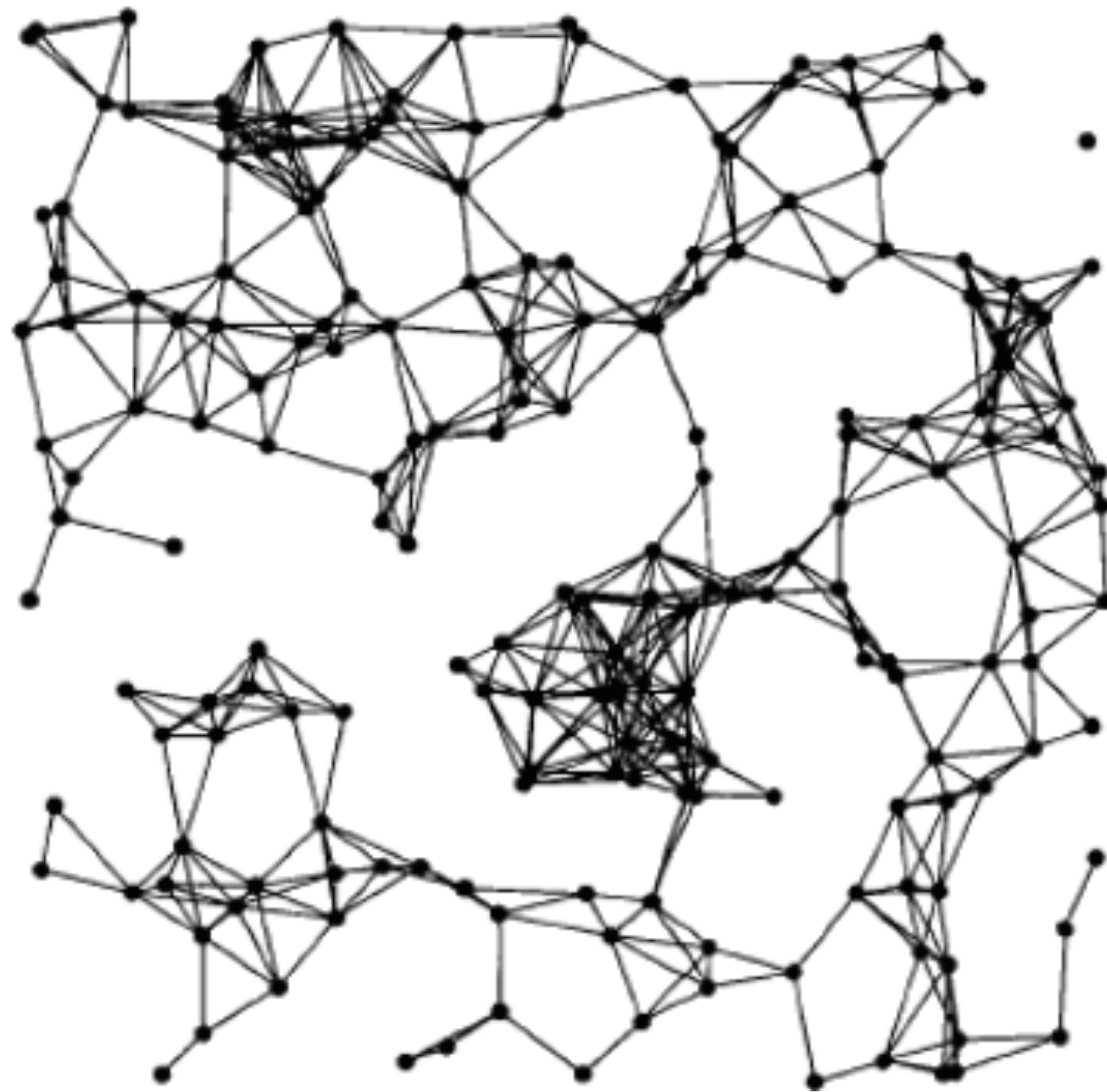
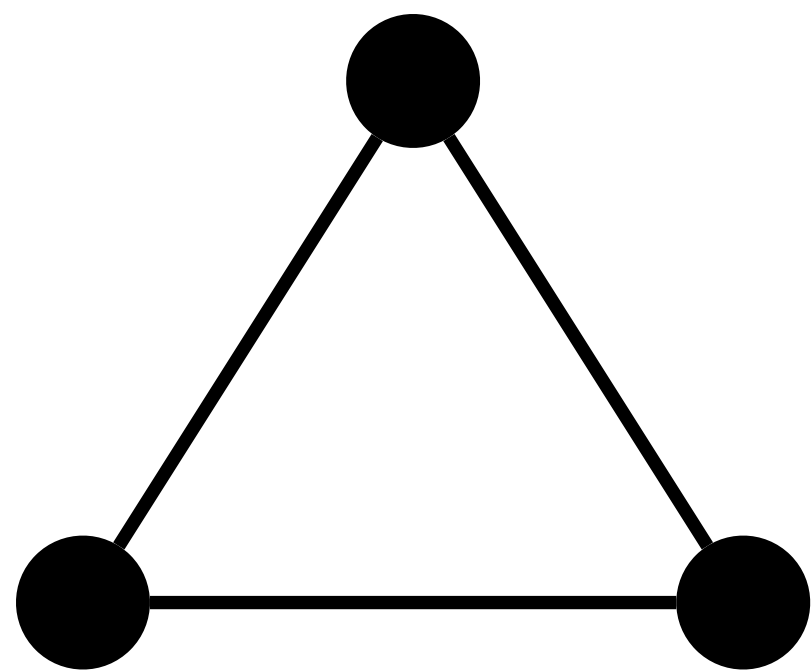


image credit: Penrose

Geometric Complexes

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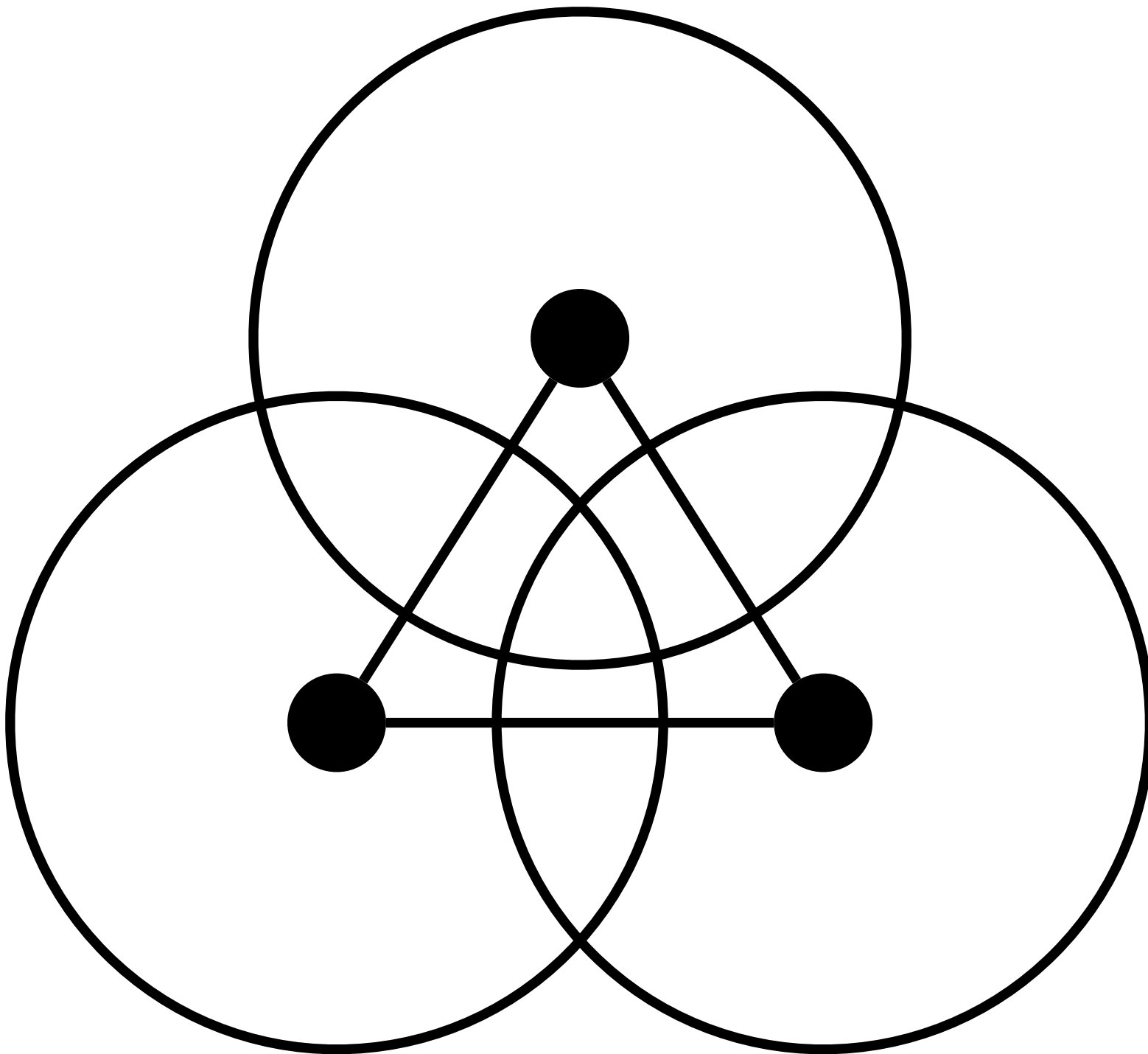


image credit: Penrose

Geometric Complexes

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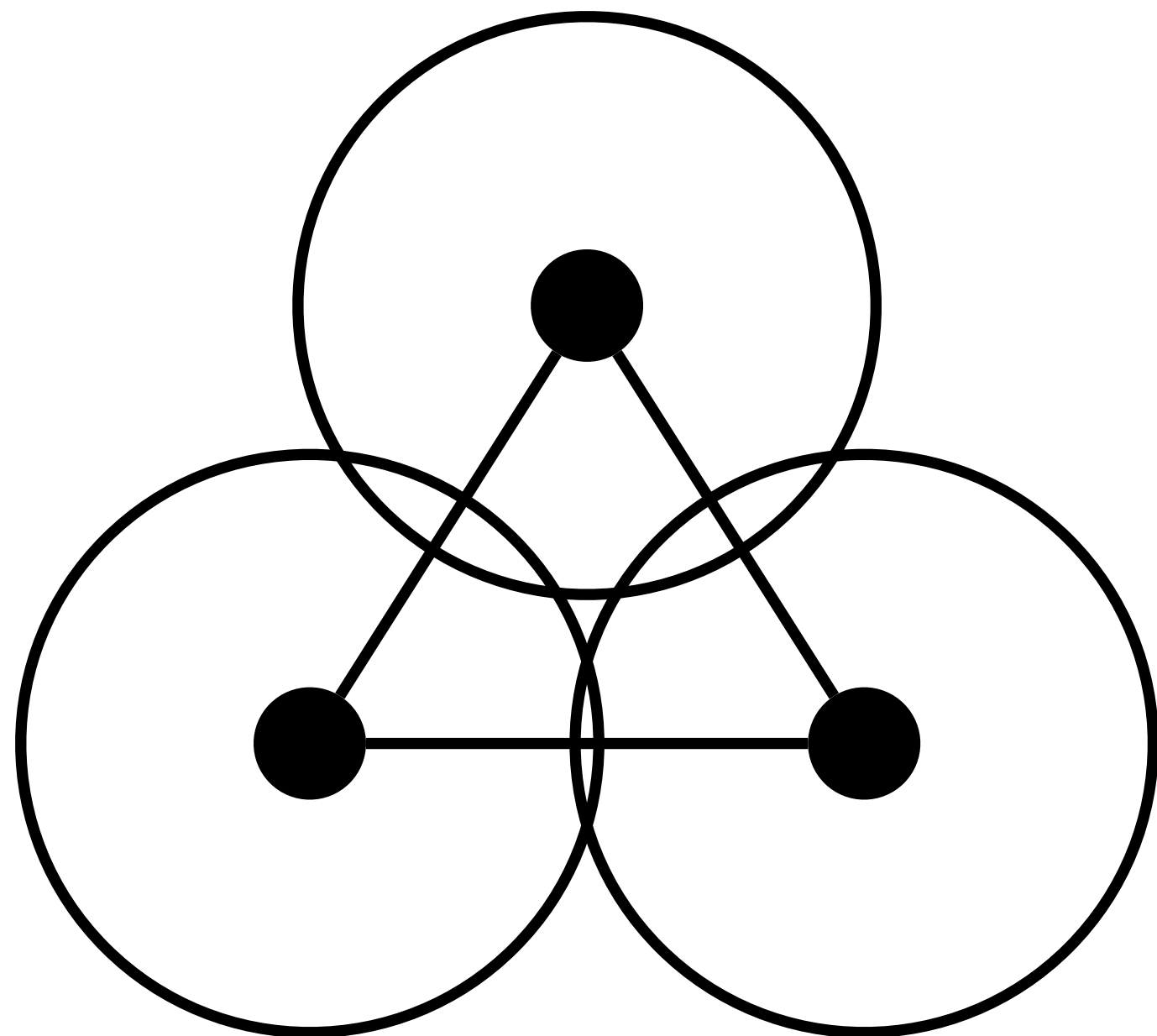


image credit: Penrose

Expected Betti numbers at dimension k

[Kahle 2011]

Expected Betti numbers at dimension k

[Kahle 2011]

- n , the number of points

Expected Betti numbers at dimension k

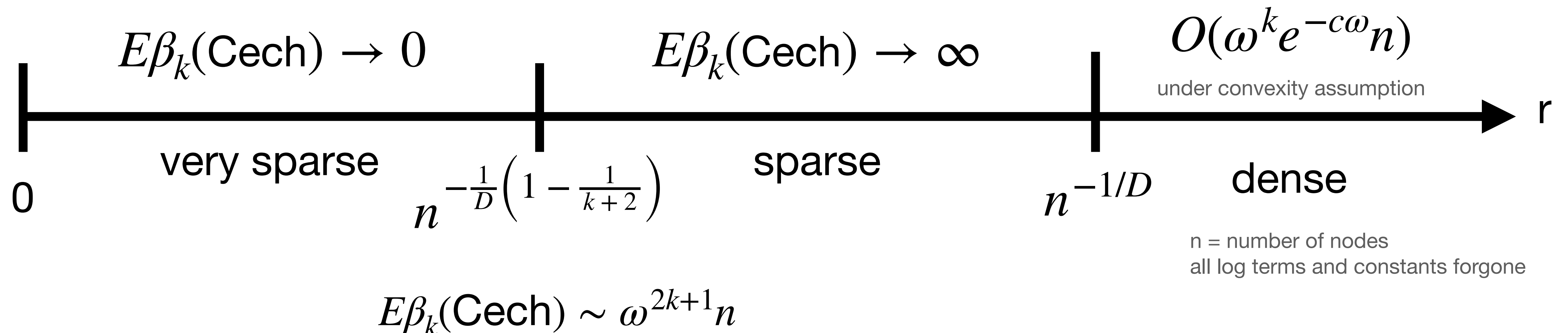
[Kahle 2011]

- n , the number of points
- $\omega = nr^D$, where D is the ambient dimension

Expected Betti numbers at dimension k

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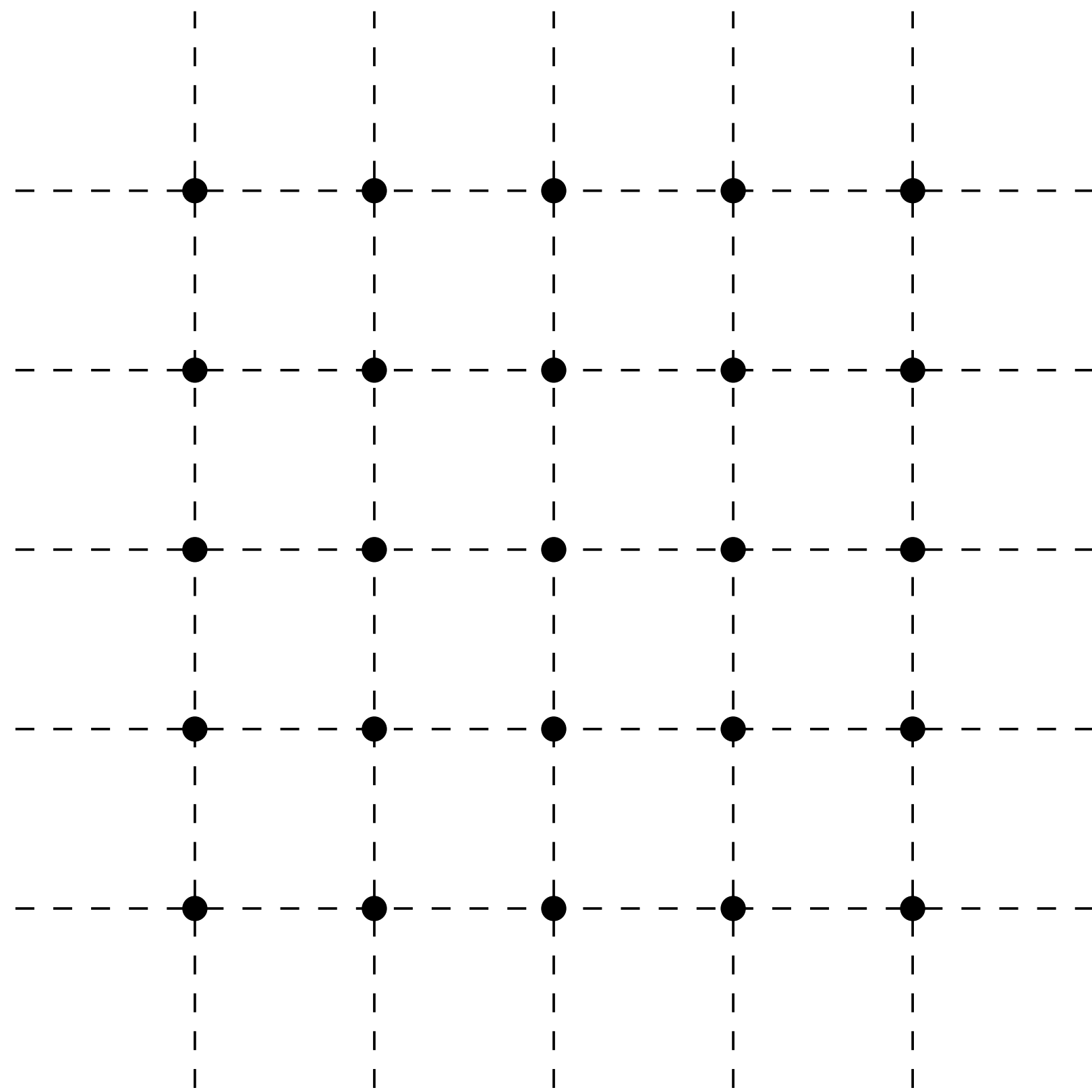
Geometric Complexes



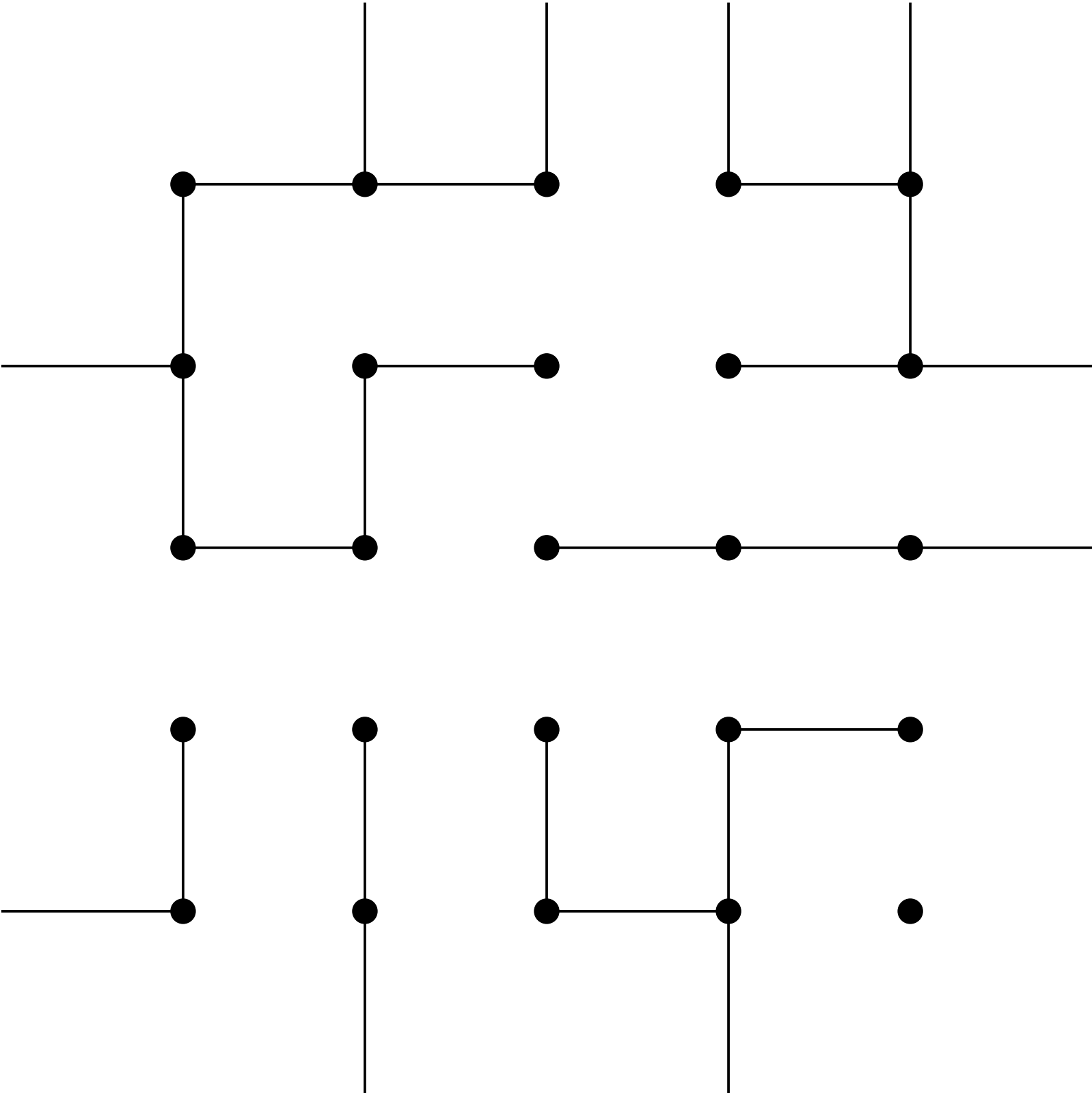
image credit: Penrose

29 min

Bernoulli Bond Percolation

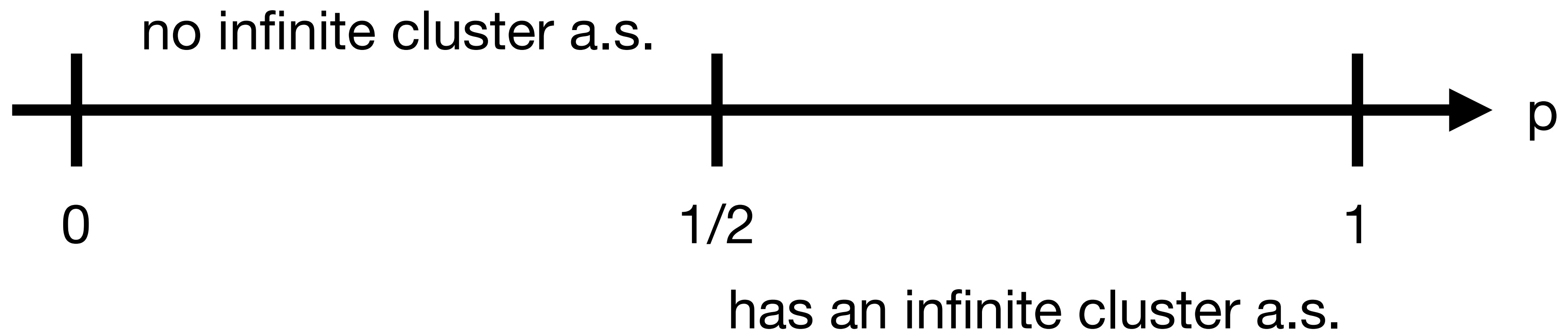


Bernoulli Bond Percolation



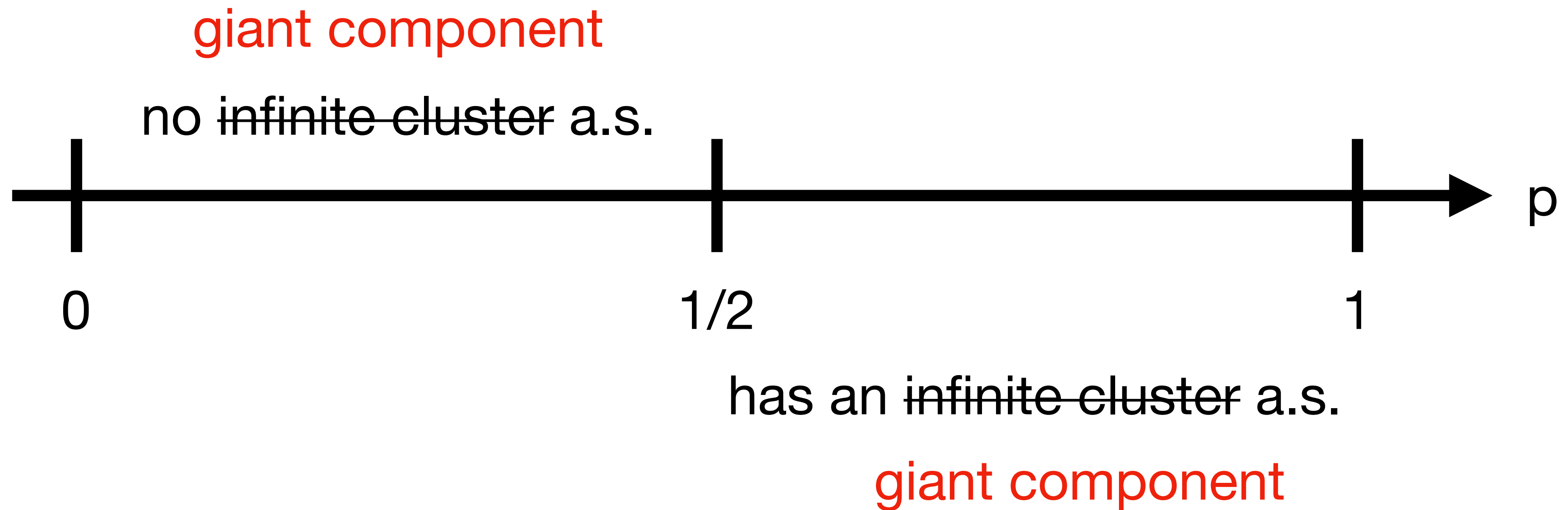
Phase Transition

[Harris 1960, Kesten 1980]



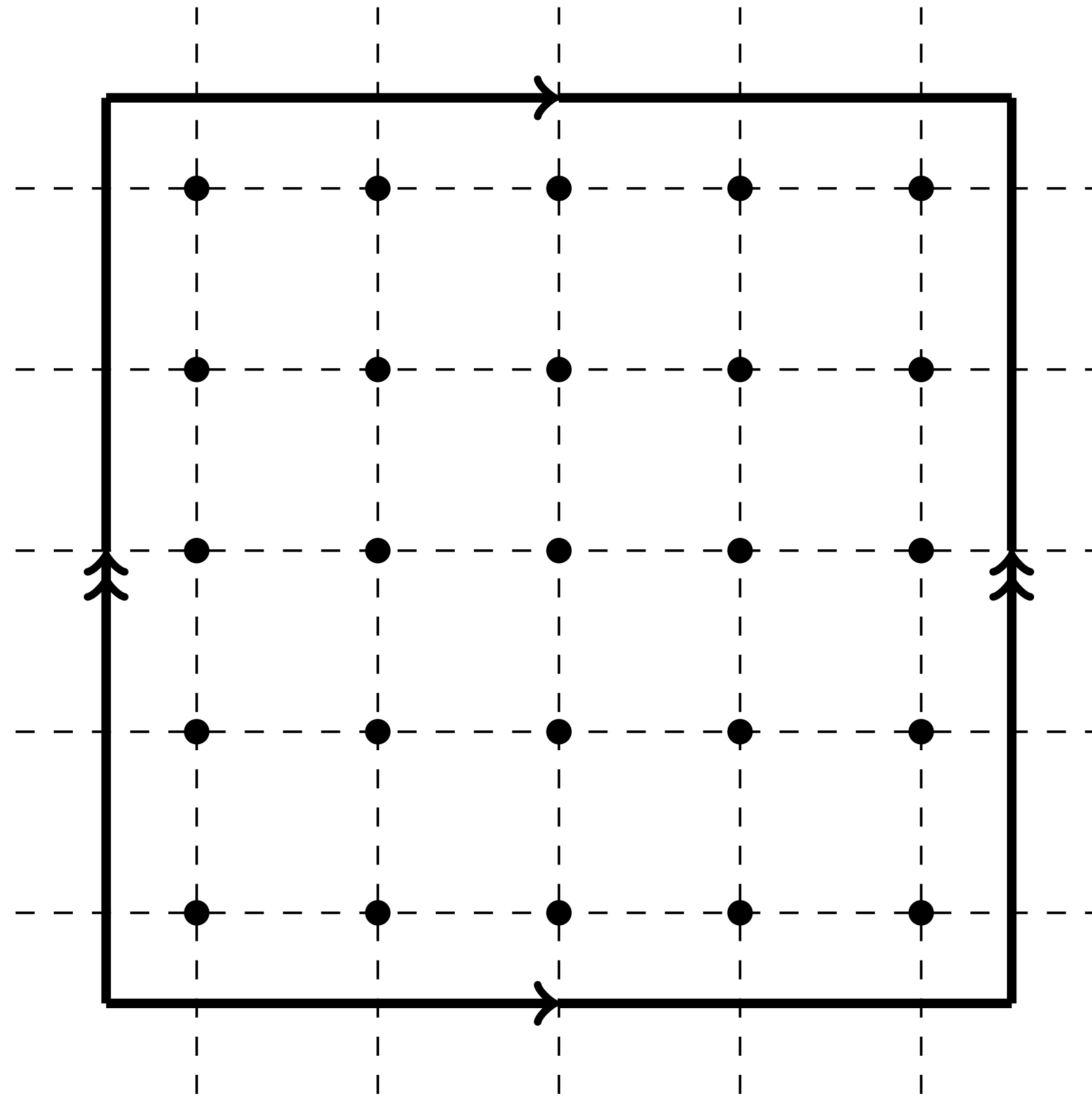
Phase Transition

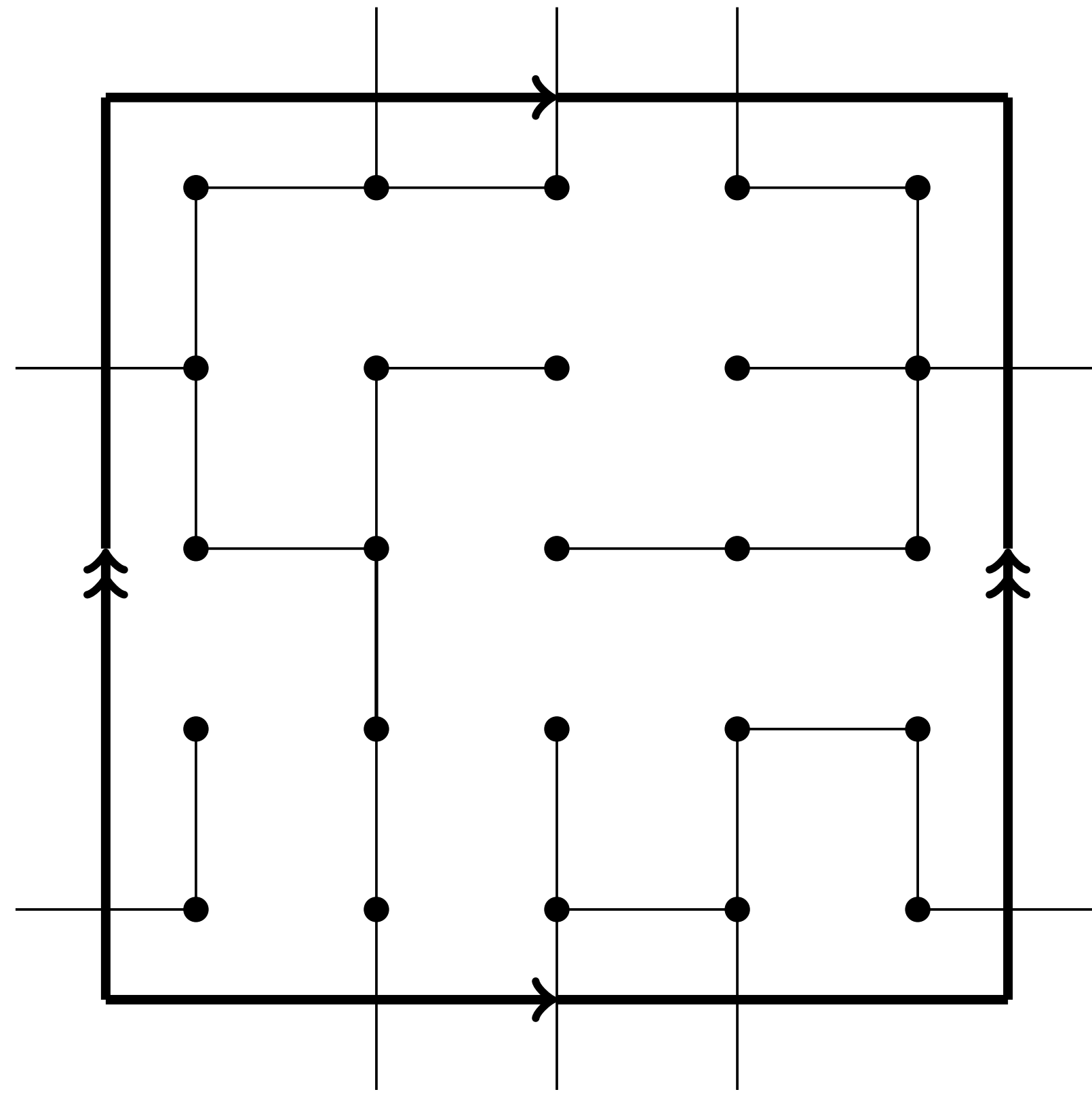
[Harris 1960, Kesten 1980]

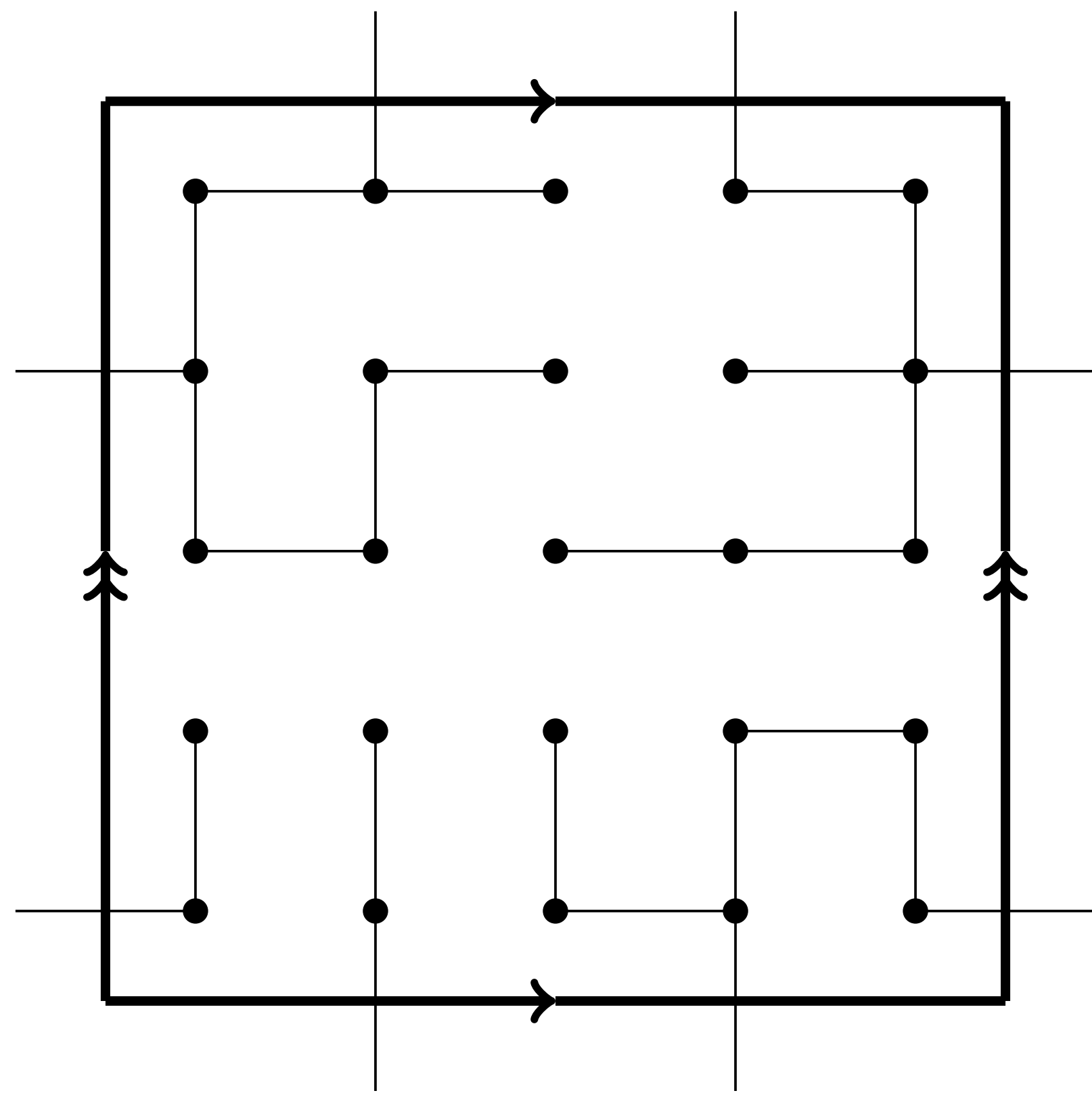


Giant Cycles?

Bernoulli Bond Percolation

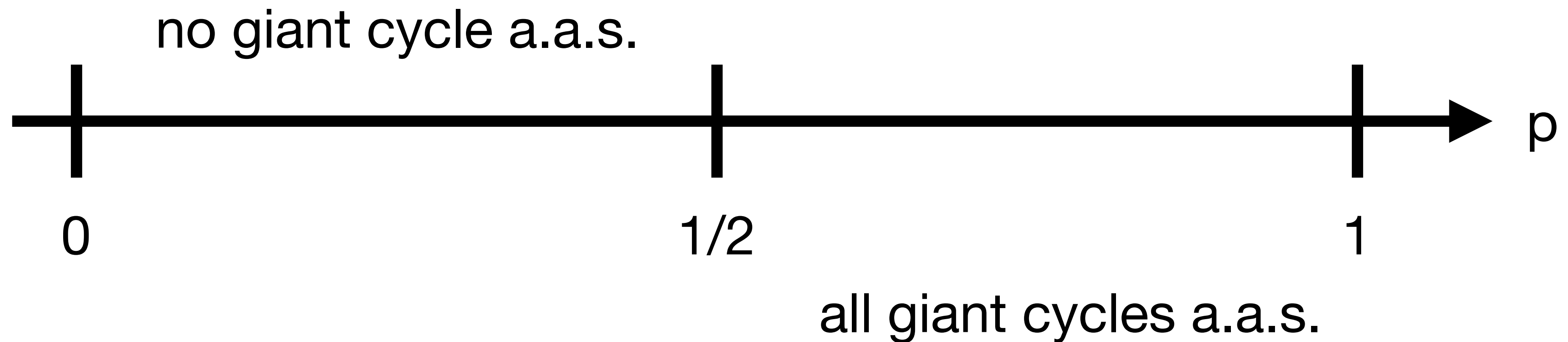




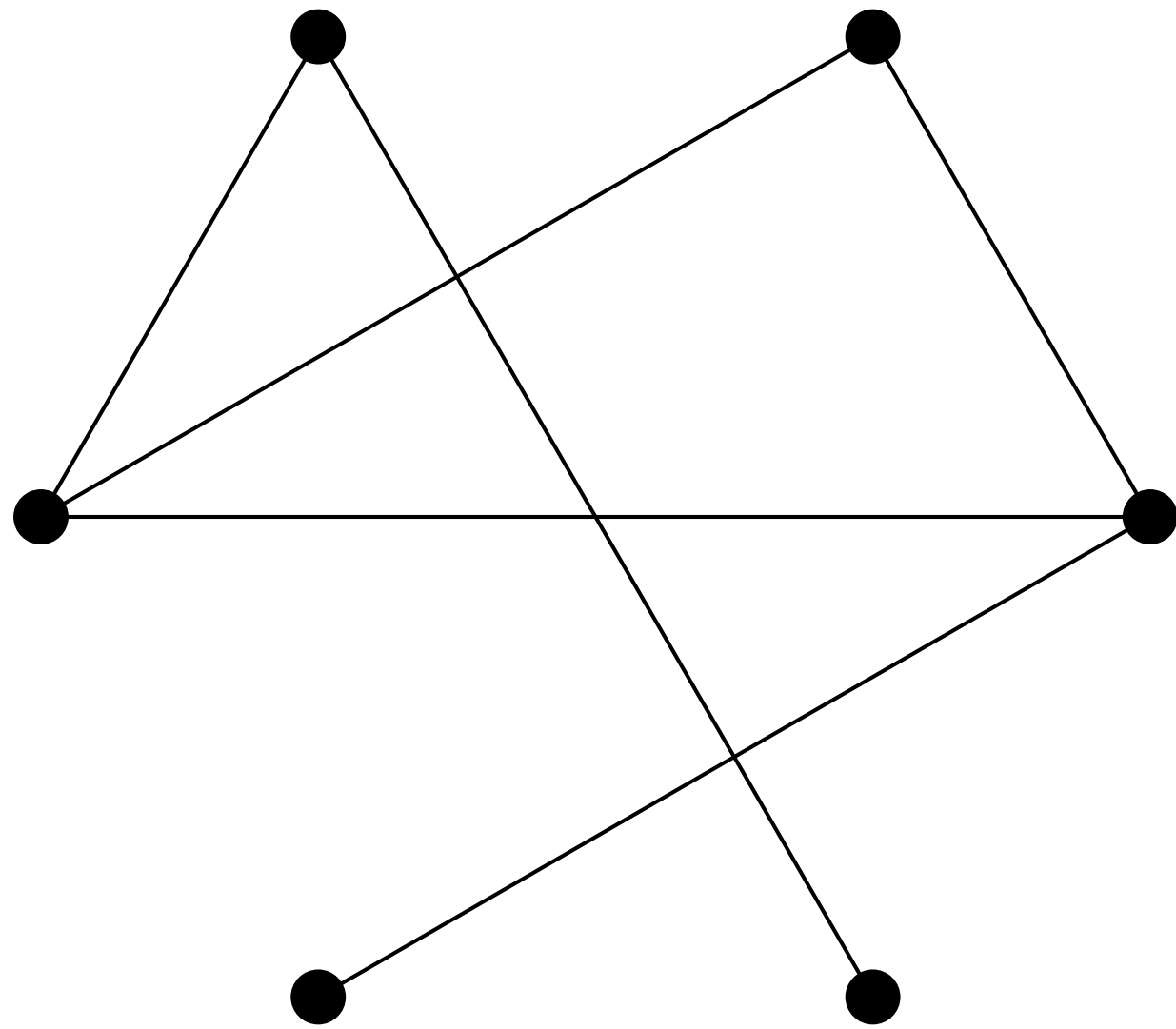


Phase Transition

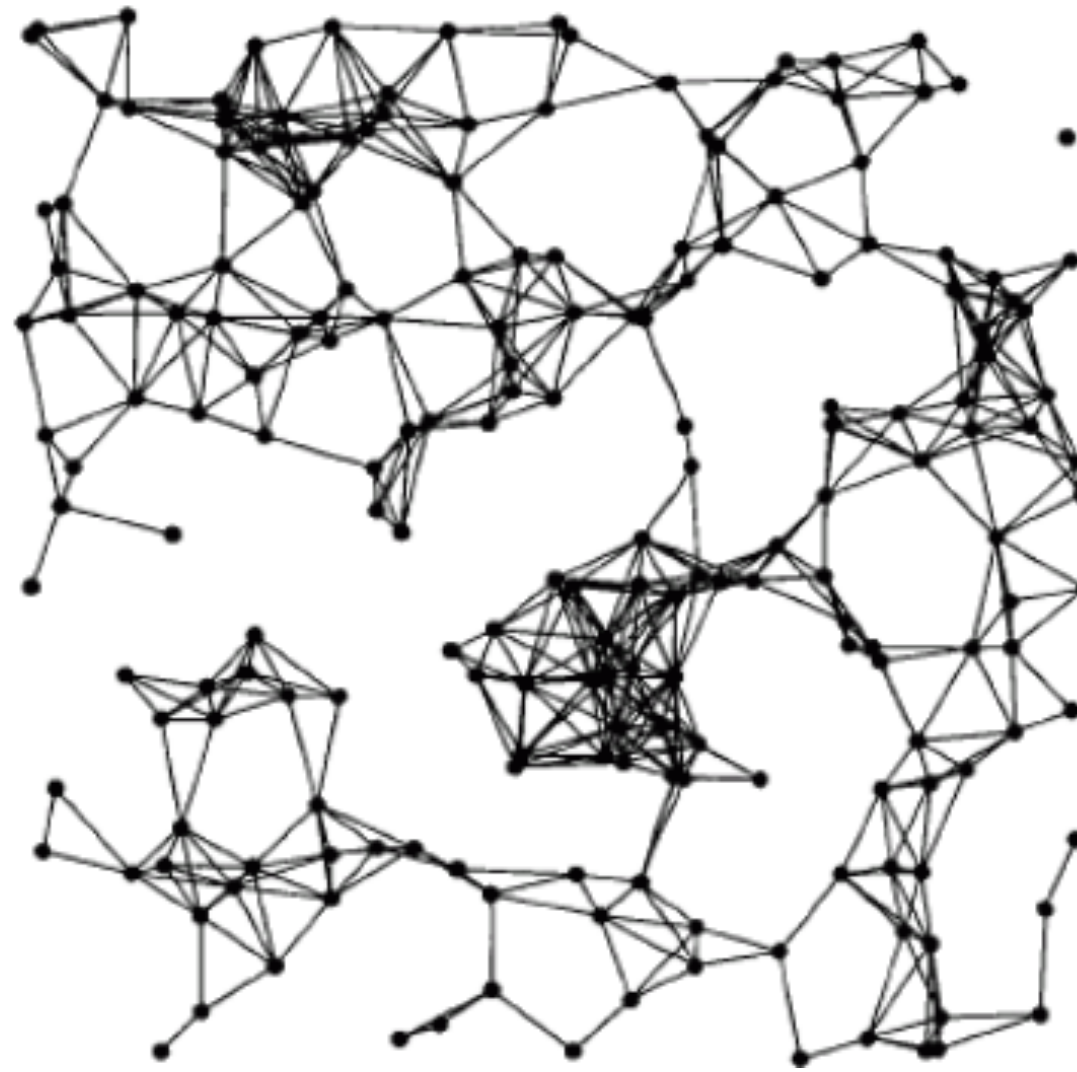
[Duncan-Kahle-Schweinhardt, 2021]



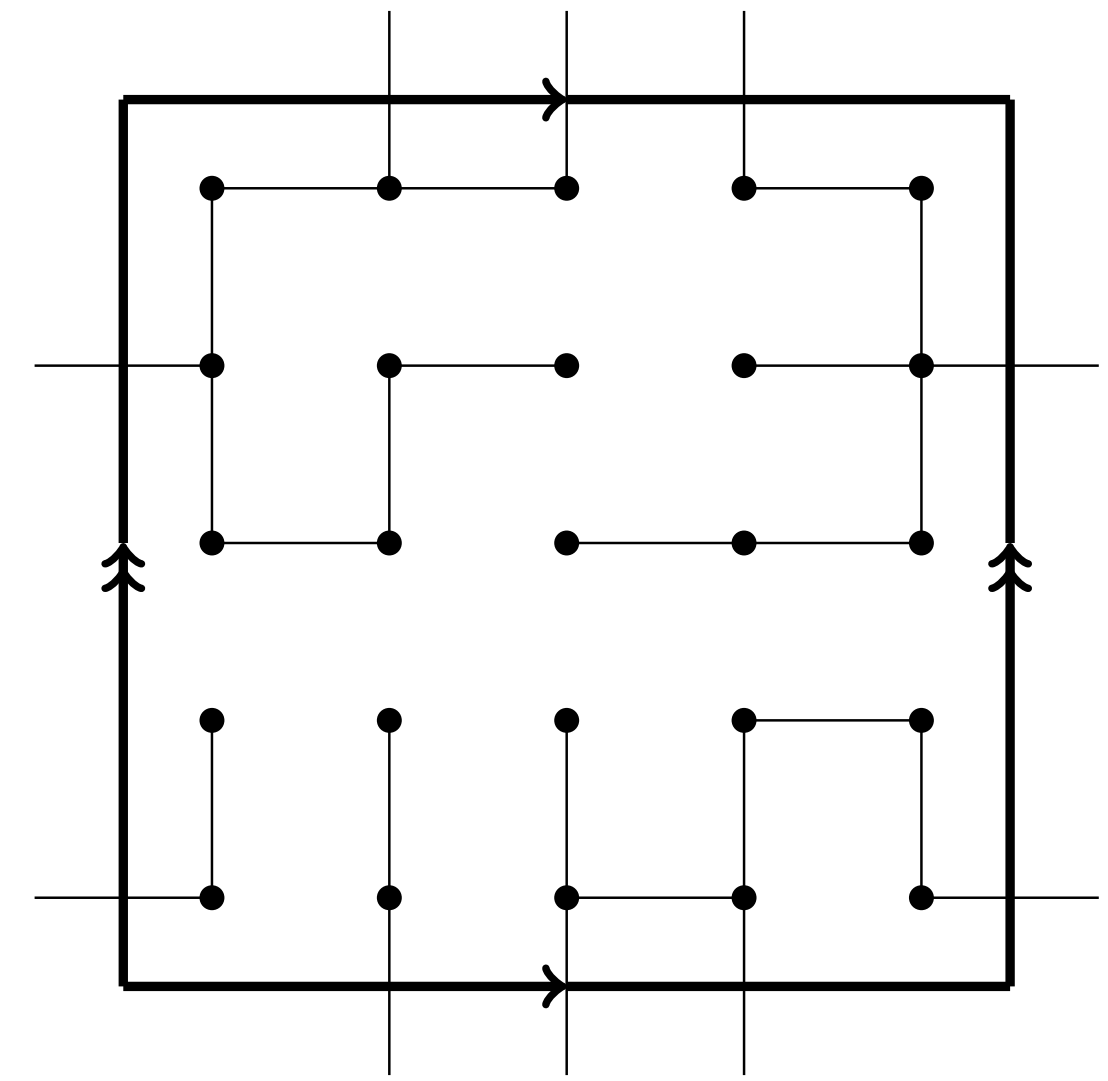
Tapas at Random Topology



Erdo-Renyi Complexes

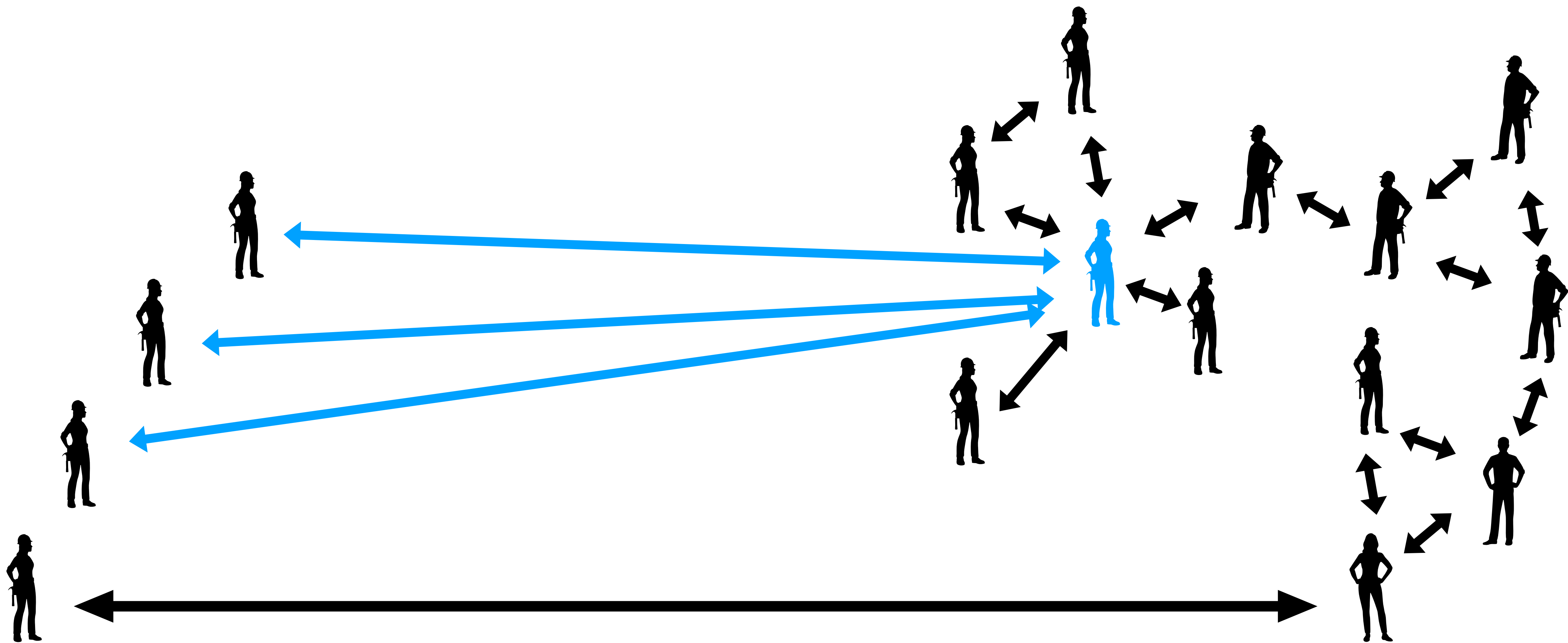


Geometric Complexes



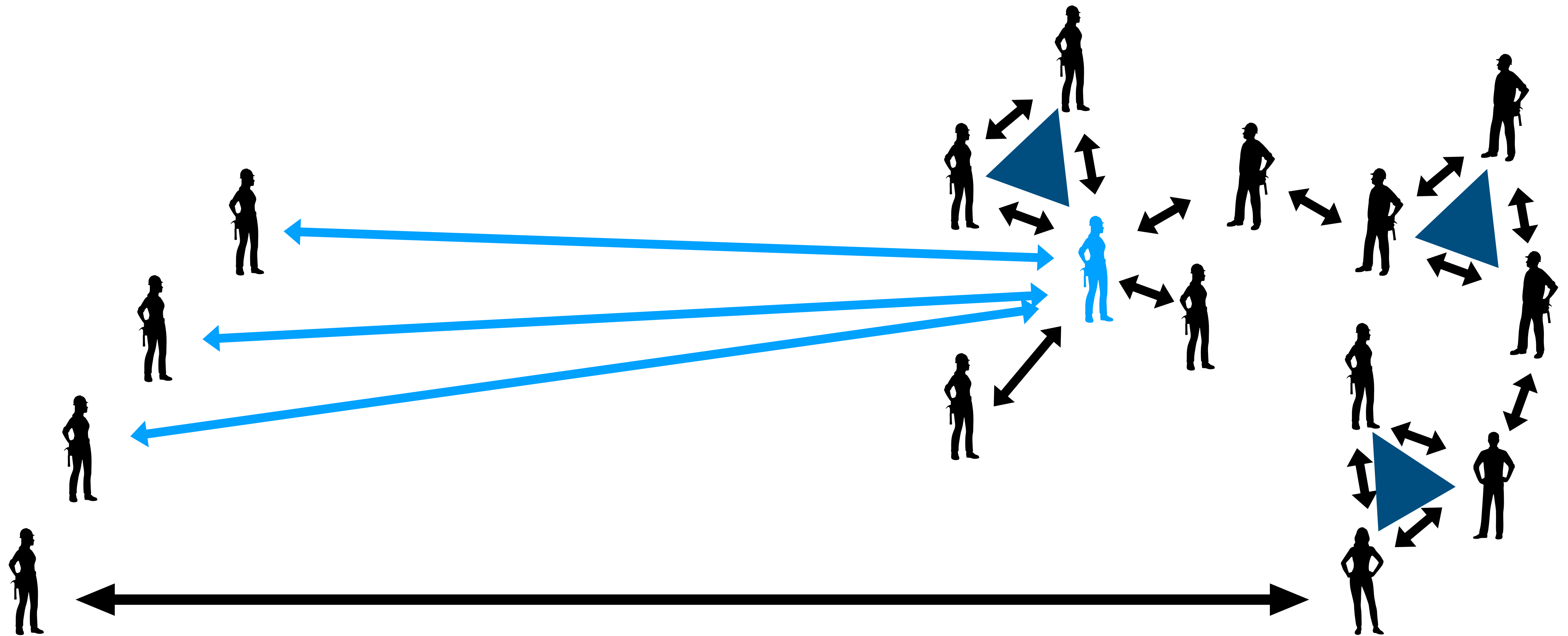
Topological Percolation

**Betti numbers count
repeated connections “in all dimensions”.**



Clique Complex

aka Flag Complex



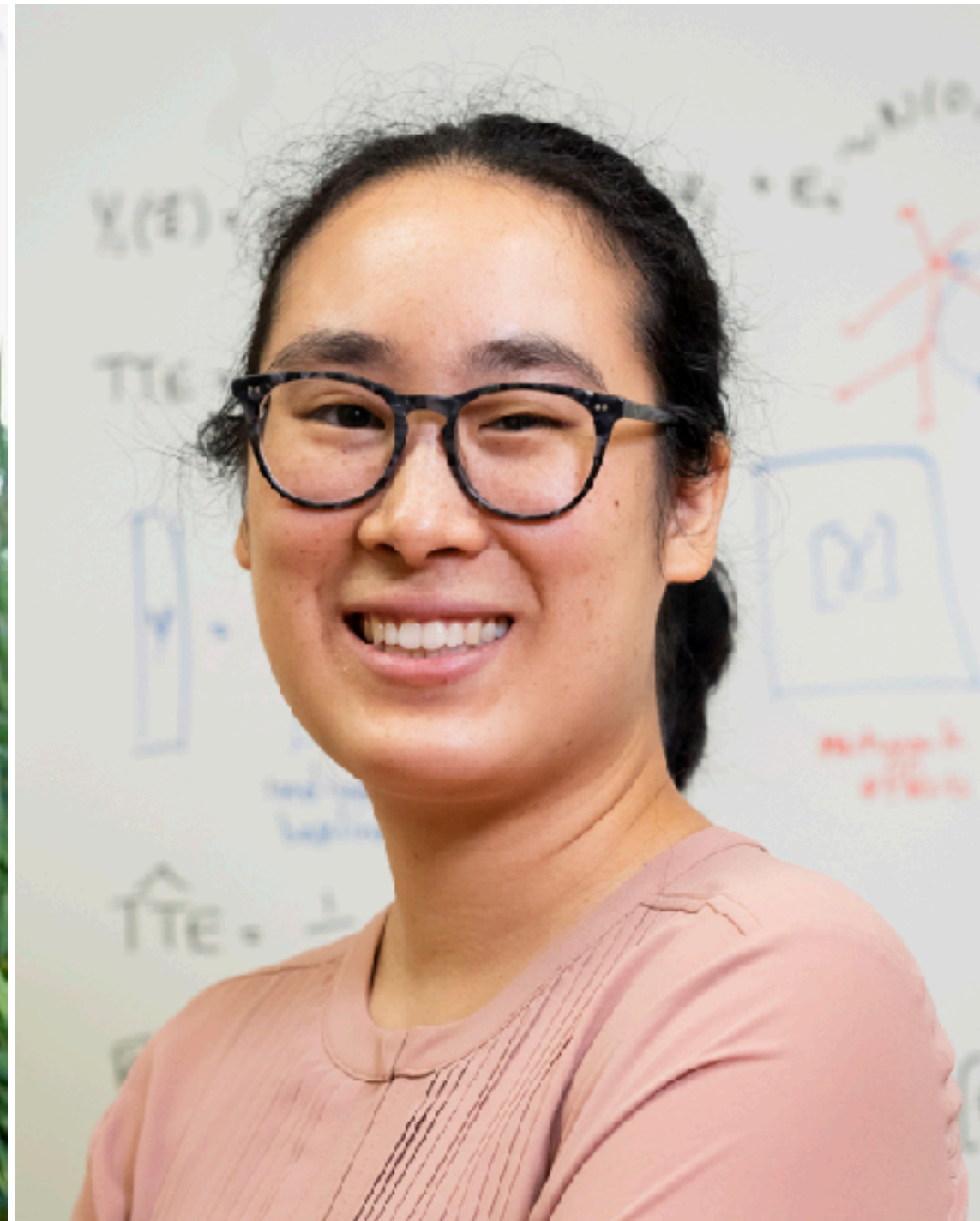
III Topology of Preferential Attachment

.....

My Lovely Collaborators



Avhan Misra



Christina Lee Yu



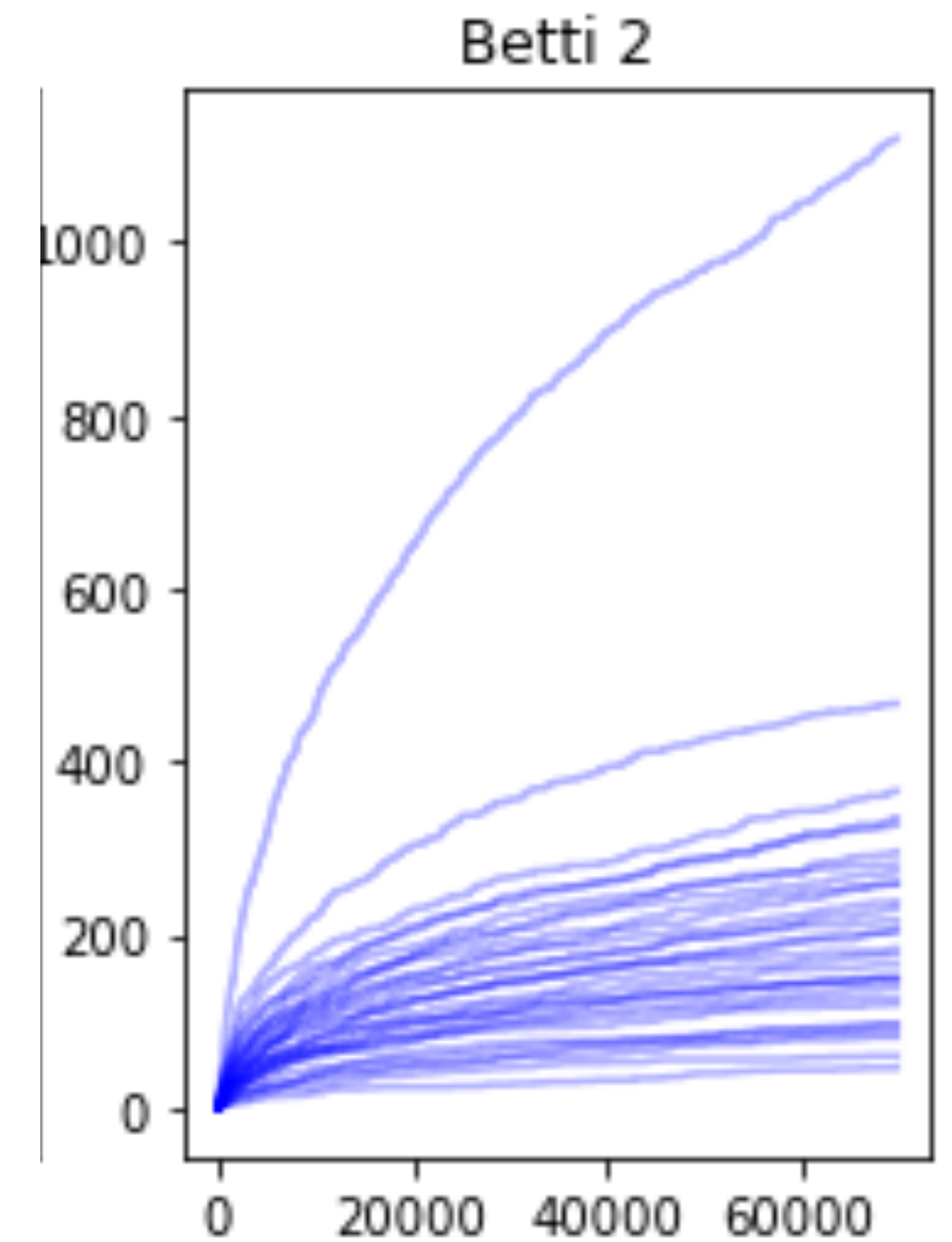
Gennady Samorodnitsky



Rongyi He (Caroline)

Betti Number β_q

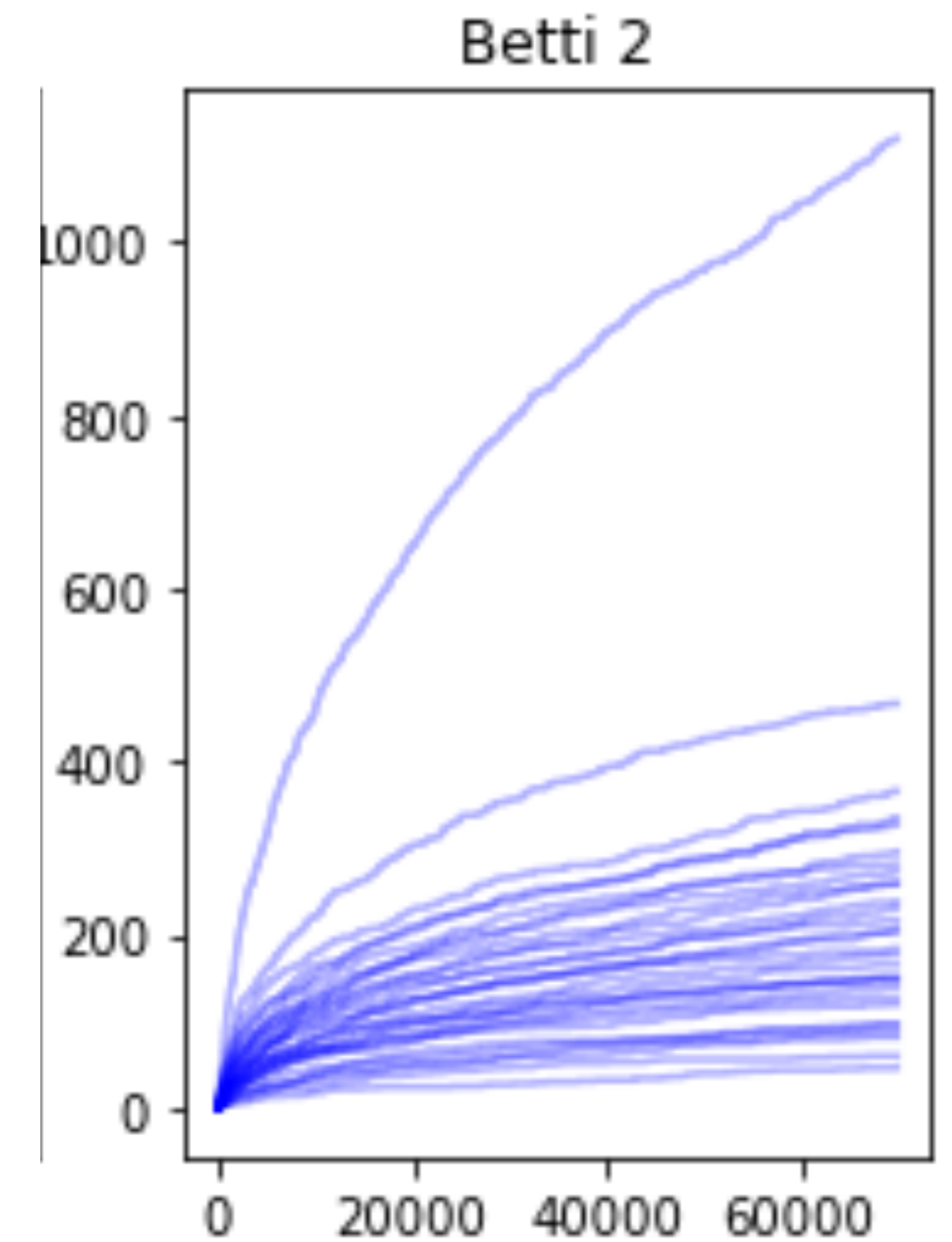
Betti Number β_q



Different curves, different random seeds.
All curves have the same model parameters.

Betti Number β_q

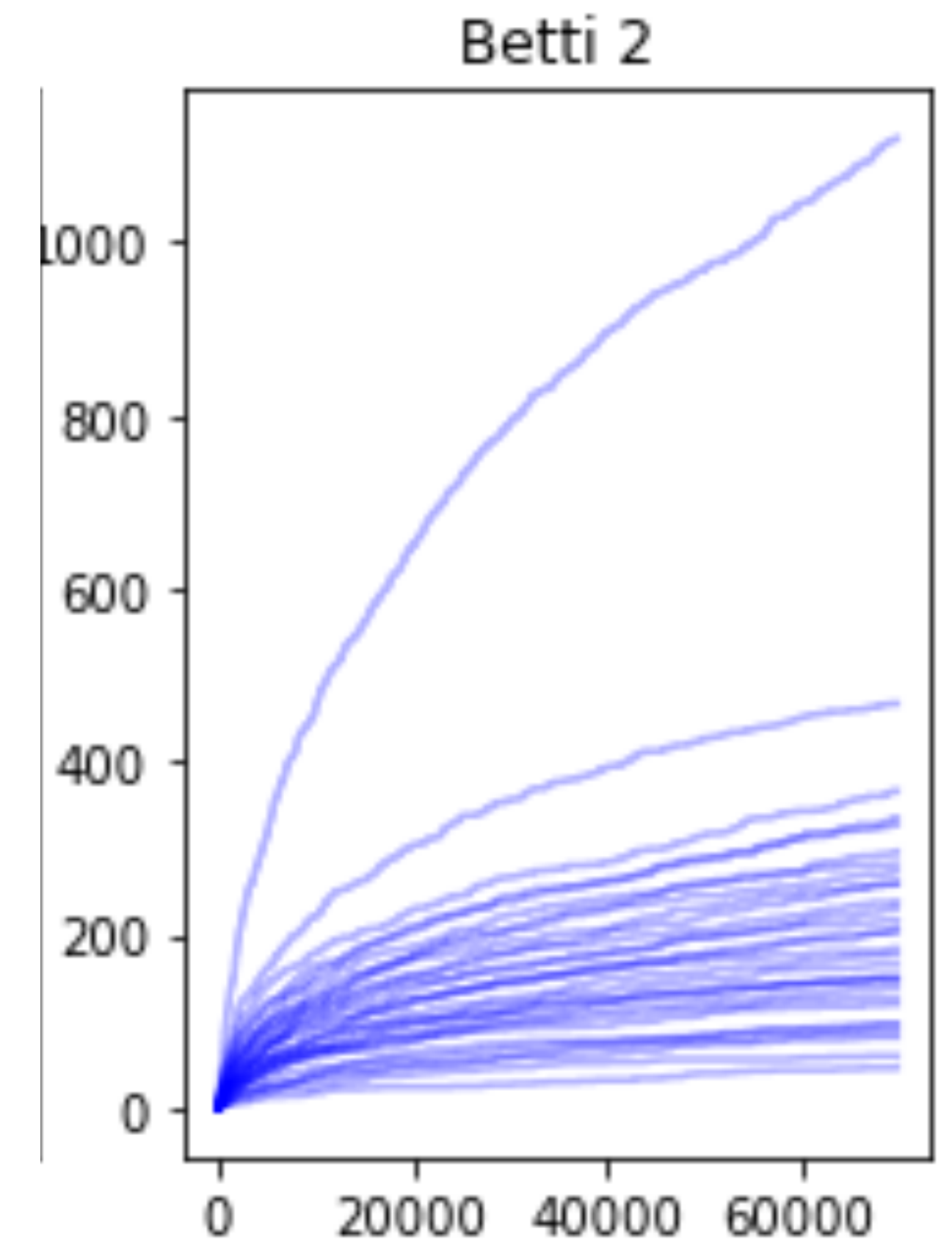
- increasing trend



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Betti Number β_q

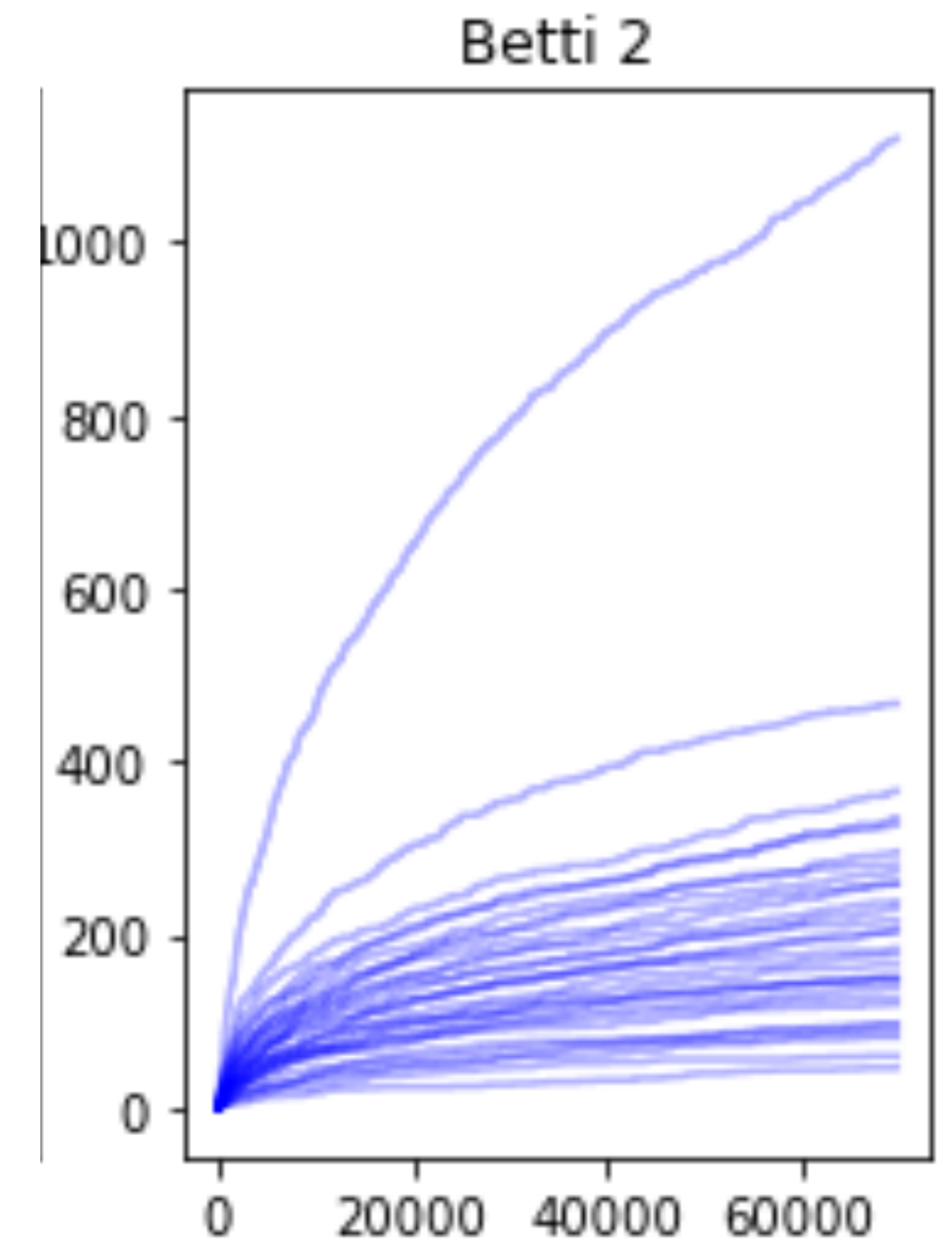
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Betti Number β_q

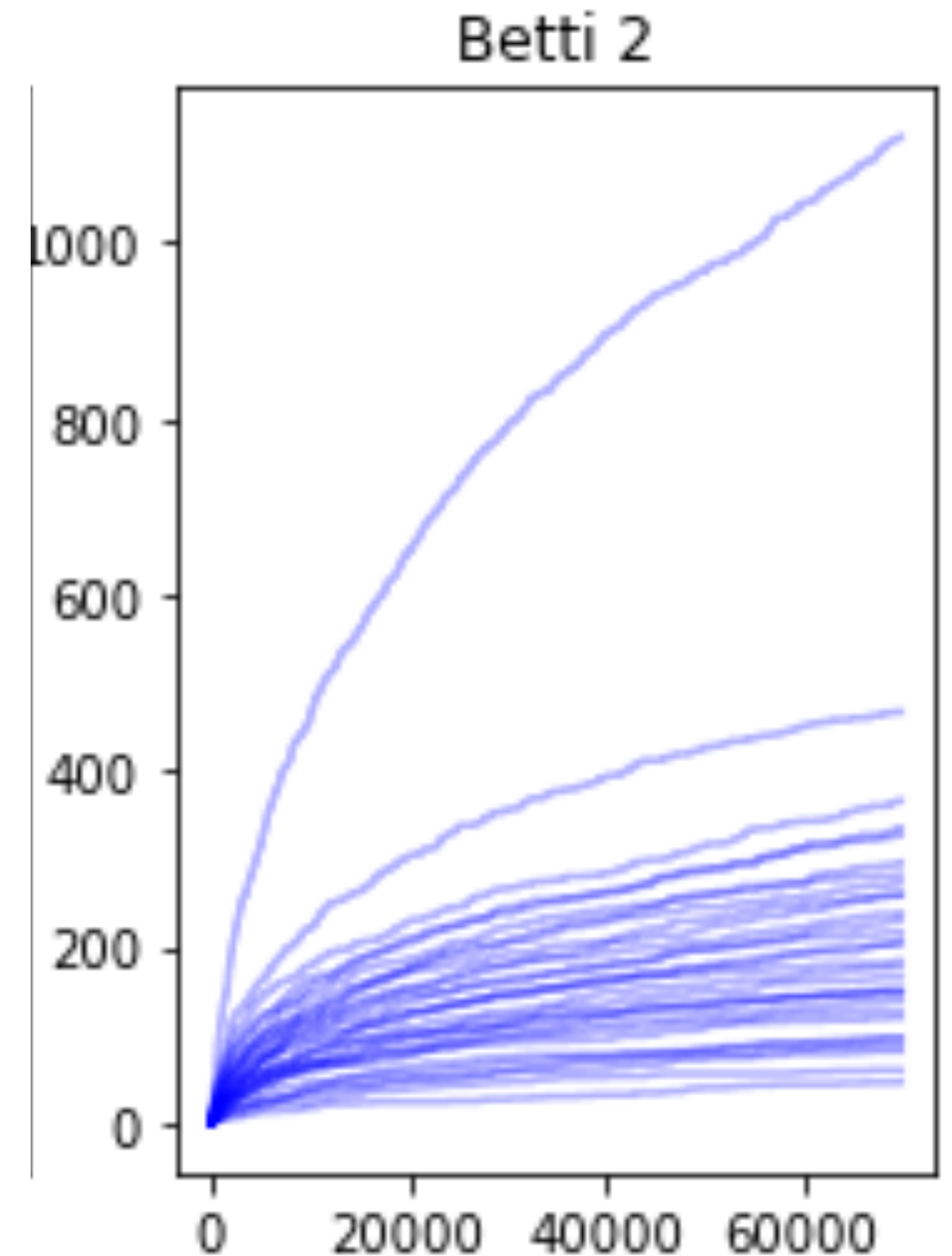
- increasing trend
- concave growth
- outlier



Different curves, different random seeds.
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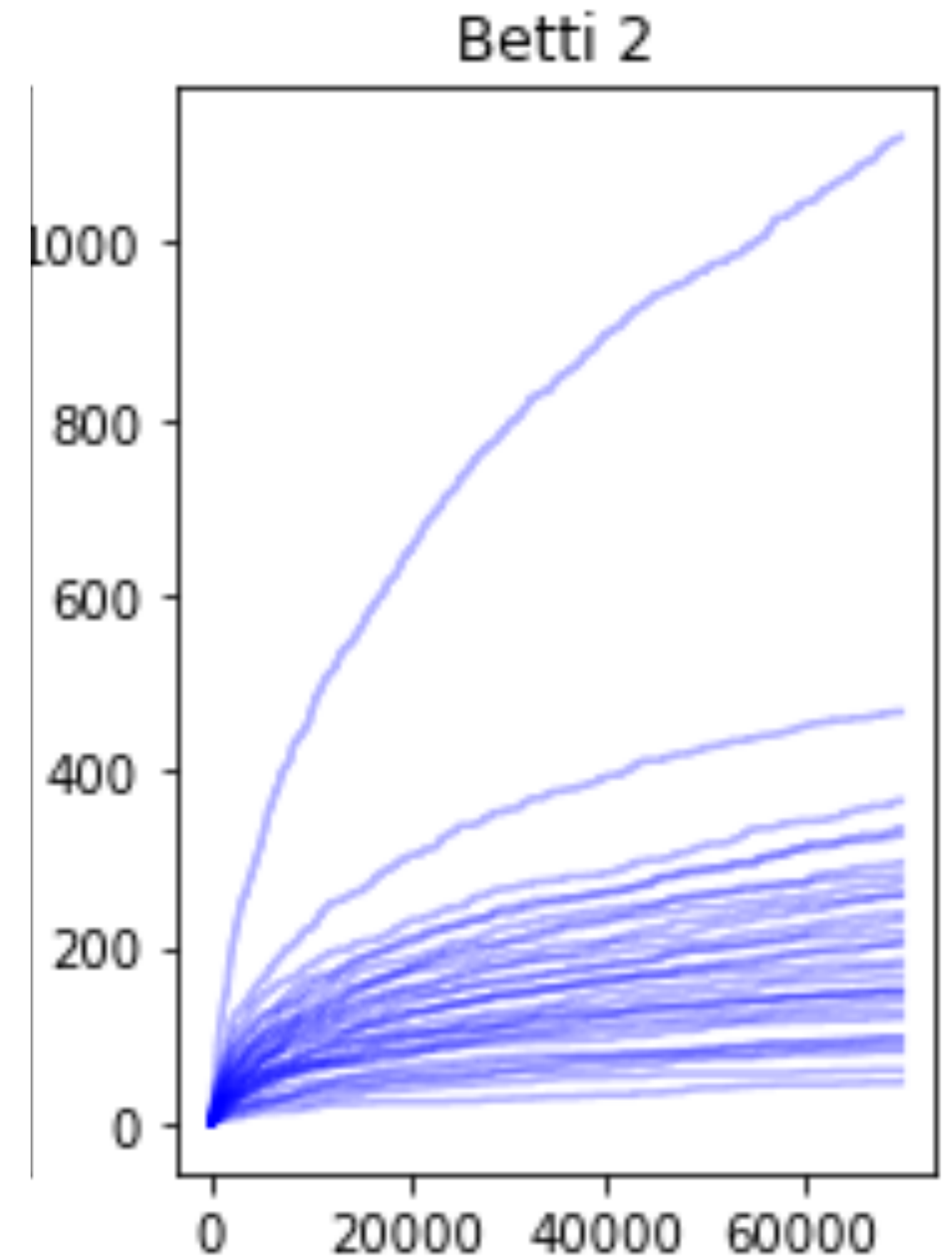
Betti Number β_q

- With probability at least $1 - \varepsilon$,
- $c_\varepsilon(\text{num of nodes}^{1-4x}) \leq \beta_2 \leq C_\varepsilon(\text{num of nodes}^{1-4x})$
- $x \in (0, 1/2)$ decreases with the preferential attachment strength
 - $P[T \text{ attaches to } i] \propto T^{-x}$



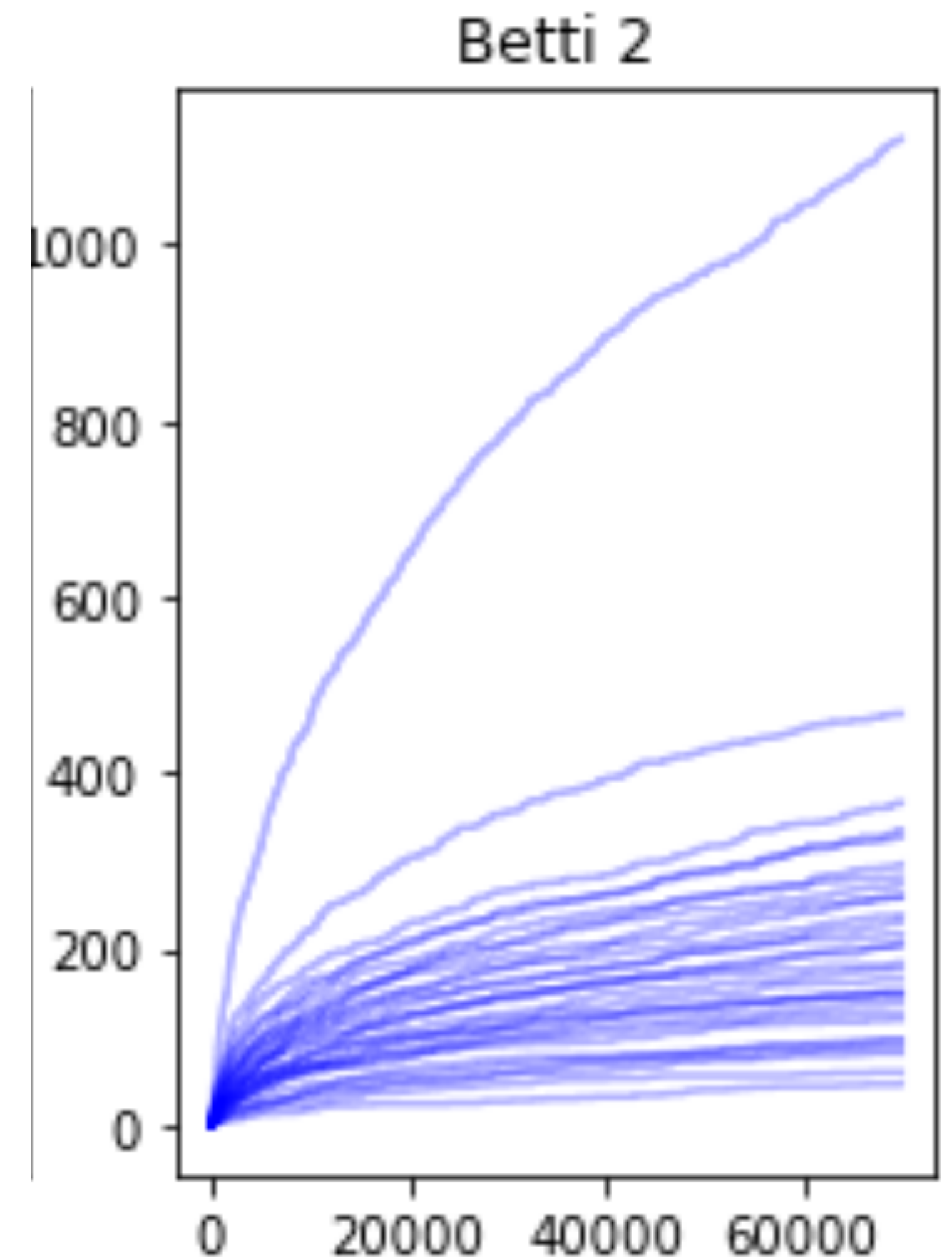
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Betti Number β_q

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- $c_\varepsilon(\text{num of nodes}^{1-\boxed{2q}^x}) \leq \beta_q \leq C_\varepsilon(\text{num of nodes}^{1-\boxed{2q}^x})$ for $q \geq 2$.

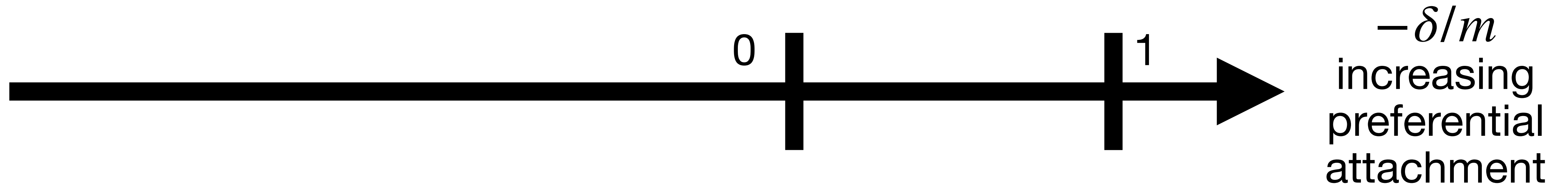


Phase transition

Recall

$P(\text{attaching to } v) \propto \text{degree} + \delta$

m = number of edges per new node



The limiting degree distribution has ...

finite variance

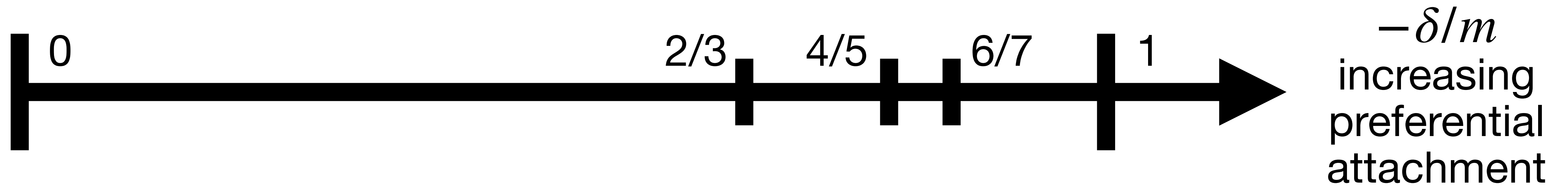
infinite variance

Phase transition

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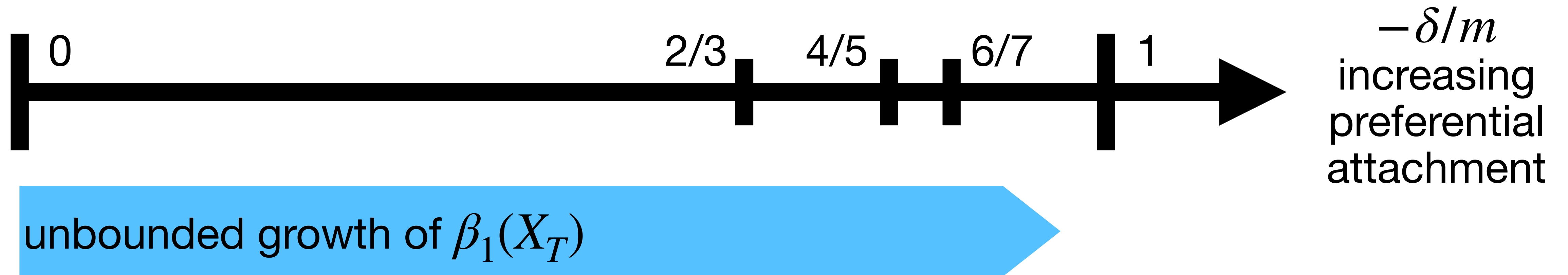


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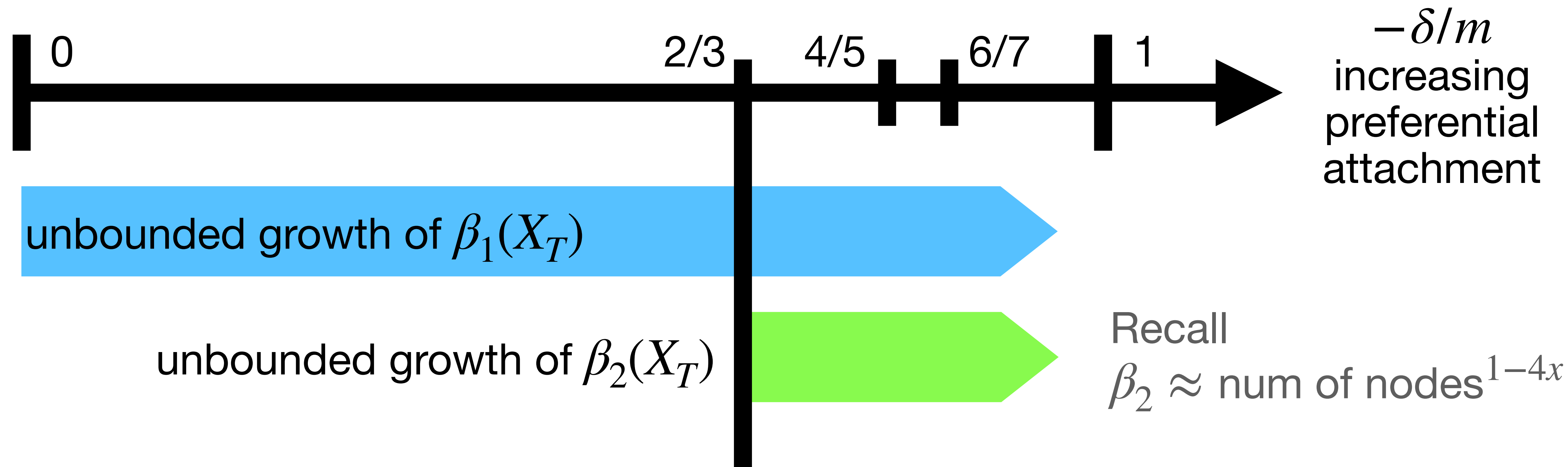
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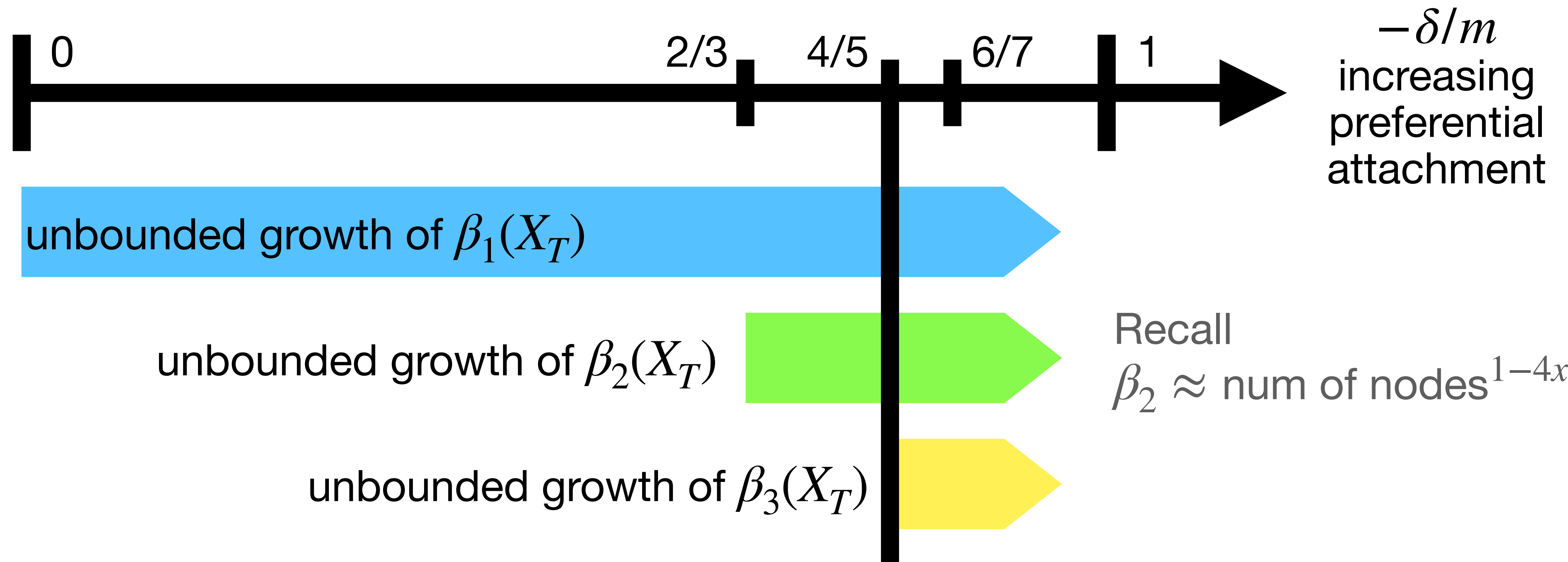
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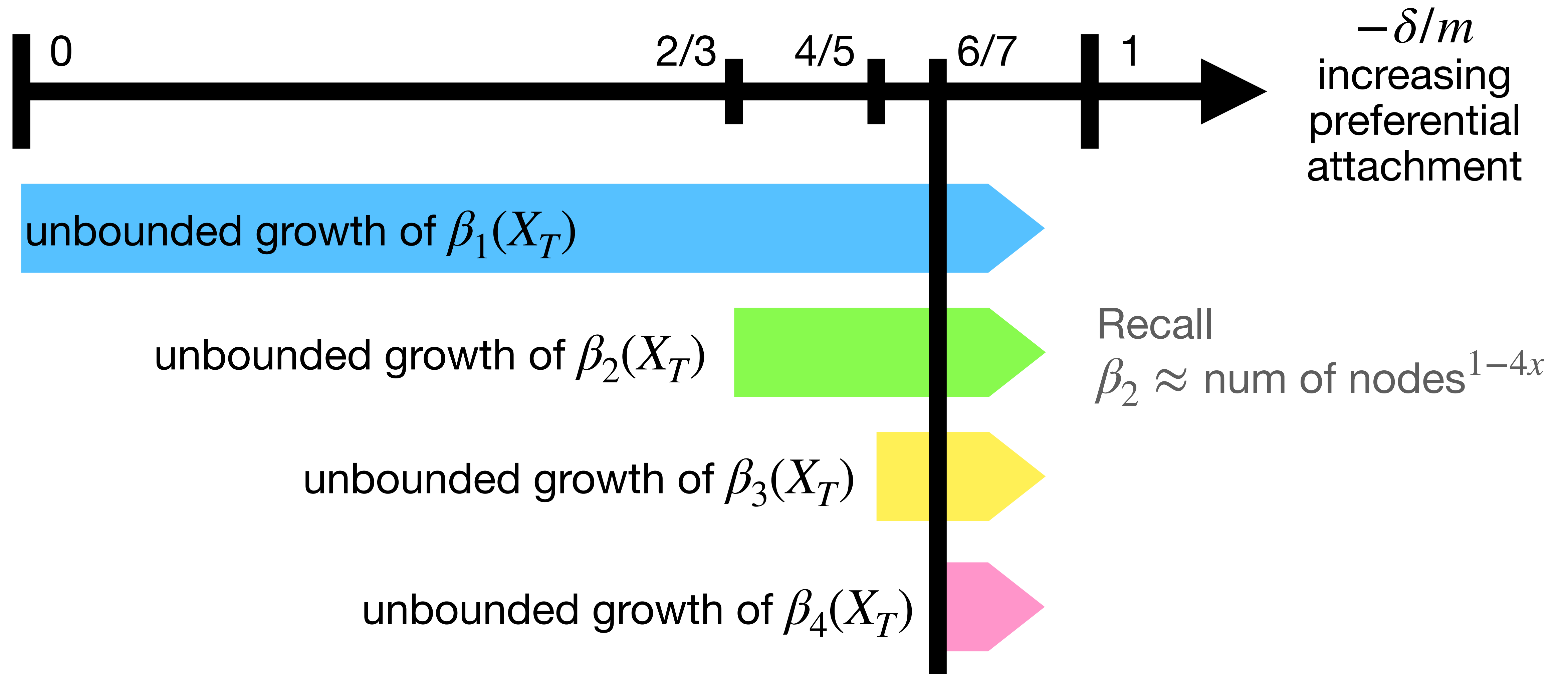
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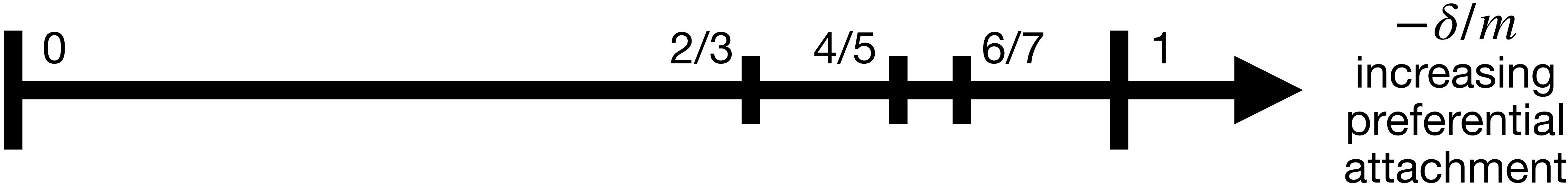
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Phase transition

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unbounded growth of $\beta_1(X_T)$

unbounded growth of $\beta_2(X_T)$

unbounded growth of $\beta_3(X_T)$

unbounded growth of $\beta_4(X_T)$

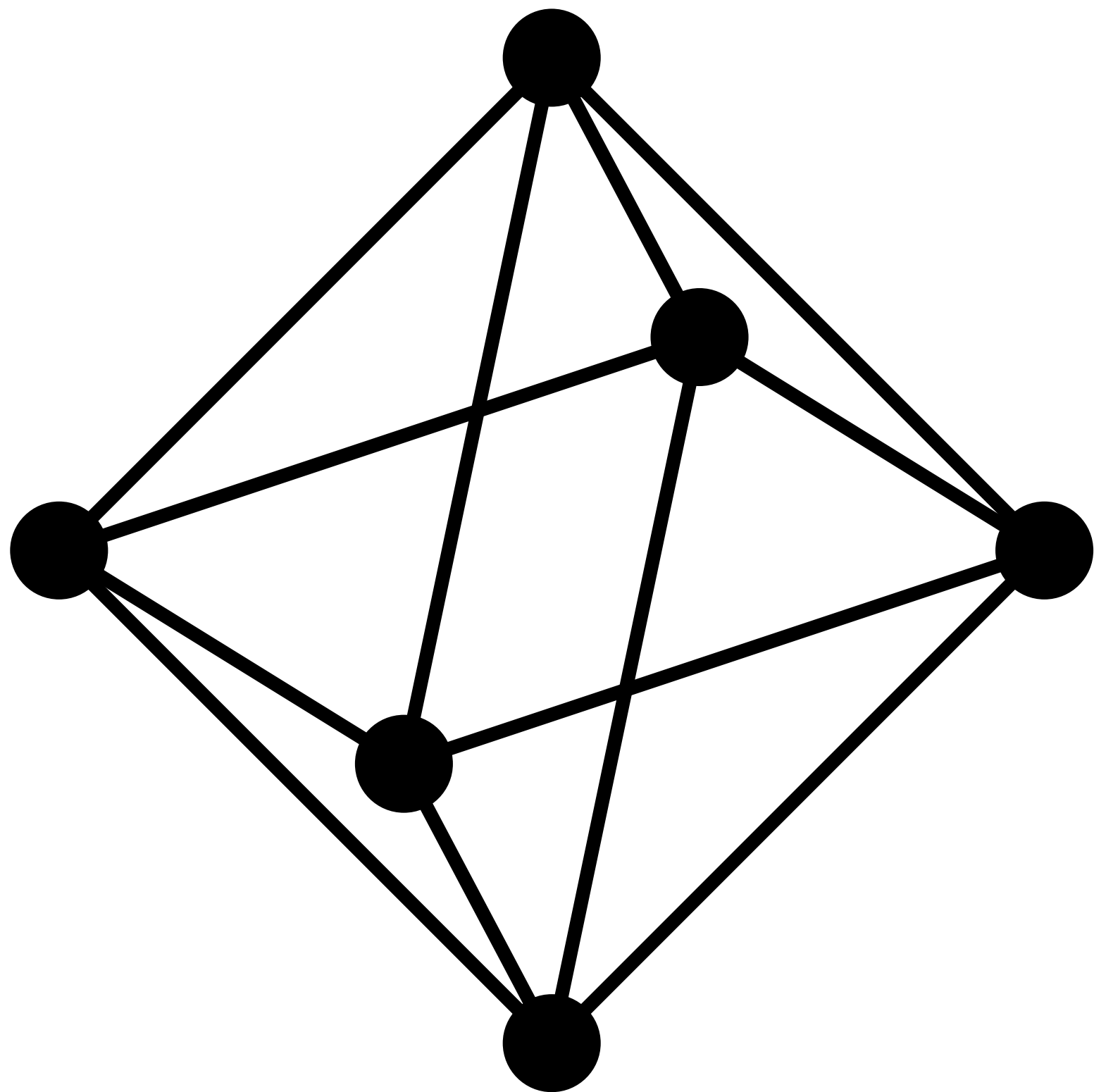
$\vdots$

Recall

$\beta_2 \approx \text{num of nodes}^{1-4x}$

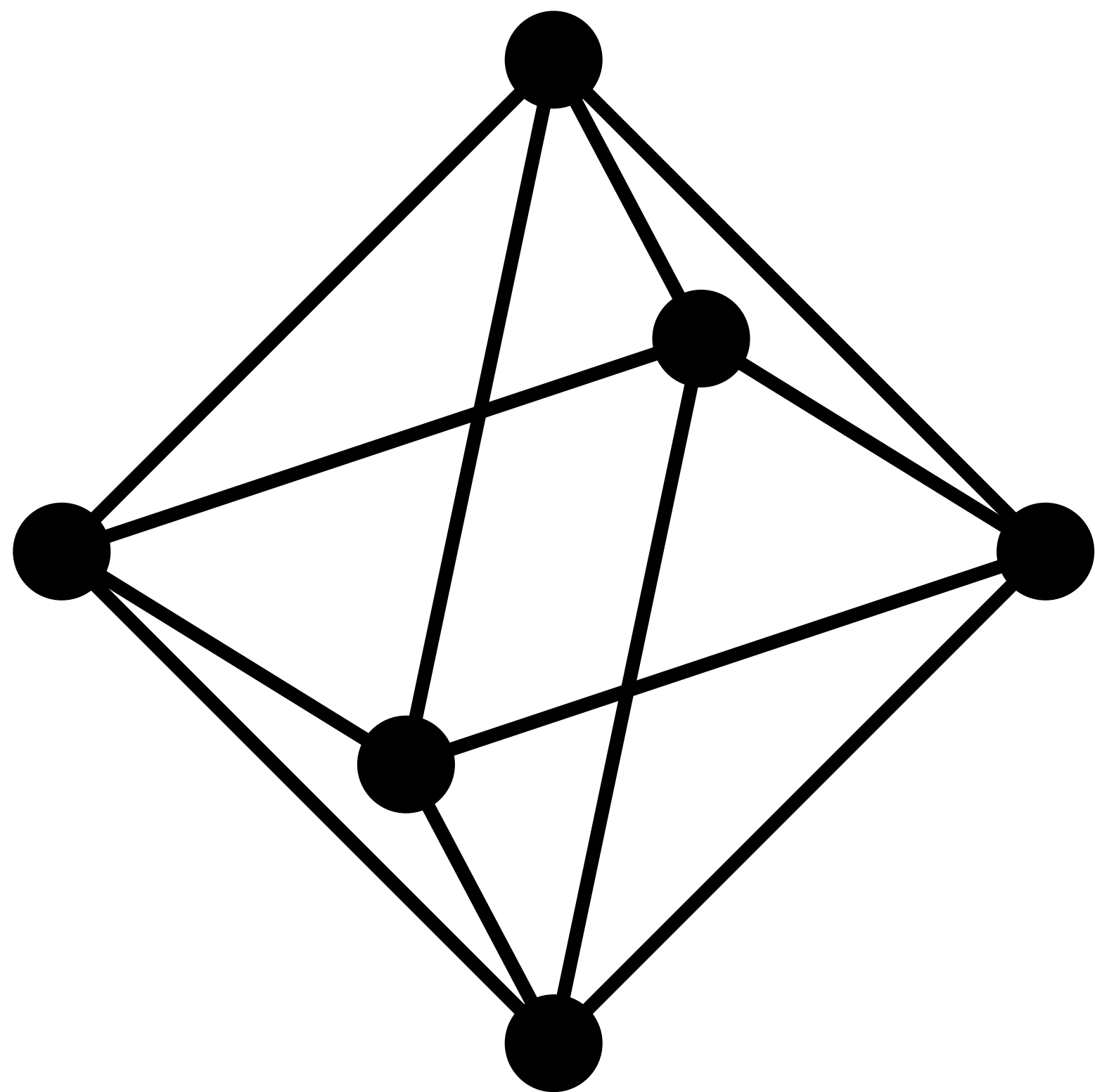
Theorem: $\beta_2 \approx \text{num of nodes}^{1-4x}$
Proof?

Proof of $\beta_2 \approx \text{num of nodes}^{1-4x}$

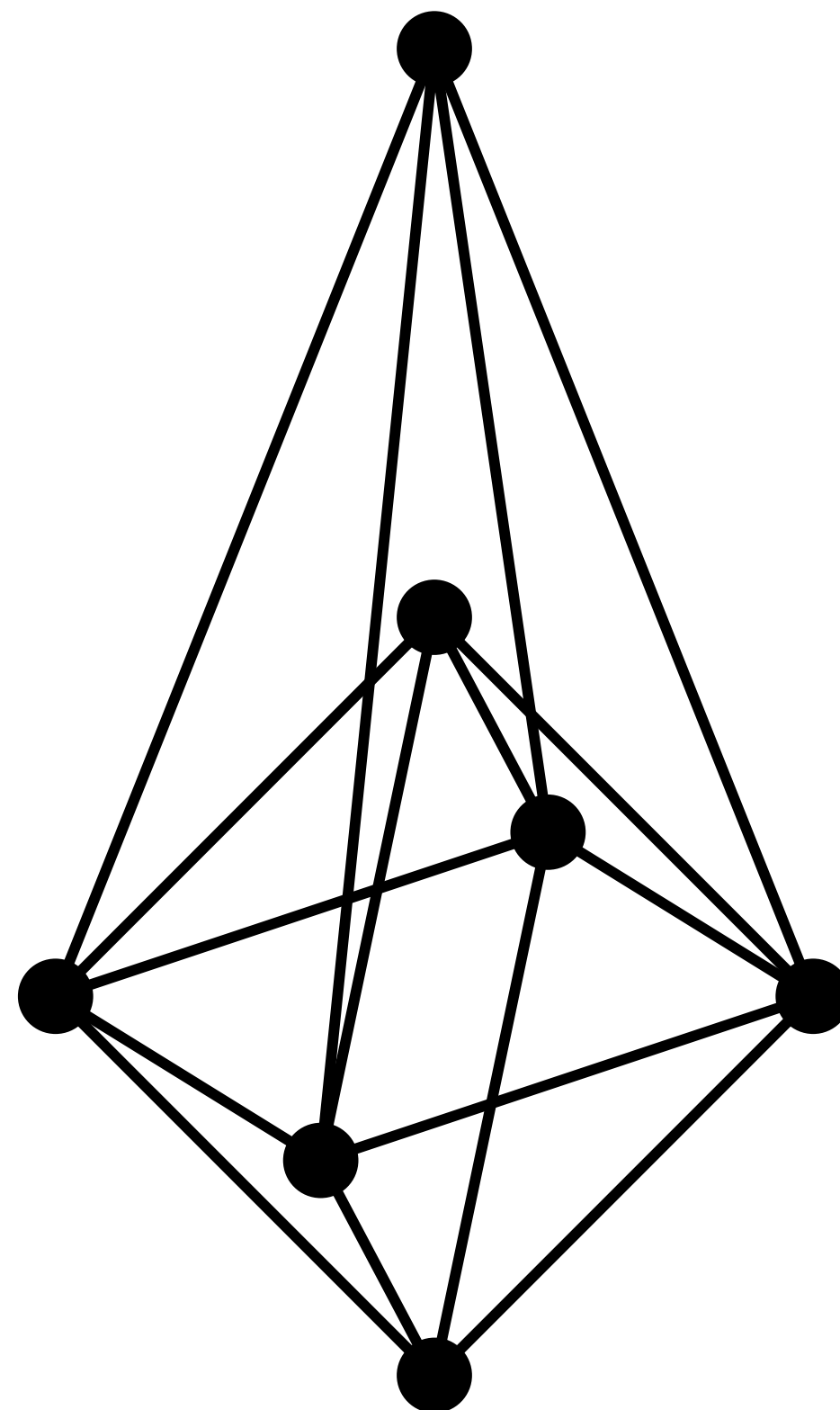


$$\beta_2 = 1$$

Proof of $\beta_2 \approx \text{num of nodes}^{1-4x}$

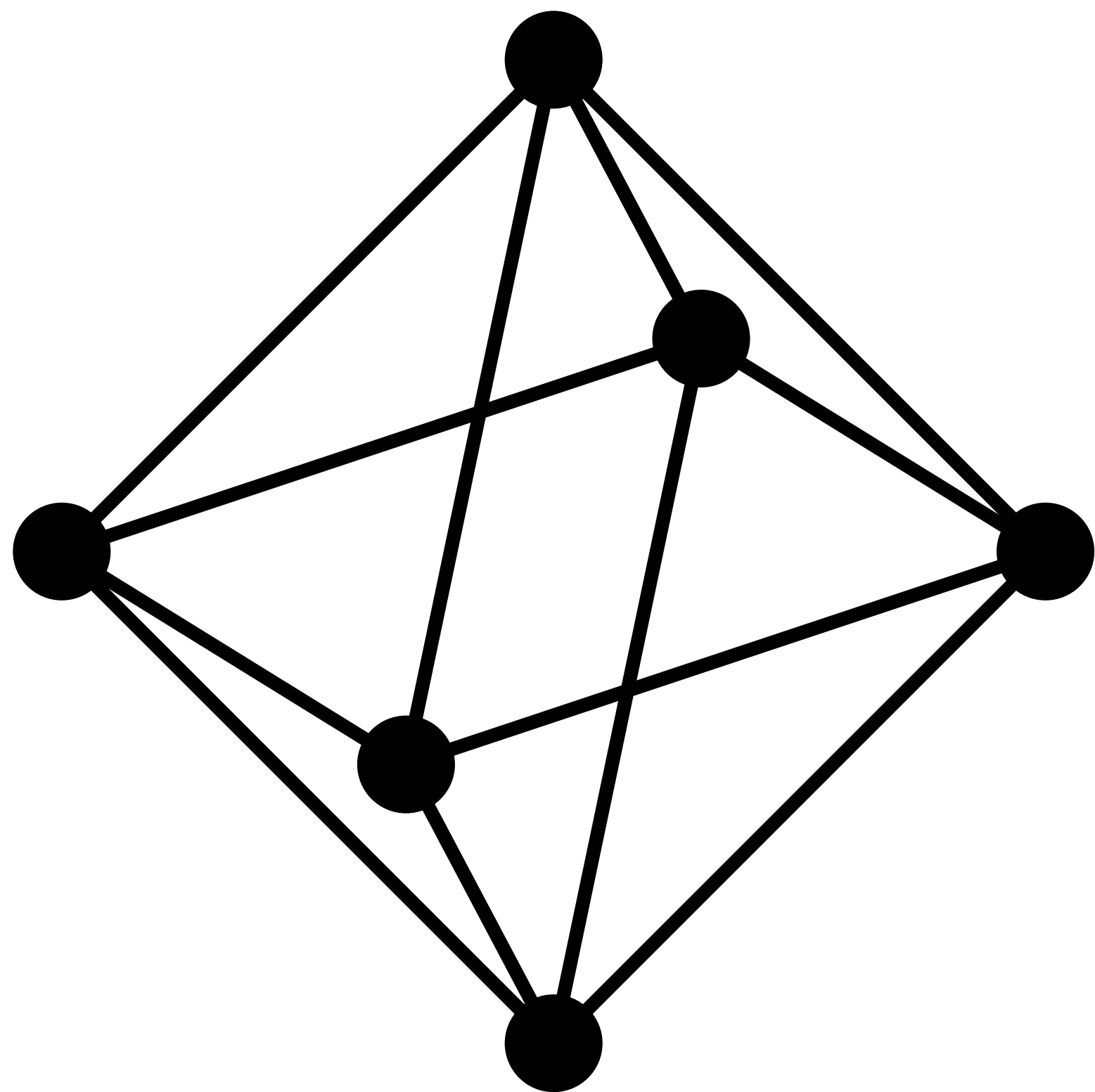


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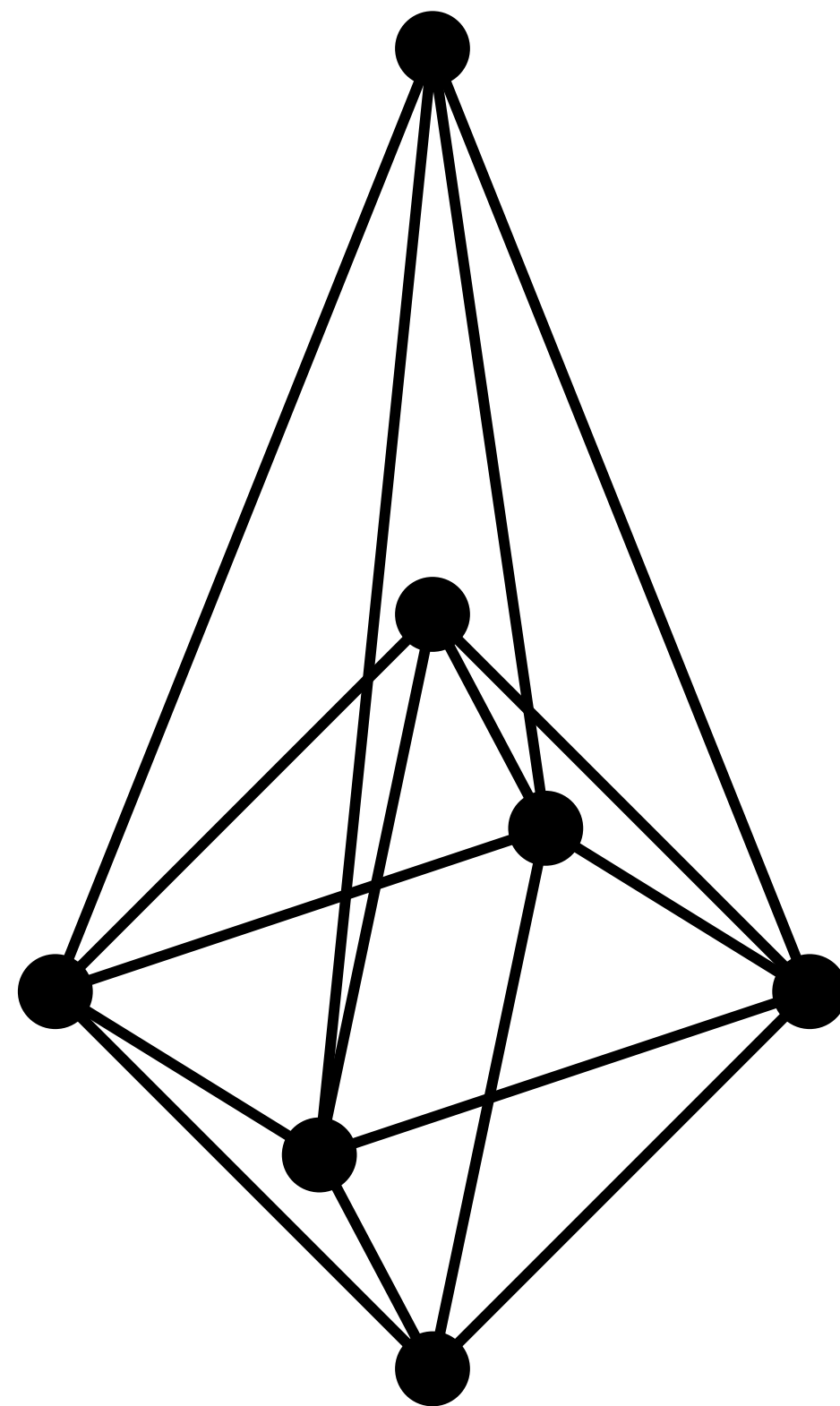


$$\beta_2 = 2$$

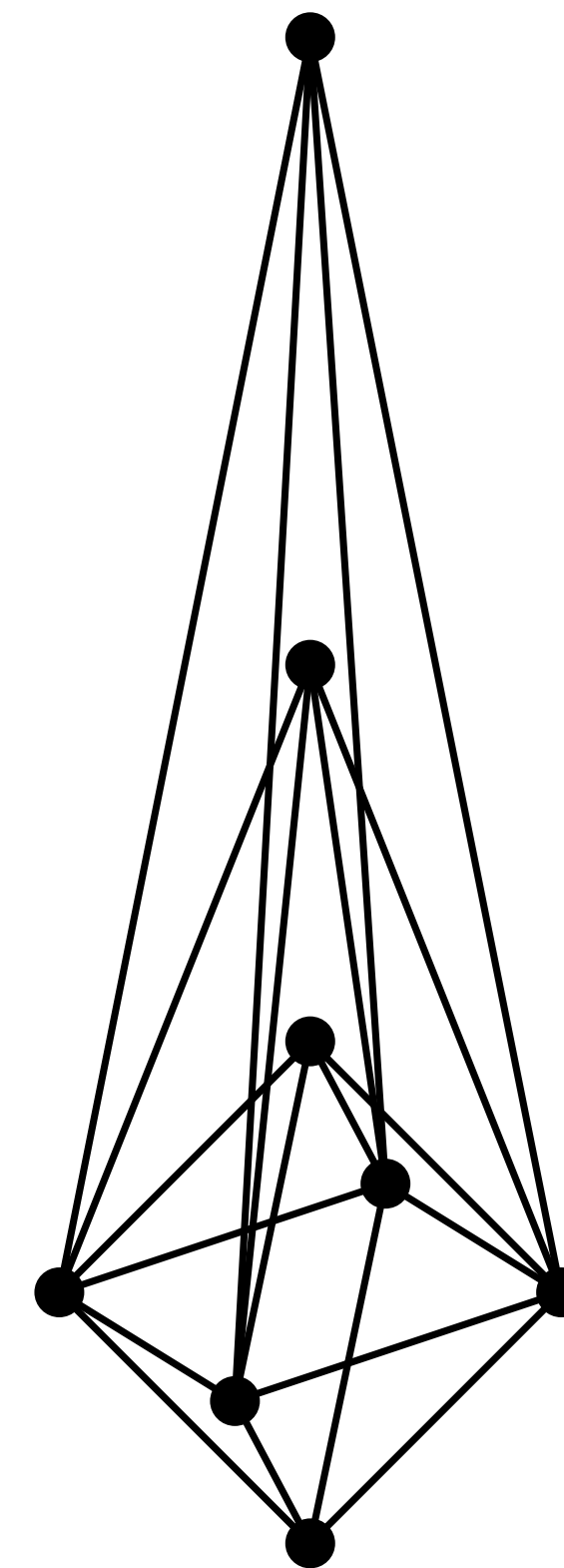
Proof of $\beta_2 \approx \text{num of nodes}^{1-4x}$



$$\beta_2 = 1$$



$$\beta_2 = 2$$



$$\beta_2 = 3$$

Subtleties

- Need homological algebra to relate Betti numbers with counts

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Subtleties

- Need homological algebra to relate Betti numbers with counts
- Identify the “square count” as the main term with minimal cycle results in [Gal 2005] and [Kahle 2009]
- Generalize minimal cycle results in the language of homological algebra
- Apply graph counting result in [Garavaglia and Stegehuis 2019] on a large class of subgraphs

Theorem: $\beta_2 \approx \text{num of nodes}^{1-4x}$

IV. What lies ahead

.....

order of magnitude of
Betti numbers

more precise estimates?

other inhomogeneous
complexes?

order of magnitude of
Betti numbers

more precise estimates?

other inhomogeneous
complexes?

order of magnitude of
Betti numbers

breakthroughs for
graphs????

The threshold for percolation in $\mathbb{Z}^3$?

Some wishful thinking...

Gwynne, 2023. private conversation

The threshold for percolation in $\mathbb{Z}^3$?

Gwynne, 2024. private conversation

Some wishful thinking...

- Maybe $\mathbb{Z}^3$ and $(\mathbb{Z}/n)^3$ are similar enough...

The threshold for percolation in $\mathbb{Z}^3$?

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The threshold for percolation in $\mathbb{Z}^3$?

Gwynne, 2024. private conversation

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- Maybe the β_1 threshold is more computable...
- Indeed, for the **site** percolation model,

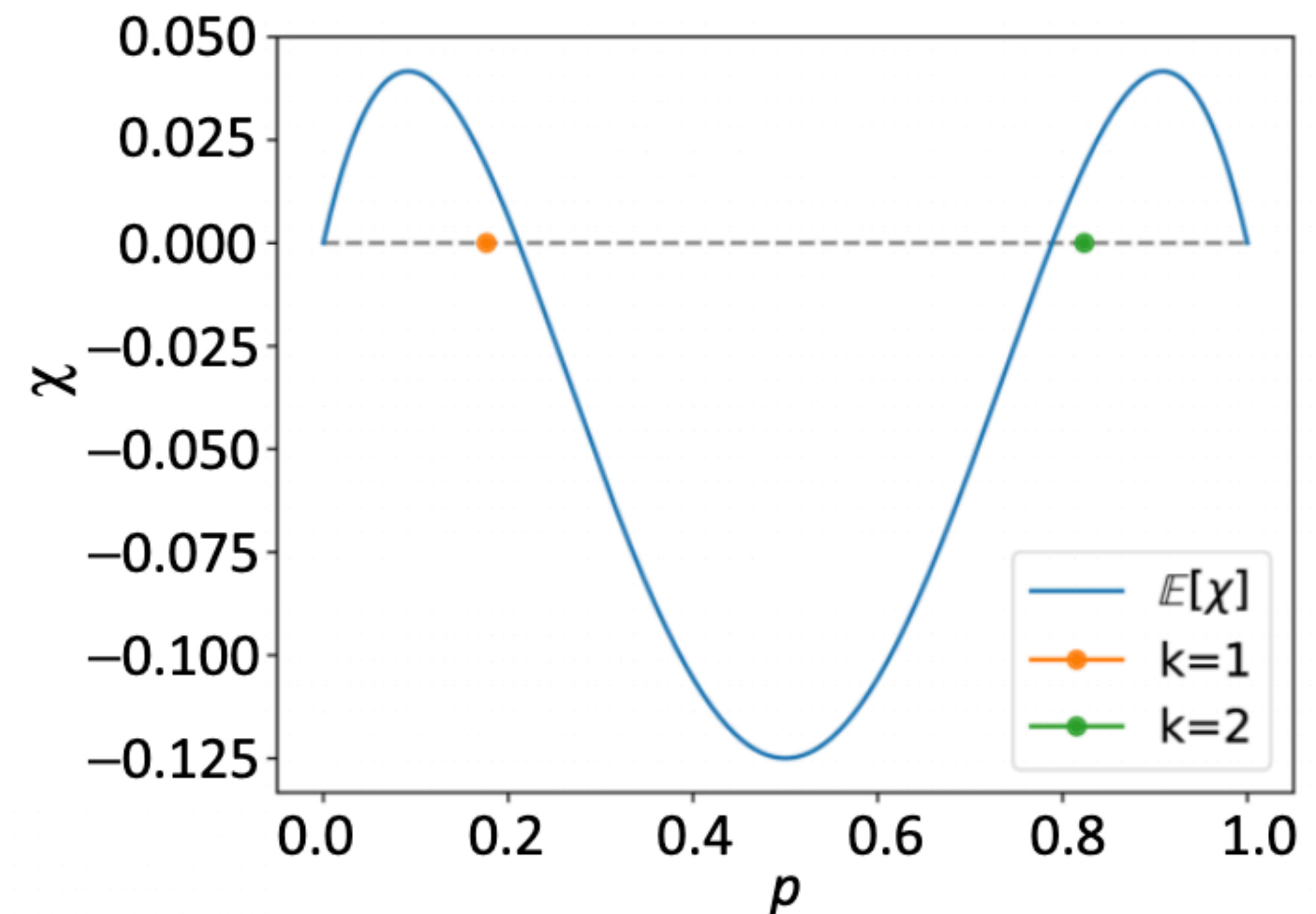


Fig 4 of [Bobrowski and Skraba, 2020]

What did we learn today?

- Random topology is cool.

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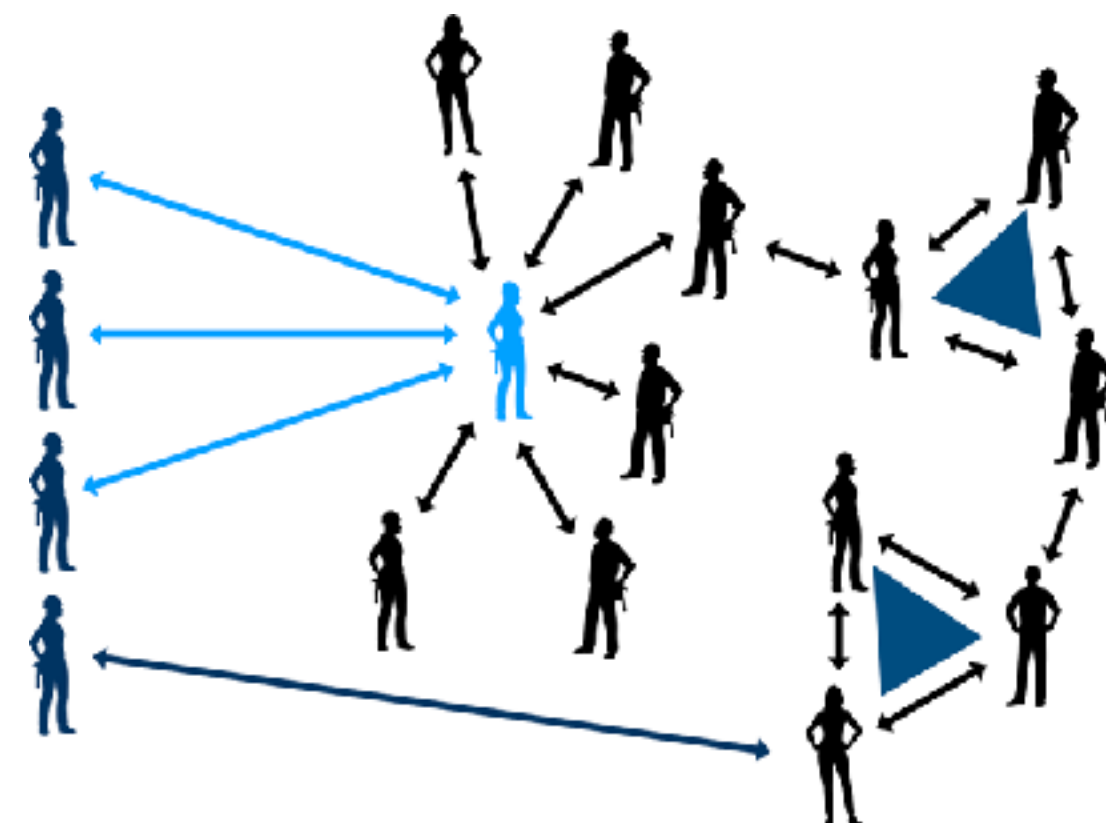
What did we learn today?

- Random topology is cool.
- Preferential attachment graph has interesting topology.
- More interesting things are waiting to be discovered.

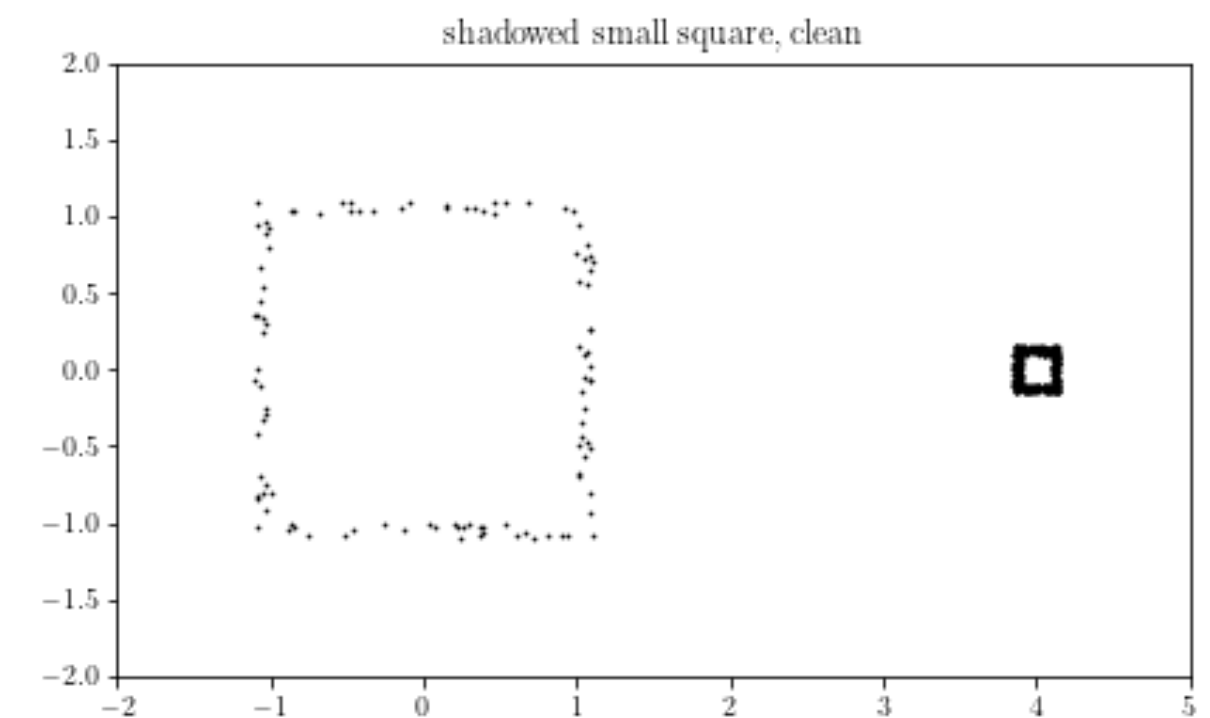
Chunyin Siu

siuc@stanford.edu

Stanford University, School of Medicine



arxiv paper



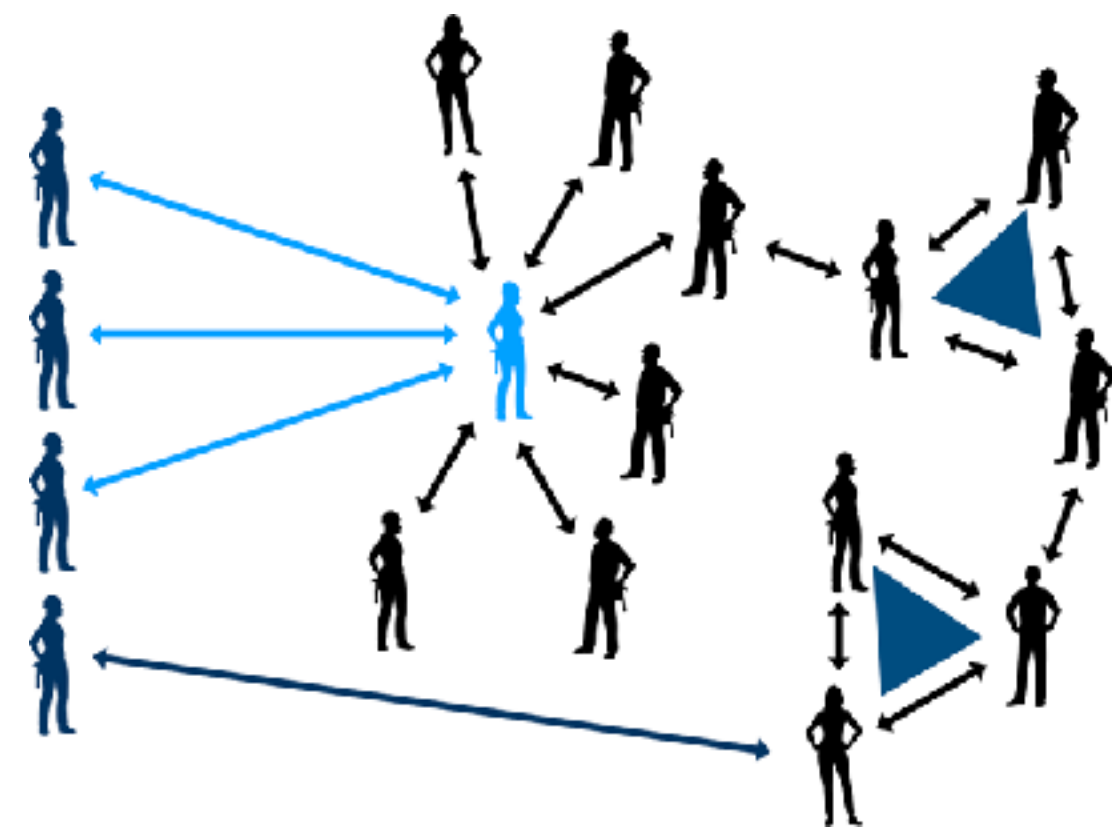
my video about small holes

Thank you!

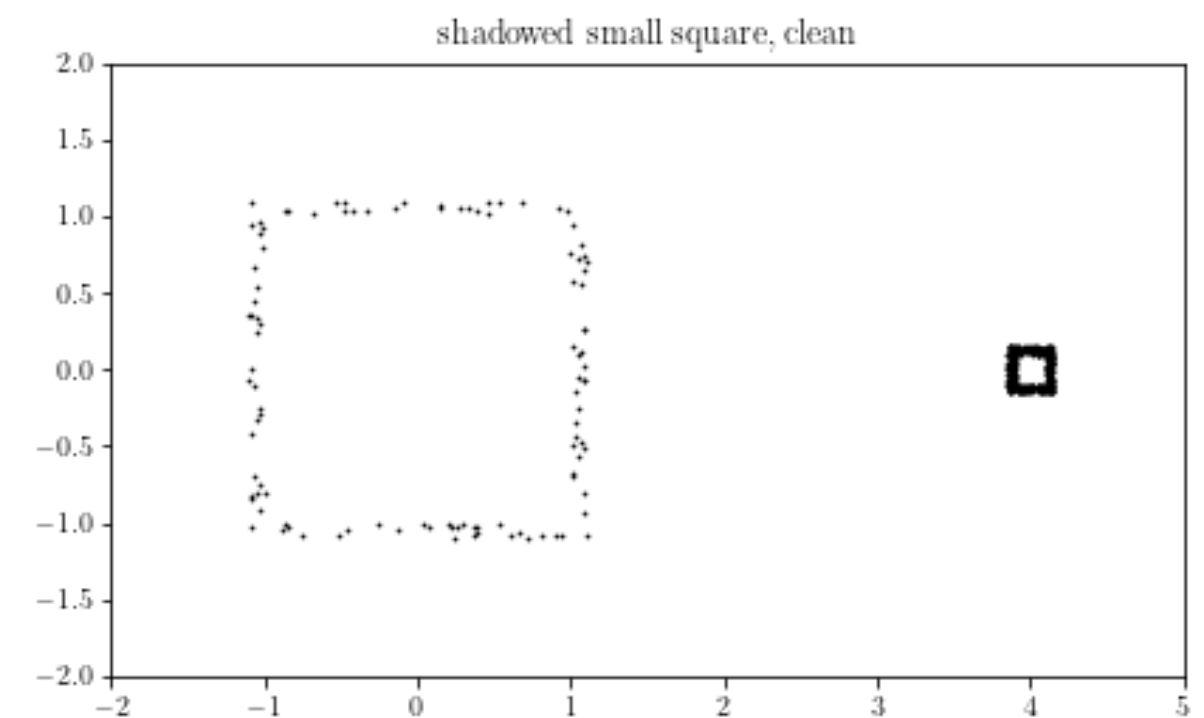
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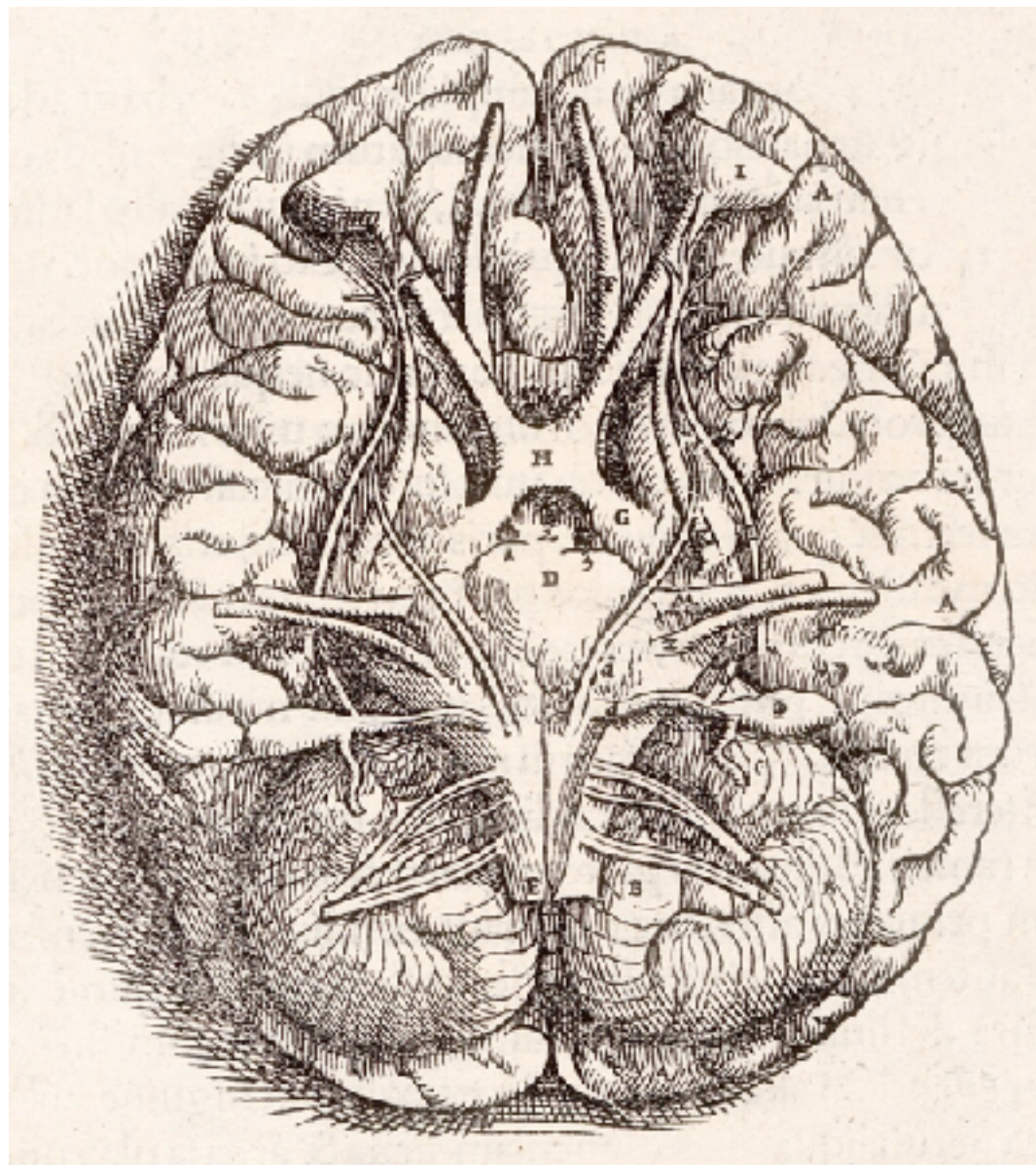
arxiv paper



my video about small holes

Applications of Topology

Topology and Networks



Neuroscience

from Andreas Vesalius' Fabrica

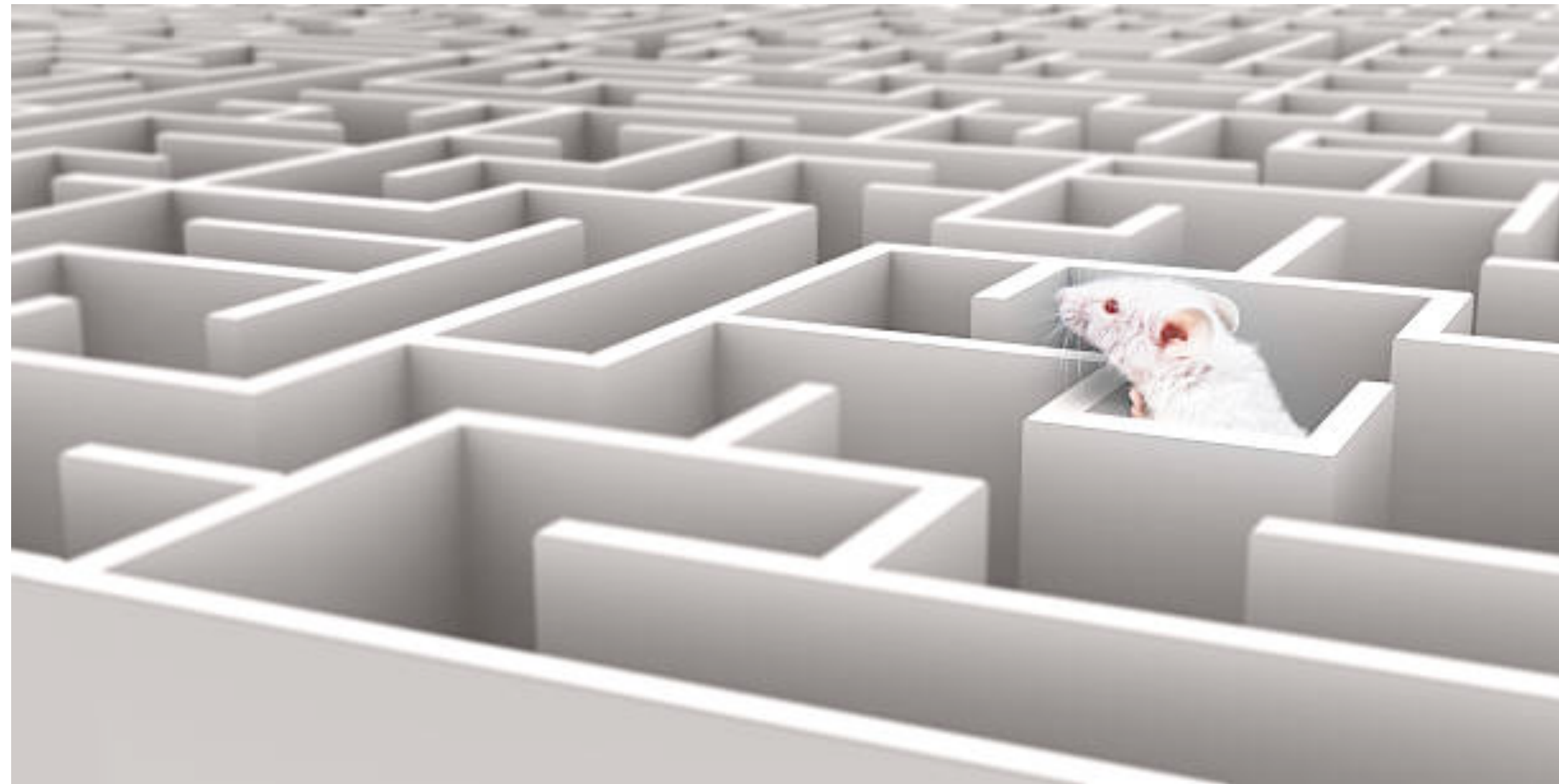
[https://en.wikipedia.org/wiki/](https://en.wikipedia.org/wiki/File:1543,_Andreas_Vesalius%27_Fabrica,_Base_Of_The_Brain.jpg)

File:1543,_Andreas_Vesalius%27_Fabrica,_Base_Of_The_Brain.jpg



Research Network

Neuroscience



by NeilLockhart, extracted from <https://media.istockphoto.com/photos/white-mouse-in-white-maze-looking-over-walls-picture-id636373812?k=6&m=636373812&s=612x612&w=0&h=iyeo-EXXgBKjfLR6sCb3K-RHotTYKqJHoxDYqP-sllg=>

Place Cell Model

[Curto and Itskov, 2008; Dabaghian et al, 2012]

- “Place cells” fire when the rat is near their locations.
- The rat remembers which cells fire together.

Connectivity of the co-firing network

v.s. connectivity of the navigation space

A. One-hole



B. Two holes



C. Quasi Linear



Figure 4 of [Dabaghian et al, 2012]

Connectivity of Real Data

[Dabaghian et al, 2014]

Connectivity of Real Data

[Dabaghian et al, 2014]

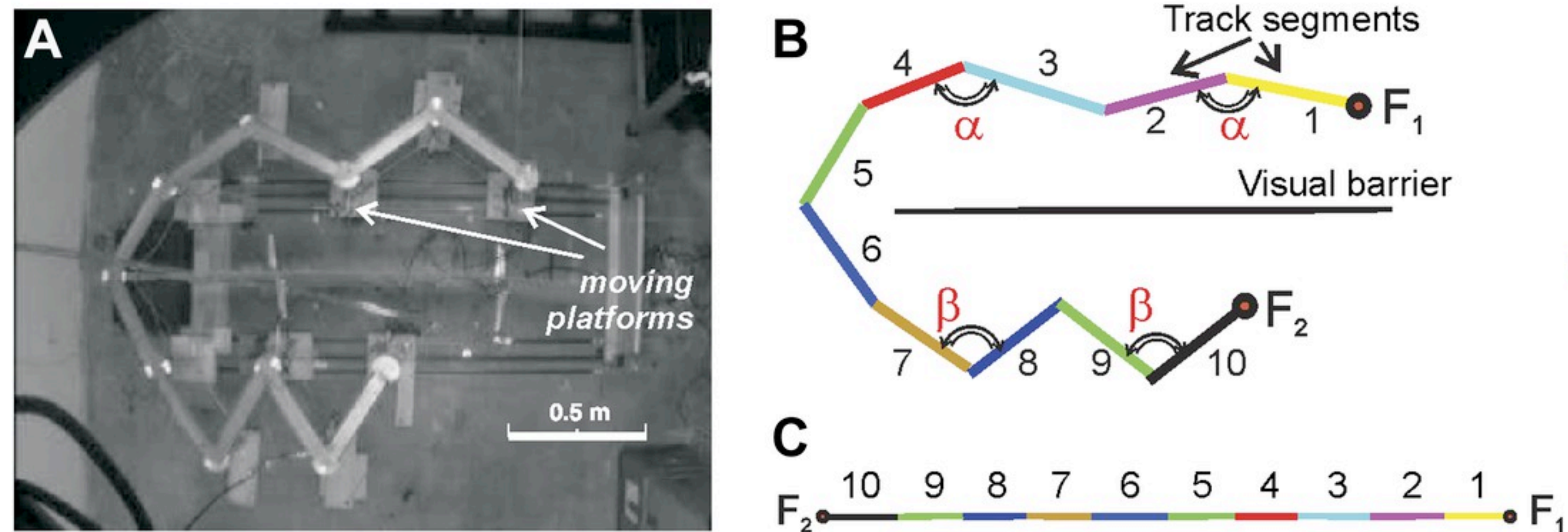


Fig 1 of [Dabaghian et al, 2014]

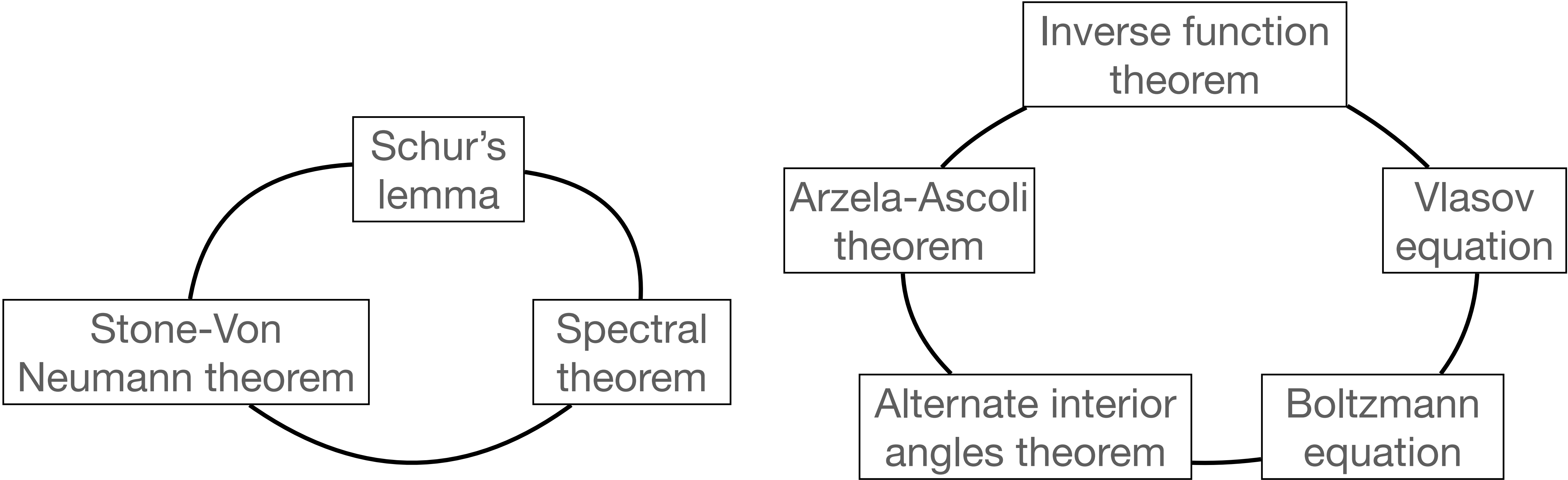
Research Network

[Salikov et al, 2018]

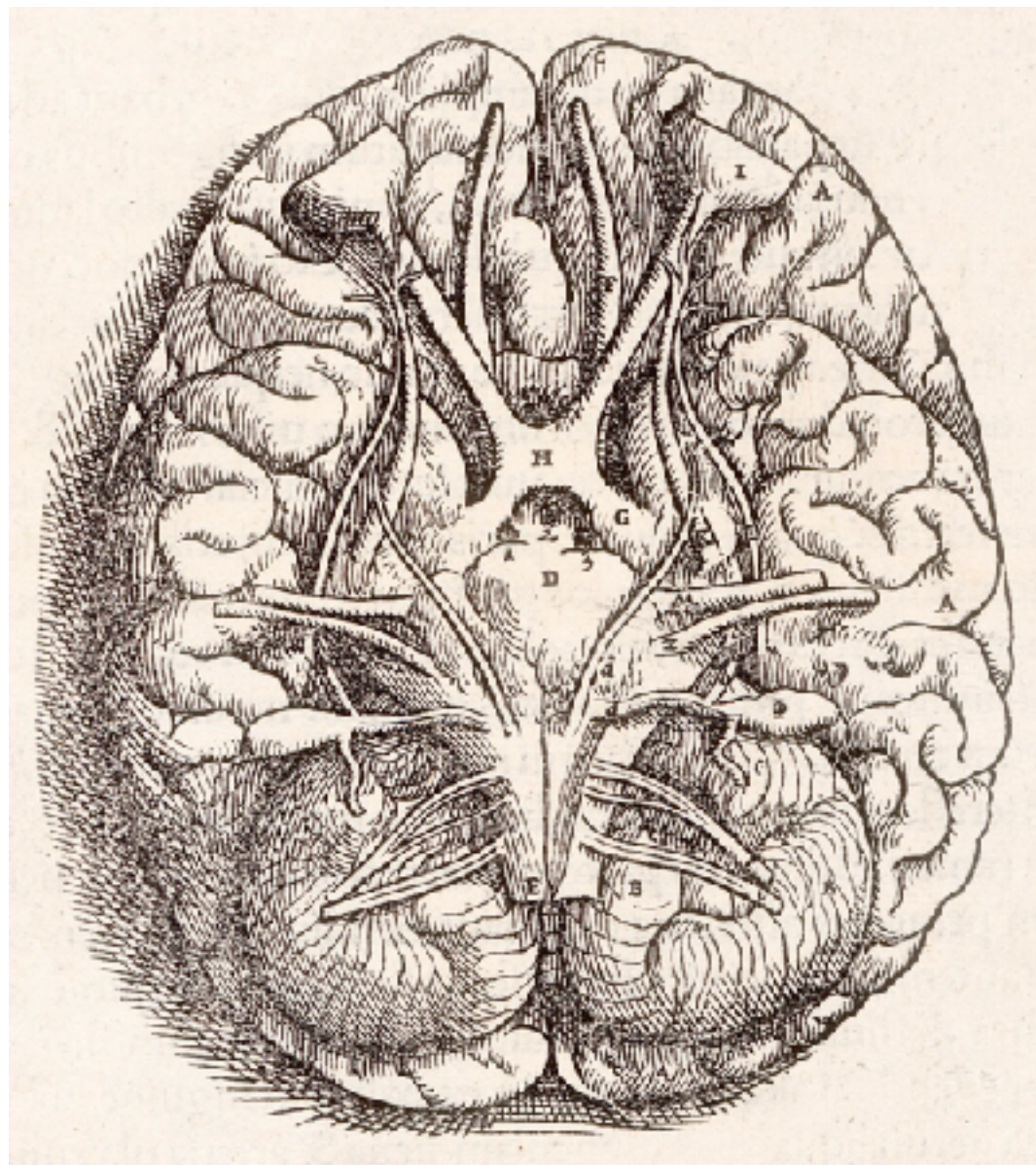
- Co-occurrence complex in Math research paper



Gap in Understanding



Topology and Networks



Neuroscience

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[https://en.wikipedia.org/wiki/](https://en.wikipedia.org/wiki/File:1543,_Andreas_Vesalius%27_Fabrica,_Base_Of_The_Brain.jpg)

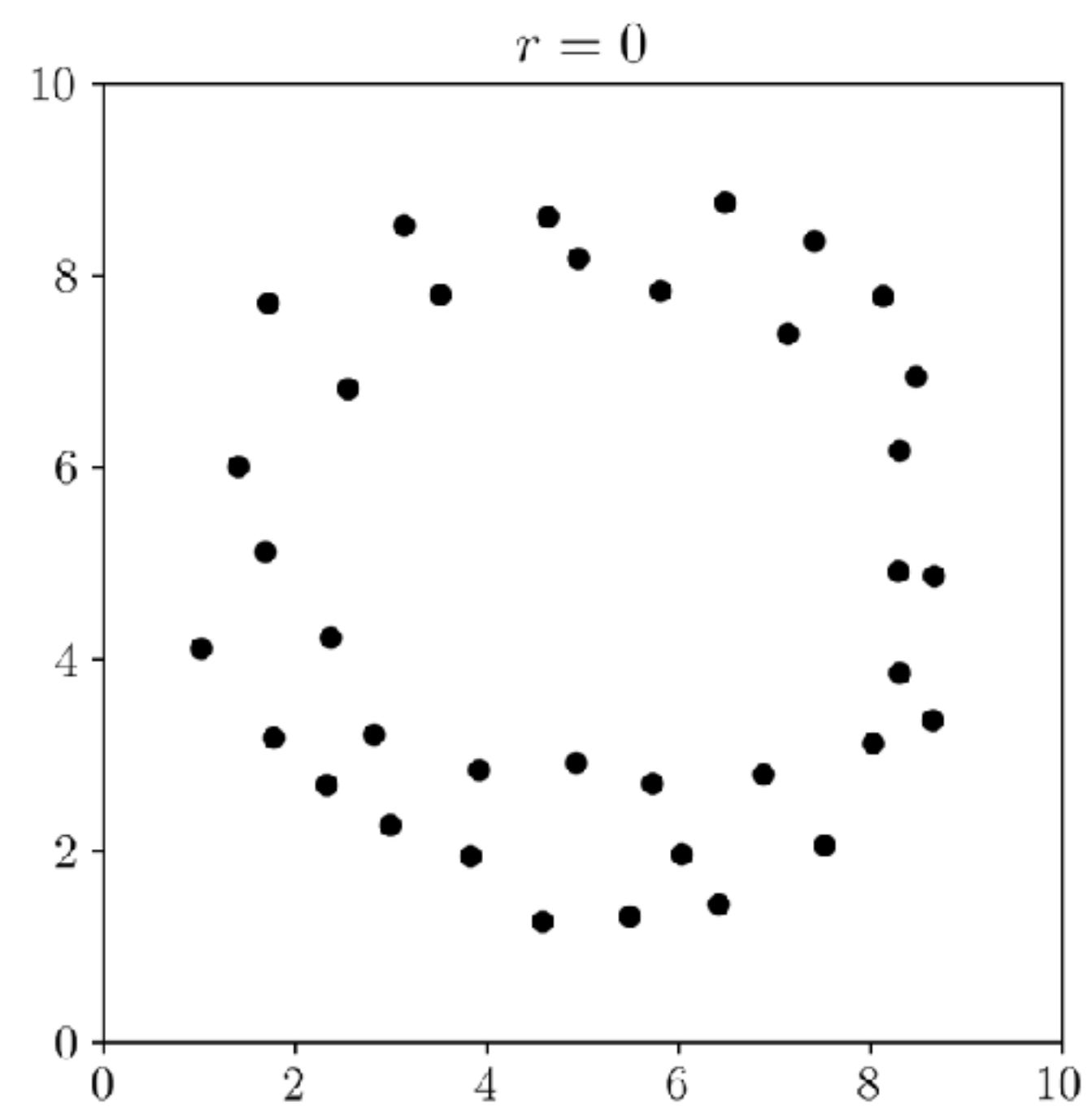
File:1543,_Andreas_Vesalius%27_Fabrica,_Base_Of_The_Brain.jpg

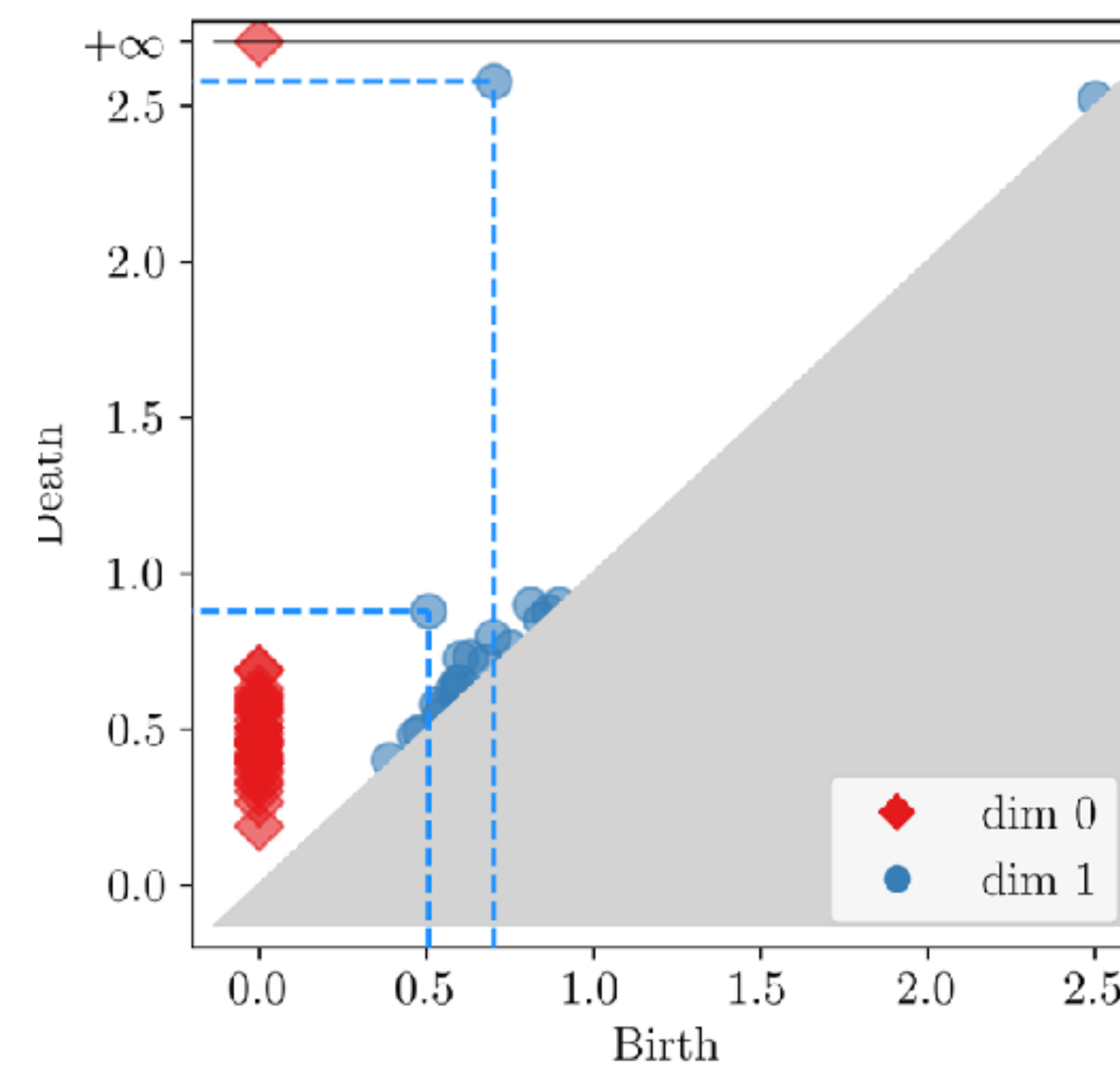
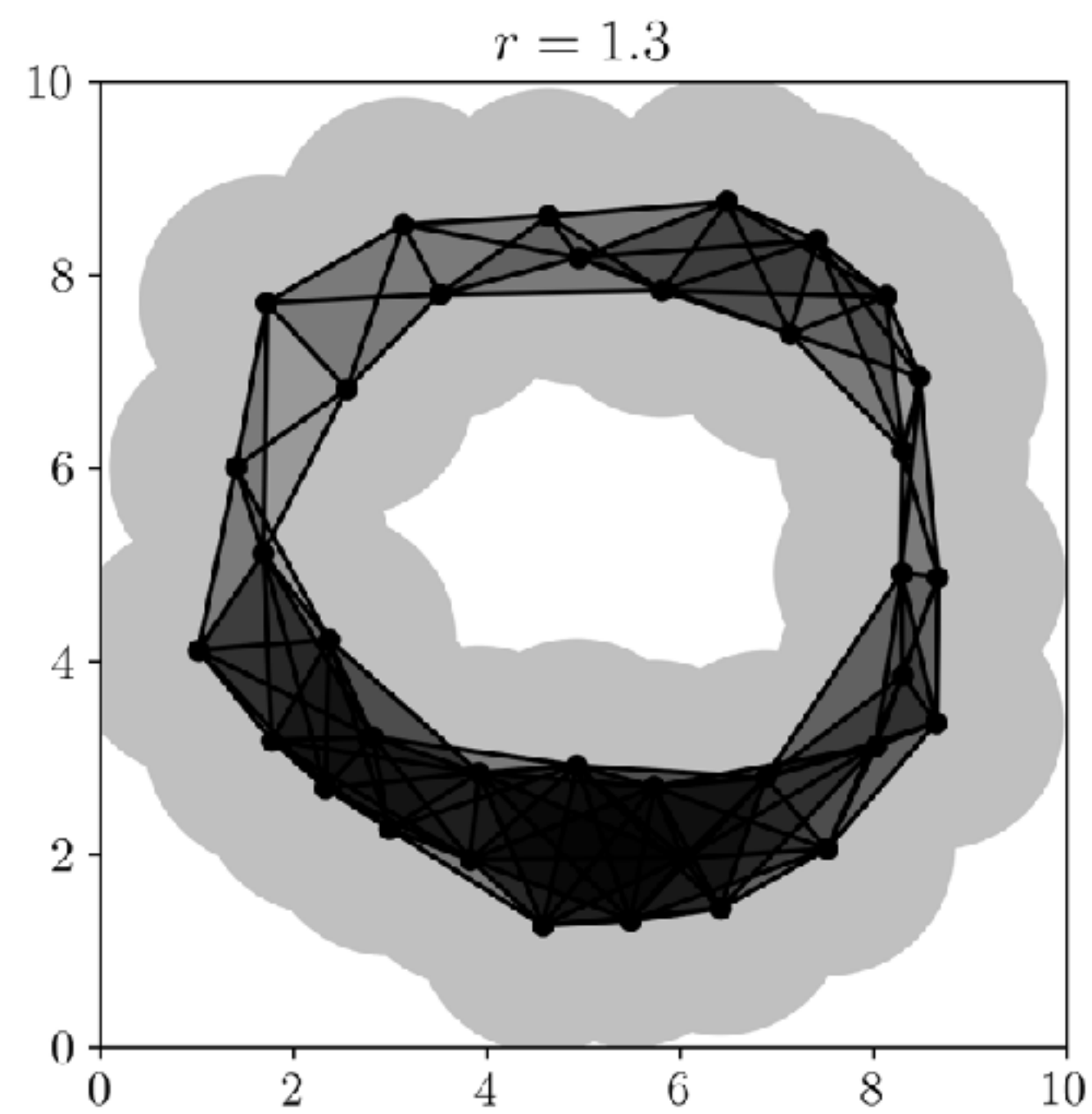
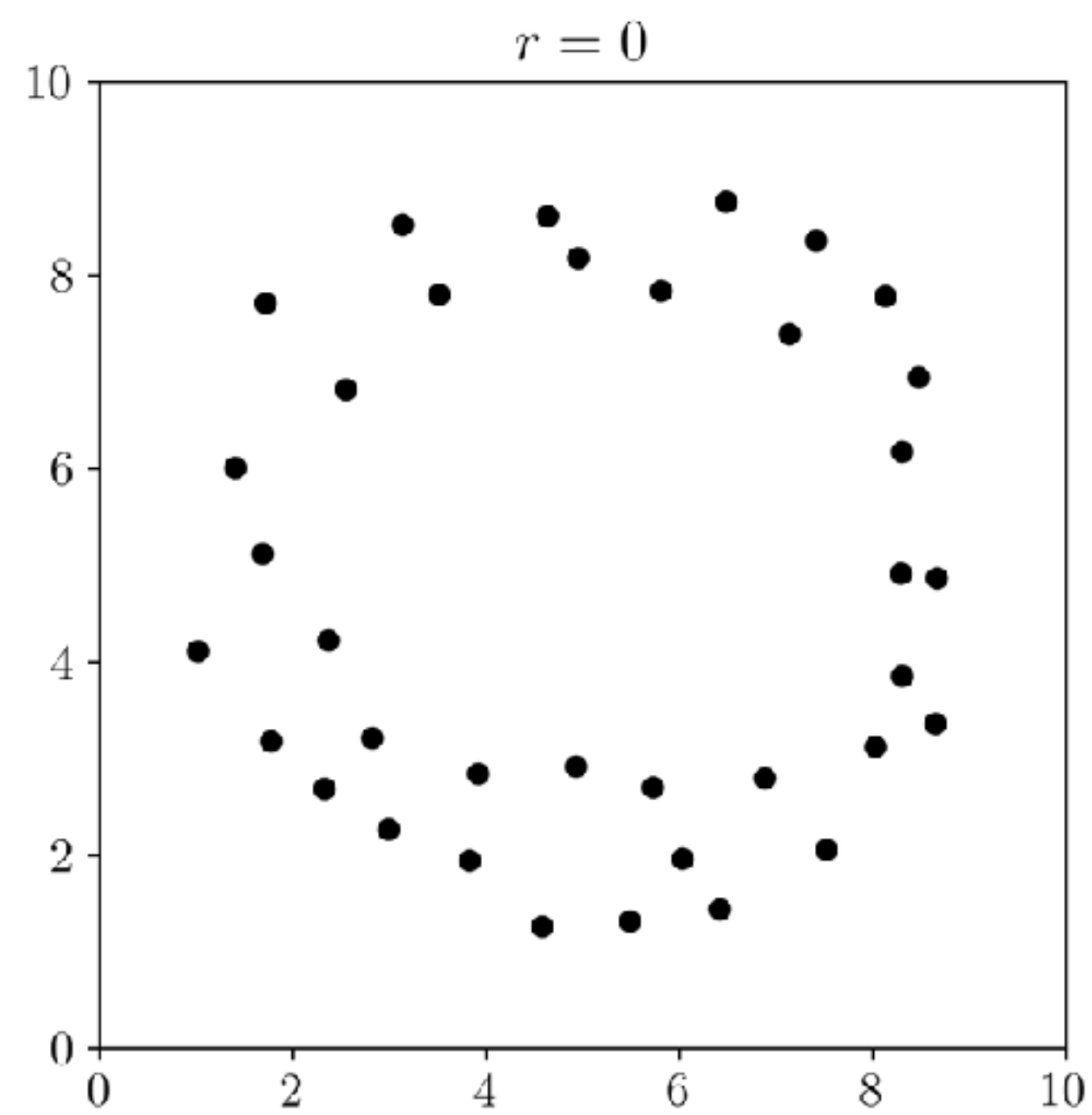


Research Network

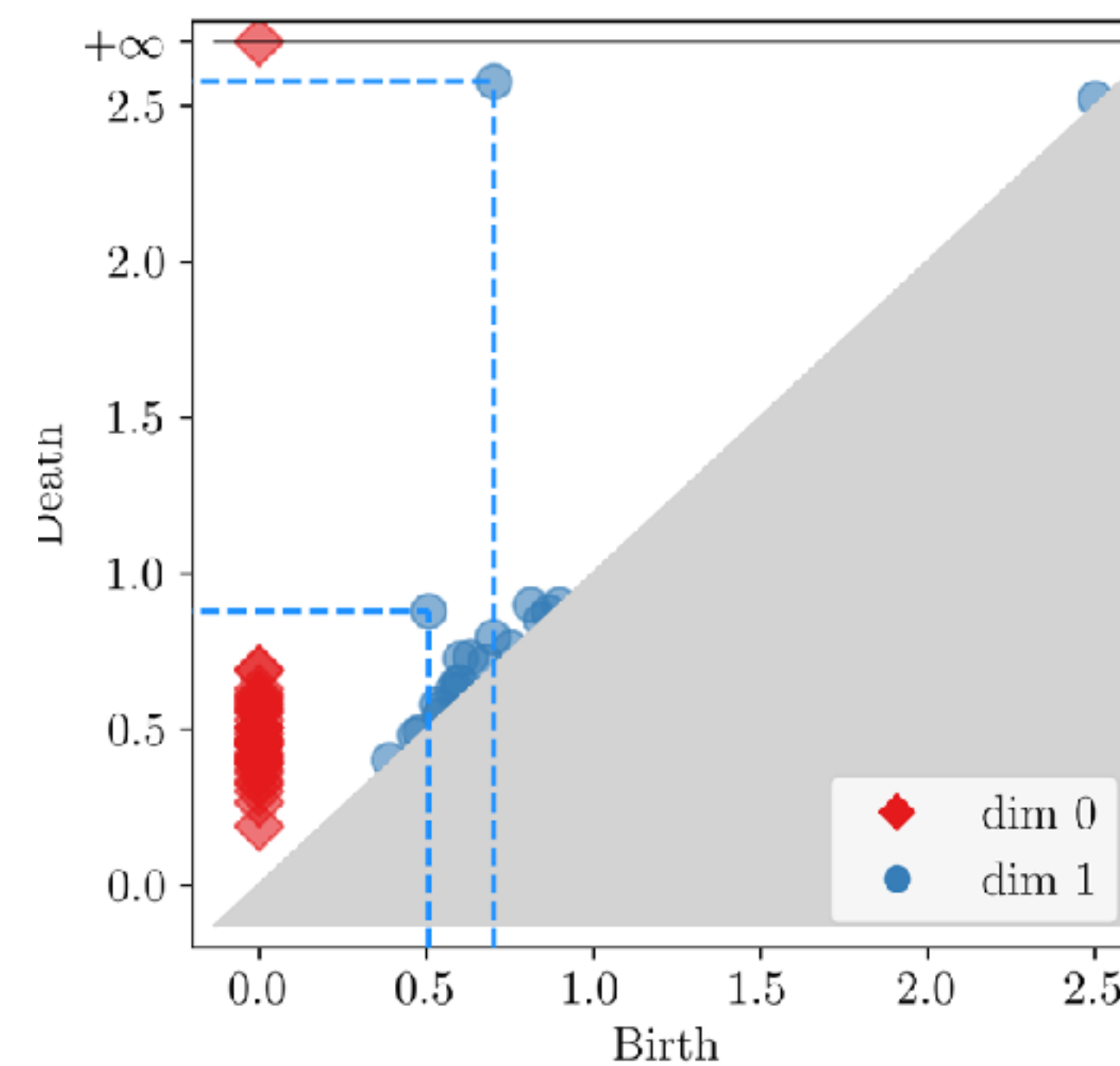
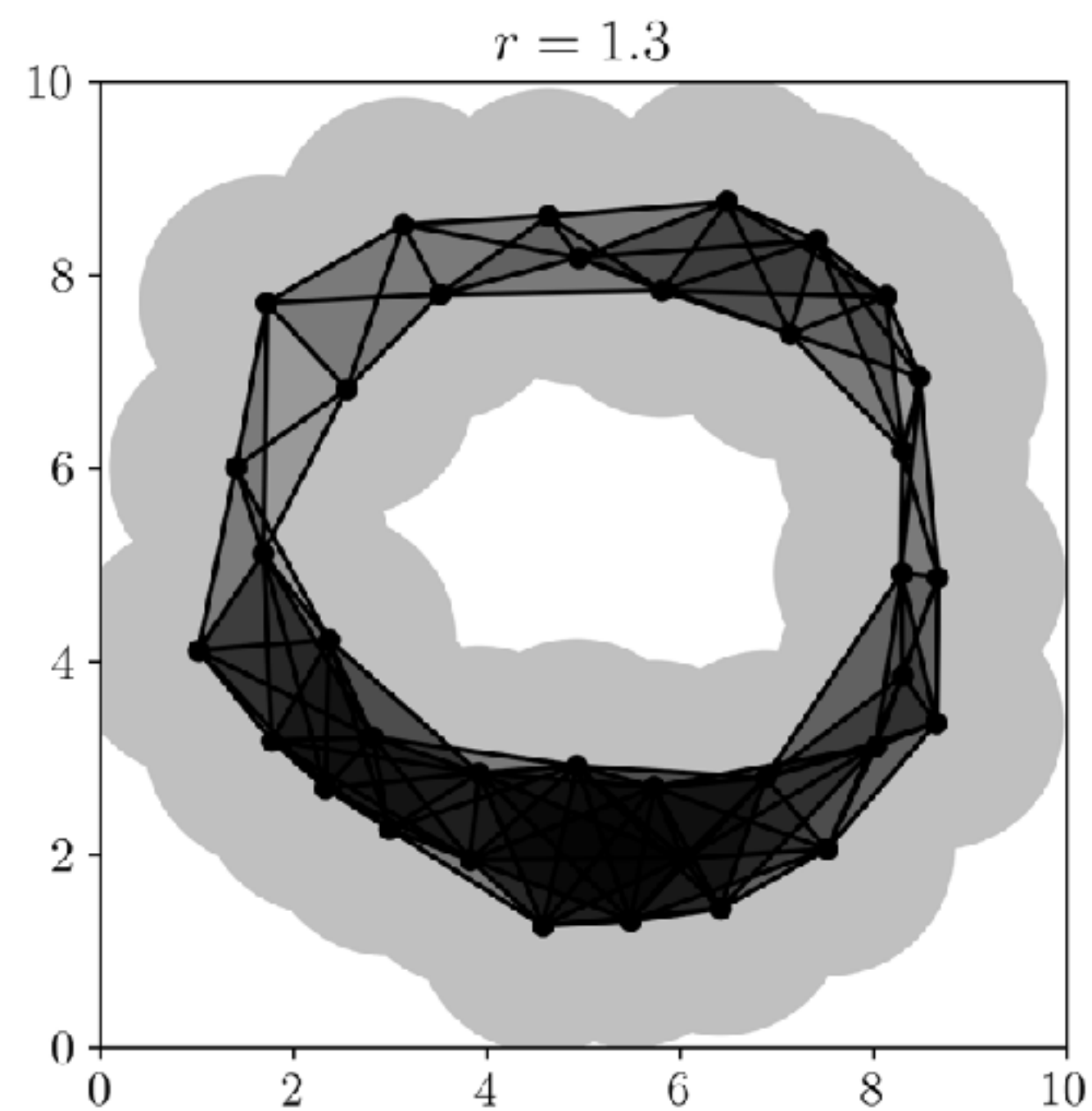
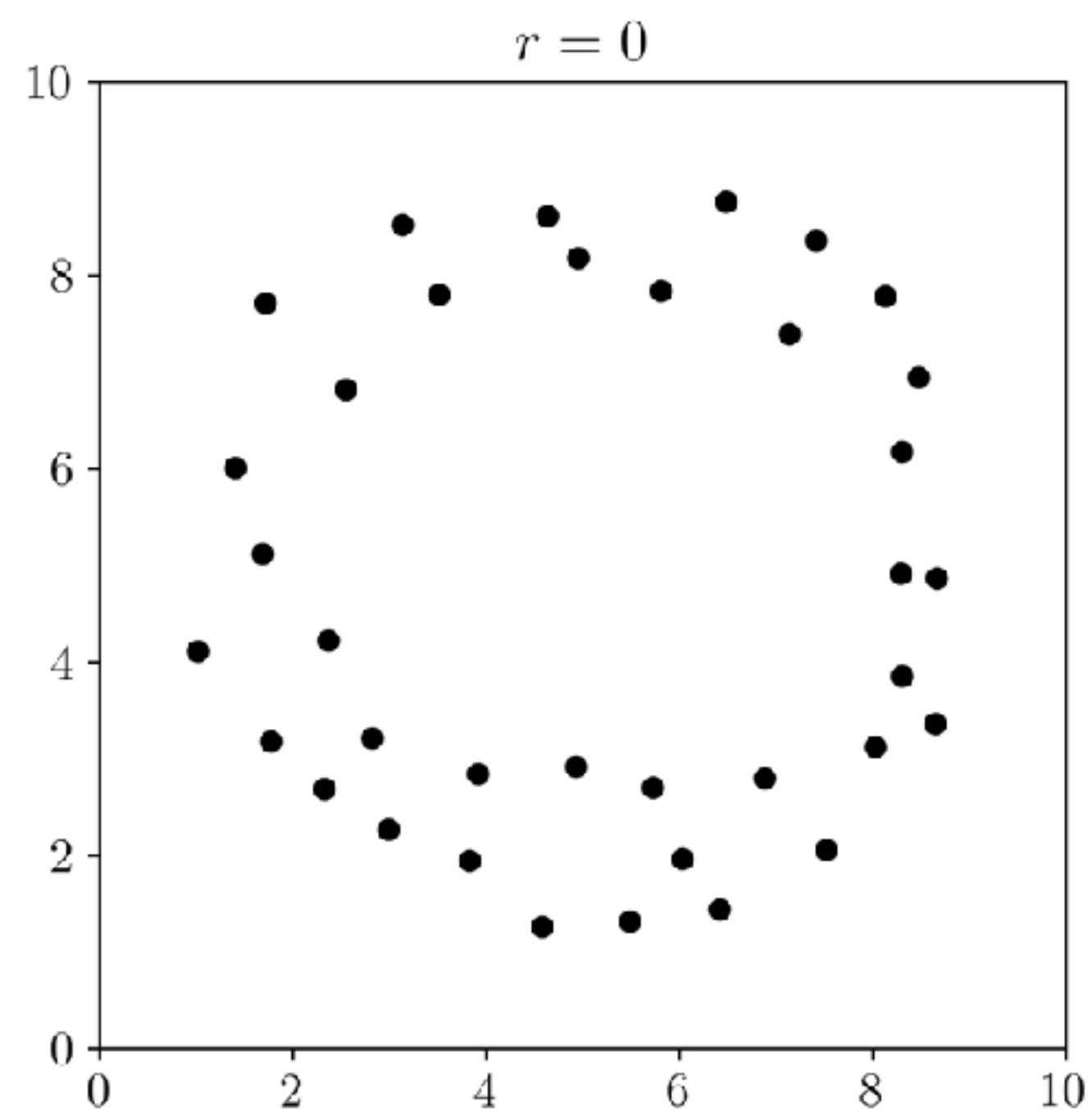
I. A Probabilist's Apology

Why Random Topology

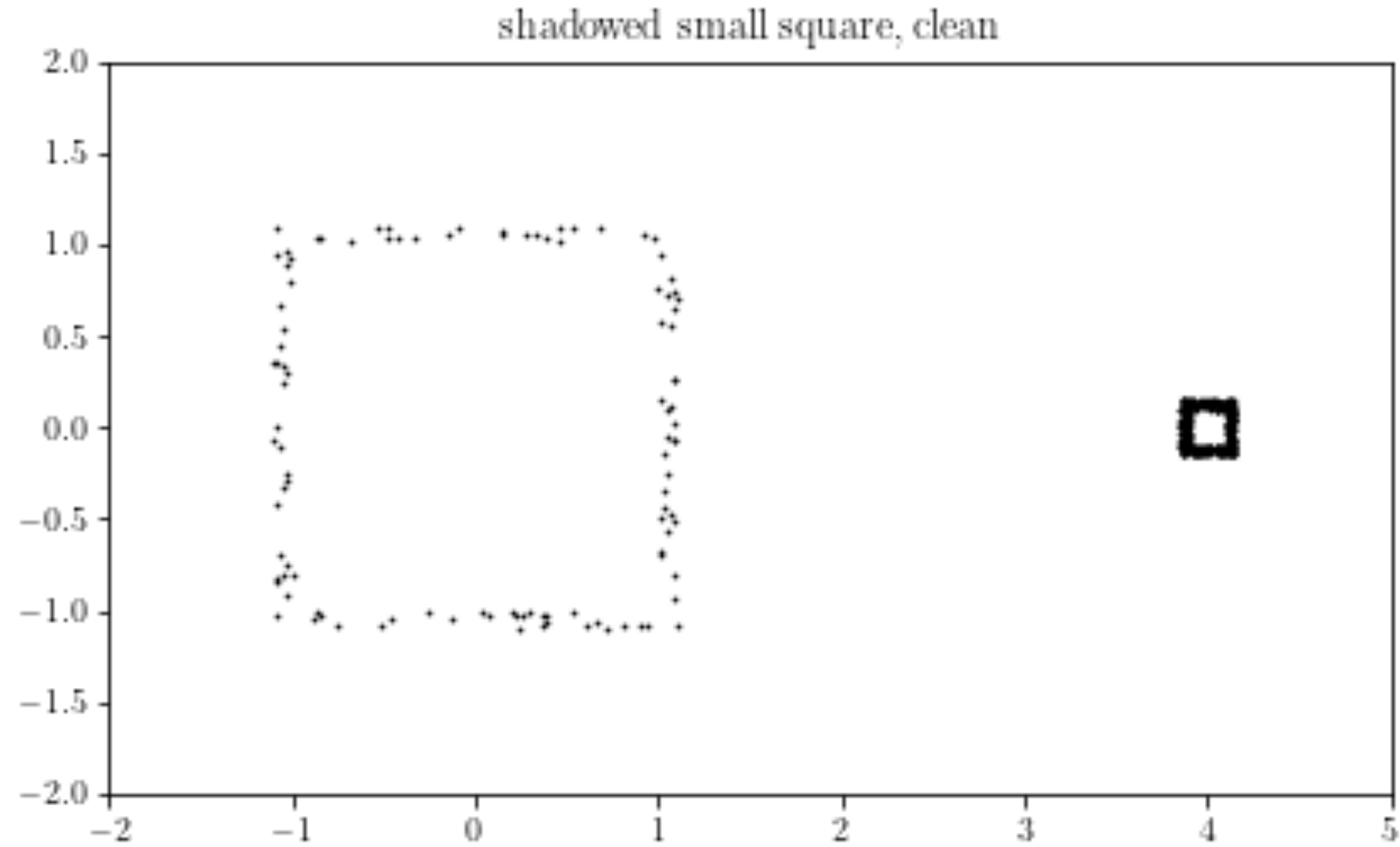




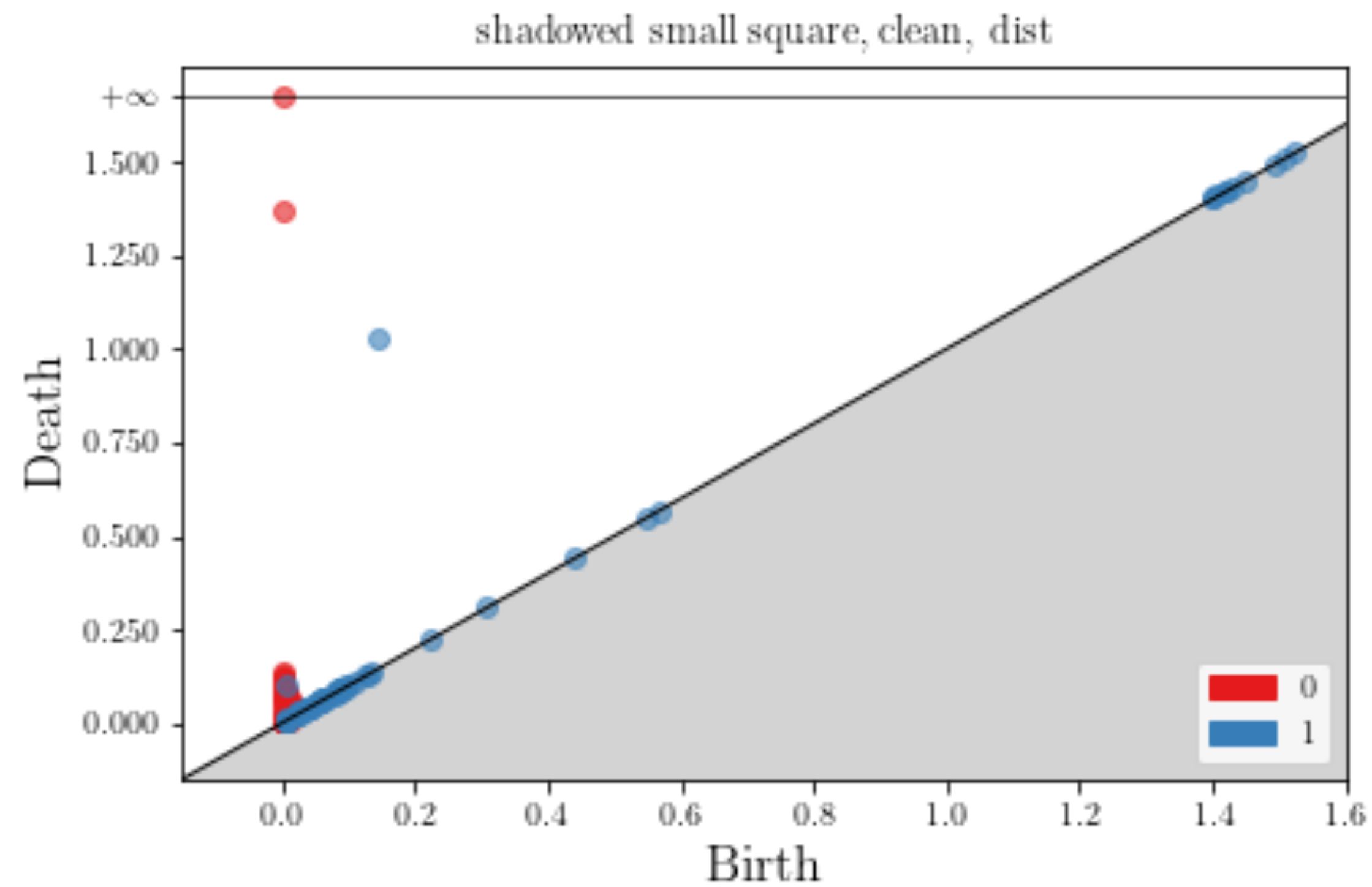
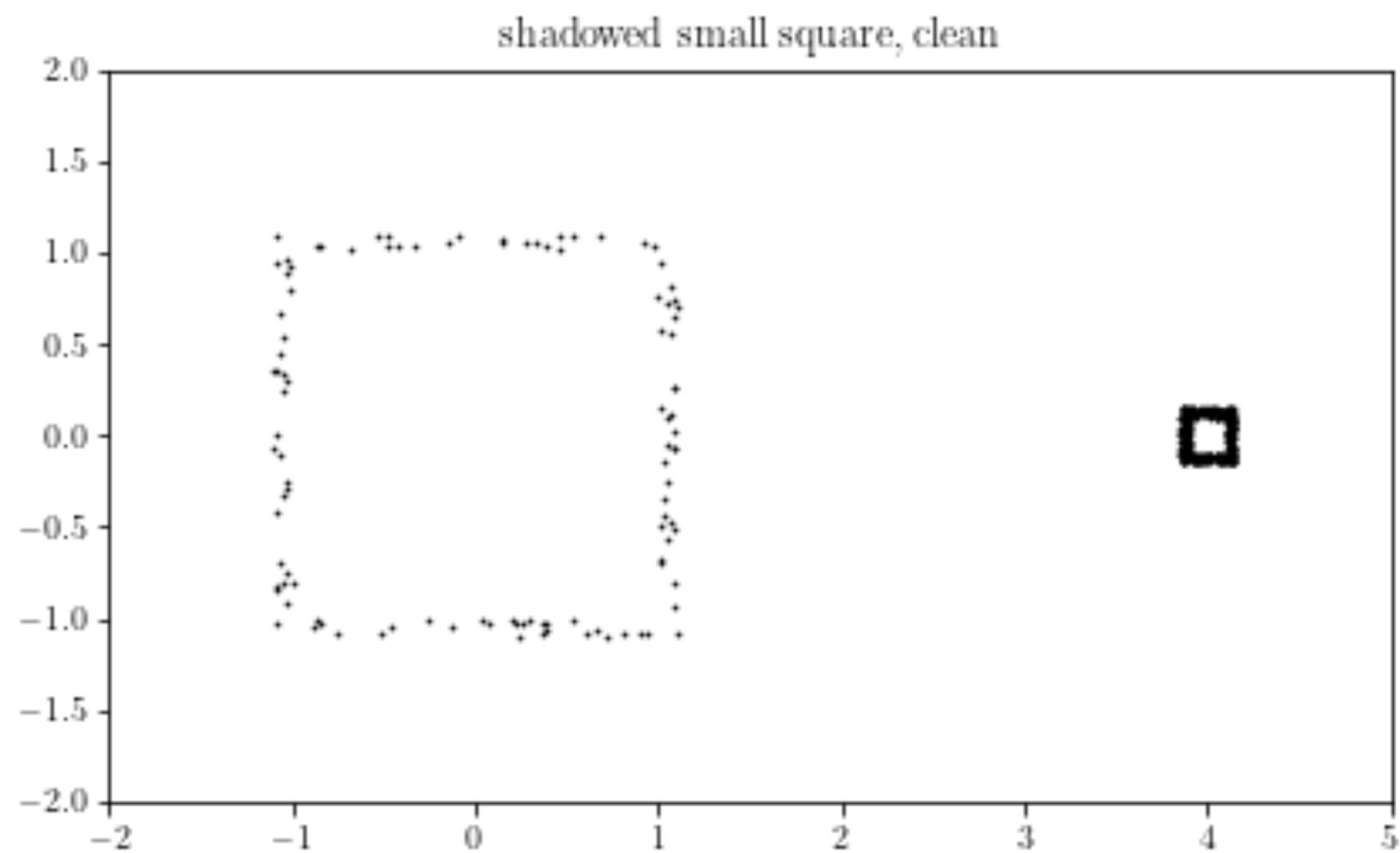
Size is Signal



Or is it?



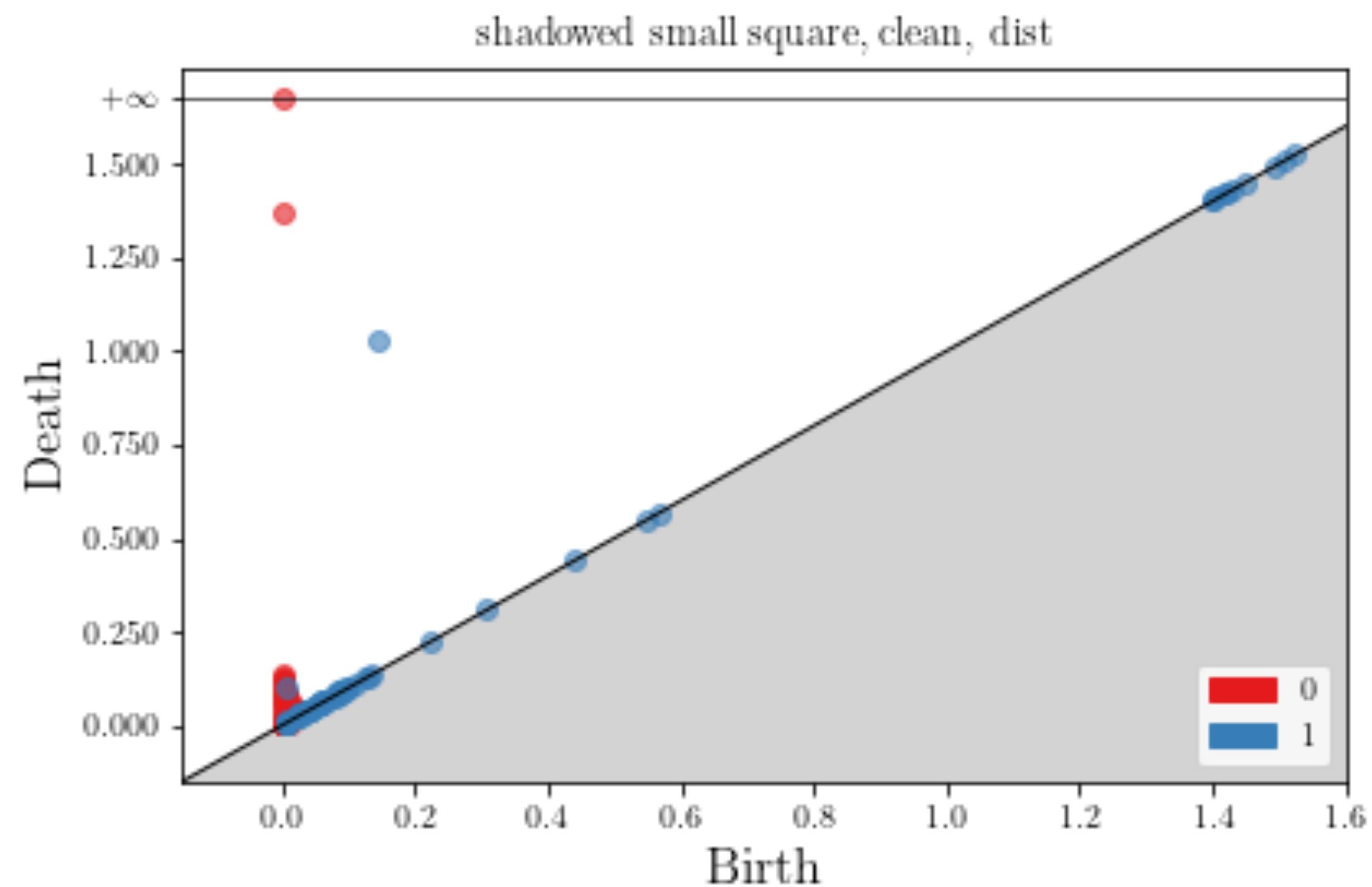
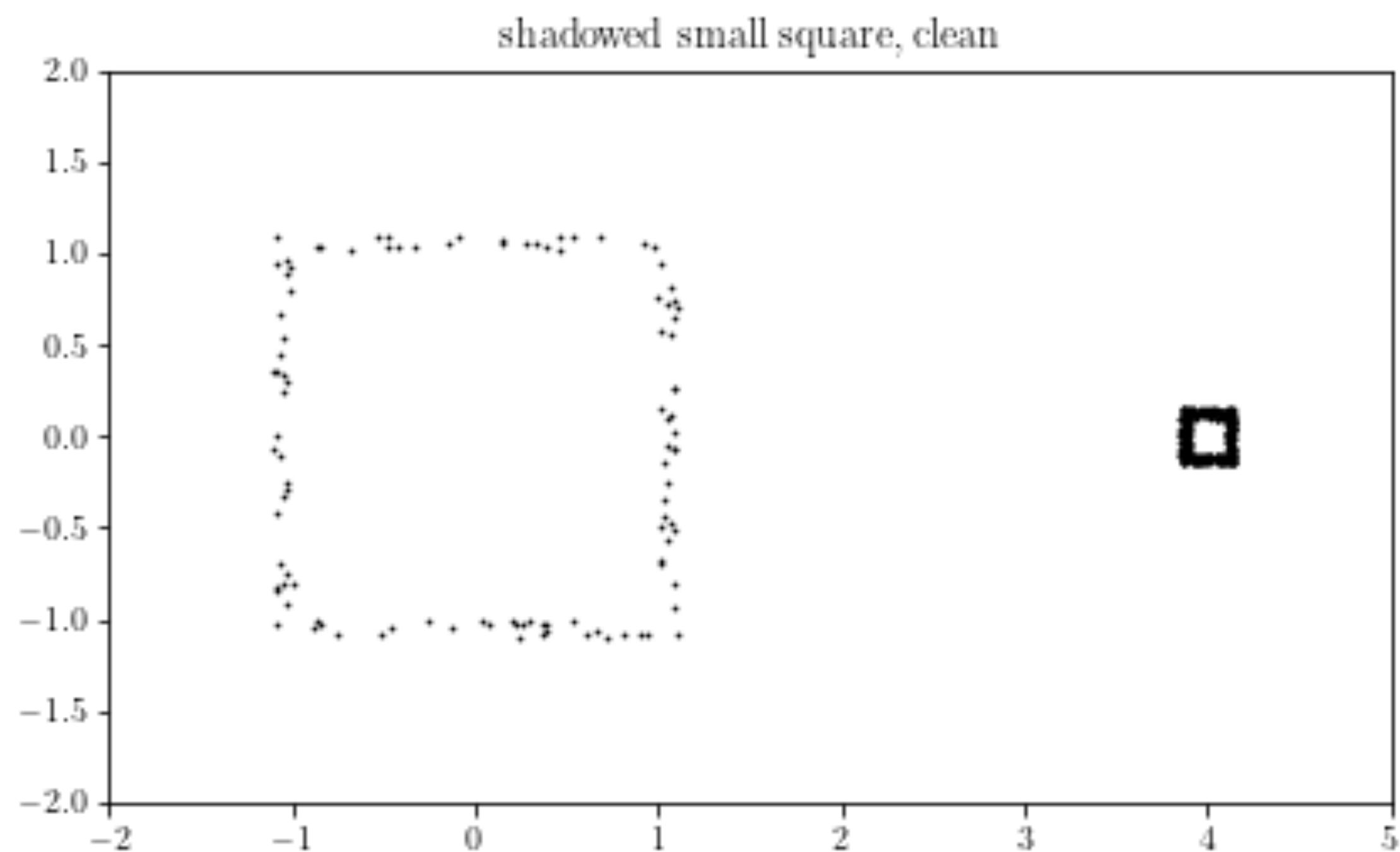
Or is it?



Size is Signal?

Surprise
~~Size~~ is Signal.

Random points don't do that.



Signal is what is not random.

**Signal is what is not random.
So what is random?**

Functional Convergence at dimension k ?

[Thomas and Owada 2020]

Functional Convergence at dimension k ?

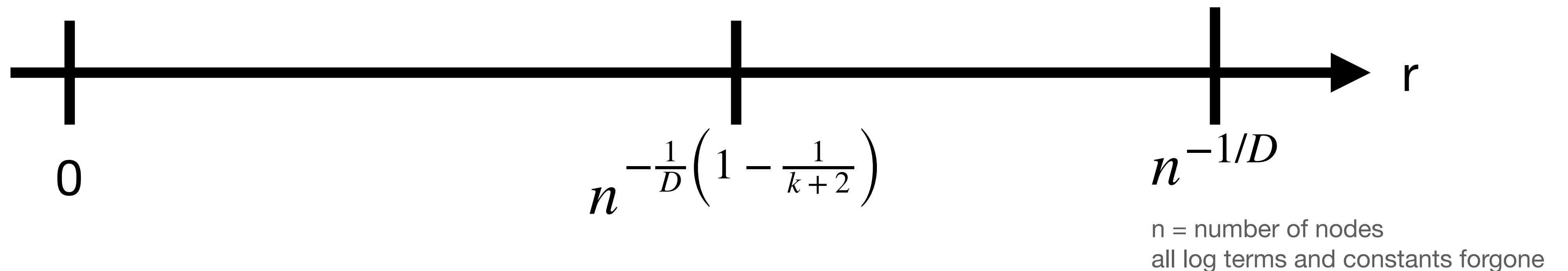
[Thomas and Owada 2020]

- Cech: weak convergence in finite-dimensional sense

Functional Convergence at dimension k ?

[Thomas and Owada 2020]

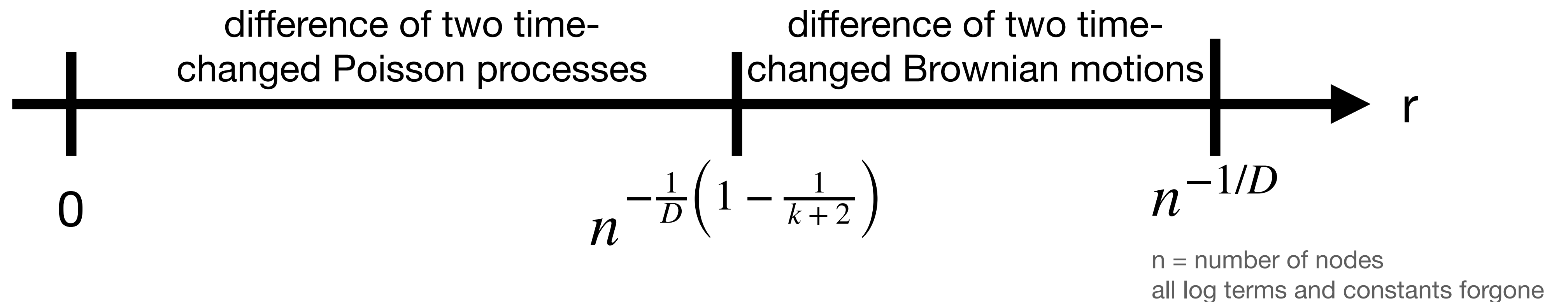
- Cech: weak convergence in finite-dimensional sense



Functional Convergence at dimension k ?

[Thomas and Owada 2020]

- Cech: weak convergence in finite-dimensional sense



Functional Convergence at dimension k?

[Thomas and Owada 2020]

- Cech: weak convergence in finite-dimensional sense

